THE STACKCALC LIBRARY • CURRICULUM SERIES — BOOK 1 OF 4

TEACHING WITH THE STACK

A Practical Concept-First Teacher's Guide to Mathematical Thinking, Physical Blocks & Four-Line Stack Verification

A unified K–12 facilitator's desk reference connecting state-mandated mathematics standards (Common Core CCSS, Texas TEKS, and district basal curricula) with our three published hands-on block studios—Place Value Studio (PLACE-01), Fraction Studio (FRAC-01), and Prime Factor Studio (PRIME-01)—our illustrated history-of-math books (From Blocks to the Stack and Thinking in the Stack), and the four-line StackCalc SC-6 and SC-32 calculators. Includes concept-first lesson plans, 30-second misconception remediations, and complete worked answer keys for all three student workbooks.

StackCalc

stackcalc.io

First Edition

BOOKS, BLOCKS & INSTRUMENTS IN THE STACKCALC LIBRARY

Illustrated Companion Trade Books (History of Mathematics & Stack Thinking)

- Volume I — From Blocks to the Stack: How Hands-On Blocks Make Numbers Click (Grades K–6 / StackCalc SC-6)
- Volume II — Thinking in the Stack: The Art of Calculating Without Parentheses (Grades 7–12, AP & Engineers / StackCalc SC-32)

K–12 Concept-First Curriculum & Student Workbooks (US Letter 8.5" × 11" Editions)

- Book 1 — Teaching with the Stack: Unified K–12 Master Teacher's Guide, Concept Lessons & Complete Answer Keys
- Book 2 — Hands-On Math Lab: K–12 Student Manipulative & Four-Line Stack Exploration Workbook
- Book 3 — Middle School Math Lab: Grades 6–8 Two-Method Student Practice Workbook (900 Problems) & 15-Page Coach Guide
- Book 4 — Stack Speed & Fluency: 15 Progressive Timed Sprints for Inside-Out Mathematical Fluency

Three Published Physical Manipulative Block Studios & Four-Line Stack Calculators (stackcalc.io)

- Place Value Studio (PLACE-01): 6×2 Whole & Decimal Trays, Digit Blocks (0–9) & Stackable Gold [+1] Carry Tokens
- Fraction Studio (FRAC-01): 84.0 mm Proportional Fraction Towers (1 Whole to 1/12) & 12-Socket Comparator Lid
- Prime Factor Studio (PRIME-01): Logarithmic Prime Blocks (2, 3, 5, 7 where equal products stand flush) & GCF/LCM Comparator
- StackCalc Calculators: SC-6 Elementary/Middle School Calculator & SC-32 Scientific Calculator (Hardware, iOS, iPadOS, watchOS & Web)

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TEACHING WITH THE STACK: A Practical Concept-First Teacher's Guide to Mathematical Thinking, Physical Blocks & Four-Line Stack Verification

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State & District Curriculum Alignment:

Designed to supplement state-mandated mathematics standards—including Common Core State Standards (CCSS-M), Texas Essential Knowledge and Skills (TEKS), Florida B.E.S.T., Virginia SOL, and AP Mathematics / Computer Science—alongside district basal textbooks (Eureka Math / EngageNY, Illustrative Mathematics / Open Up, enVision Mathematics, Big Ideas Math, HMH Into Math, McGraw-Hill Reveal, Bridges in Mathematics, and Singapore Dimensions Math).

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Preface: Why Teach Mathematics with a Stack?

How visible four-line stack thinking, physical blocks, and inside-out evaluation cure the black-box calculator problem.

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1. THE TWO CLASSROOM FRUSTRATIONS THIS PROGRAM WAS BUILT TO SOLVE

Every math teacher from Kindergarten through AP Calculus has watched students struggle with two connected habits:
• Fragile Rule Memorization Without Quantity Sense: Students memorize mechanical steps ('cross-multiply,' 'flip and multiply,' 'FOIL') without picturing the quantities—leading to classic errors like adding denominators ($1/3 + 1/4 = 2/7$) or losing negative signs.
• The 'Black-Box' Calculator Habit: When handed a standard 1-line calculator with an [=] key, students stop thinking about mathematical structure. They type long strings of buttons and parentheses from left to right—e.g., (8 + 4) ÷ (9 − 3) =—and blindly copy whatever decimal appears on the screen, with no way to see intermediate steps or spot a missing parenthesis.

2. WHY THE STACK IS NOT A CALCULATOR GIMMICK — IT IS PAPER-AND-PENCIL MATH MADE VISIBLE

Nobody goes looking for an 'RPN calculator curriculum' at first—until they see what happens when the [=] key and parentheses keys are removed. Think about how you already teach math by hand on paper or with blocks: you never operate on a number before the second number exists! You write the first quantity on a top Waiting Line (Line Y), write the second quantity below it on your active Workbench (Line X), and then apply the operation ($+, -, \times, \div$) to combine them.
In computer science and engineering, this visible step-by-step workspace is called a Stack (or Reverse Polish Notation, adopted by Hewlett-Packard for scientists and engineers in 1968). Because our 4-line screen (T, Z, Y, X) has no parentheses keys, a student cannot type an expression blindly from left to right. To evaluate $(8 + 4) \div (9 - 3)$, they must think mathematically from the inside out: compute $8 + 4 = 12$ first, watch 12 wait visibly on Line Y while computing $9 - 3 = 6$ on Line X, and then press [÷] to divide 12 by 6. The calculator stops being a shortcut around thinking and becomes an X-ray machine for order of operations.

3. HOW EVERY LESSON FLOWS: 1. CONCEPT (STACK) → 2. BLOCKS (WHEN APPLICABLE) → 3. CALCULATOR

Step 1 — Explain the Mathematical Concept in Stack Terms (Never Start by Pushing Buttons)

Every lesson begins with the pure mathematical concept: how a quantity is composed, factored, or compared, which value waits on Waiting Shelf Y, and which value is built or transformed on Workbench X.

Step 2 — Model with Our Three Published Block Kits (Only Where Physical Blocks Truly Fit)

We publish three precision-engineered classroom block kits: Place Value Studio (0–9 digit blocks & [+1] carry tokens on a 6×2 tray mirroring Lines Y and X), Fraction Studio (84 mm proportional fraction towers where common denominators stand dead-flush), and Prime Factor Studio (logarithmic prime blocks 2, 3, 5, 7 where equal products have equal tower height). For upper-level topics like trigonometry where plastic blocks would feel contrived, we use honest geometric diagrams instead.

Step 3 — Predict by Hand, Then Verify on the SC-6 or SC-32 Calculator & Companion Books

Students write their paper-and-pencil prediction first, then verify step-by-step on the SC-6 Elementary/Middle School Calculator (Grades K–8) or SC-32 Scientific Calculator (Grades 7–12 & AP). Each unit also connects to a true historical story in our illustrated companion trade books, From Blocks to the Stack and Thinking in the Stack.

Using This Guide with Your Mandated Curriculum

How public, charter, private, and homeschool teachers plug stack concepts into their required textbook without changing their pacing calendar.

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1. YOU DO NOT HAVE TO CHANGE YOUR DISTRICT TEXTBOOK OR PACING CALENDAR

Most school teachers do not get to choose their curriculum: you are required by your district and state to teach Common Core (CCSS), Texas TEKS, Florida B.E.S.T., or Virginia SOL using an adopted basal series—such as Eureka Math / EngageNY, Illustrative Mathematics / Open Up, enVision Mathematics, Big Ideas Math, HMH Into Math, McGraw-Hill Reveal, or Bridges in Mathematics.

Because every state standard and district textbook teaches the same 12 foundational mathematical concepts (simply in different chapter orders), Part I of this Teacher's Guide is organized by Universal Concept Domain (with a complete 45-section Middle School companion in Part II). Whatever chapter your district pacing calendar mandates this week, you can open directly to that concept domain in seconds.

2. FOUR PRACTICAL WAYS TO PLUG STACK LESSONS INTO YOUR EXISTING CLASSROOM

Way A: The 10-Minute Concept Launch (Before Opening Your District Textbook)

Start your daily lesson with the Domain's Classroom Hook, physical block setup, and 4-line stack trace on the board. Once students can see why the operation works on Waiting Shelf Y and Workbench X, transition directly into your mandated district textbook problems.

Way B: The Paper-First + Stack Verification Routine (During Daily Homework & Classwork)

Have students solve their assigned district textbook problems by hand first, then use the SC-6 or SC-32 calculator to check intermediate steps (e.g., checking remainders with [÷R], common denominators with Two-Dot fractions, or equation balance with [!=]).

Way C: 20-Minute Tier-2 Intervention & Small-Group Math Station

When students struggle with abstract rules during a unit, pull the Domain's Target Misconceptions & Hands-On Block Remediations box to run a targeted 20-minute small-group station using Place Value, Fraction, or Prime Factor blocks.

Way D: Dedicated Friday STEM Math Lab or Homeschool Core Sequence

Pair each concept domain with our three student workbooks: Book 2 (Hands-On Math Lab, 12 guided K–12 investigations), Book 3 (Middle School Math Lab, 90 two-method worksheets), and Book 4 (Stack Speed & Fluency, 15 timed 10-minute sprints).

3. COMPLETE 4-BOOK CURRICULUM LINEUP & COMPANION STORYBOOKS AT A GLANCE

Book 1 (This Guide) • Teaching with the Stack:	Master K–12 Teacher's Guide by Concept (Part I: 12 Domains; Part II: 45 Middle School Sections; Part III: All Keys).
Book 2 (Student) • Hands-On Math Lab:	K–12 Student Exploration Workbook (12 two-page concept labs with story openers, block sketches & stack traces).
Book 3 (Student) • Middle School Math Lab:	Grades 6–8 Daily Practice Workbook (10 units, 45 sections, 90 Practice A & B worksheets, 900 two-method problems).
Book 4 (Student) • Stack Speed & Fluency:	Grades 3–12 Timed Proficiency Trials (15 UIL-style 10-minute RPN speed sprints, tactics & progress graph).
Companion Trade Books (Amazon / stackcalc.io):	From Blocks to the Stack (Grades K–6 / SC-6) & Thinking in the Stack (Grades 7–12 / SC-32) illustrated manuals.

Why There Is No [=] Key: You Already Teach RPN

If you grew up with a standard algebraic calculator, here is why stack-based math matches your classroom teaching.

IF YOU GREW UP WITH AN [=] BUTTON, READ THIS FIRST!

Almost every teacher grew up using an algebraic calculator where you type 8 + 5 =. When educators first see Reverse Polish Notation (RPN)—an [ENTER] key and no [=] button—their natural reaction is hesitation: 'Why is the plus sign at the end? I don't have time to learn or teach a strange calculator system!' Here is the reassuring truth: RPN is not a new way to do math. It is the exact order in which you already teach math with physical blocks and on the whiteboard every day.

1. THE 'NOUN, NOUN, VERB' SECRET: PHYSICAL BLOCKS & PAPER ARE ALREADY RPN

How You Teach with Physical Blocks

1. 'Put 8 blocks on the tray.'
2. 'Now bring 5 more blocks.'
3. 'Now push them together (+)!'

You can't push blocks together before the second group is on the desk!

How You Teach Vertical Paper Math

1. Write 8 on the top line (Line Y).
2. Write 5 on the line below (Line X).
3. Draw the line and add (+ → 13).

Both numbers exist on paper before the operation happens!

How the 4-Line Stack Calculator Works

1. Type [8] [ENTER] → 8 sits on Line Y.
2. Type [5] → 5 sits on Line X.
3. Press [+] → adds Y + X = 13!

Line Y is the Waiting Shelf; Line X is the Workbench. Zero hidden state!

2. HOW THE STACK CURES THE #1 CLASSROOM CALCULATOR HEADACHE (ORDER OF OPERATIONS)

STANDARD [=] CALCULATOR ON (8 + 4) / (9 - 3):

• The Trap: A student types 8 + 4 / 9 - 3 = left-to-right and gets 5.444... (WRONG!) because the calculator silently divided 4 / 9 first.
• Or Parentheses Soup: Students blindly copy ((8+4)/(9-3)) without knowing which operation happens first.
• Hidden Mistakes: A 1-line screen erases each number as soon as + is pressed, so teachers cannot see where a student miskeyed.

4-LINE STACK CALCULATOR ON (8 + 4) / (9 - 3):

• True Inside-Out PEMDAS: With no parentheses keys, the student MUST decide what to solve first!
• Room 1: [8] [ENTER] [4] [+] → 12 waits visibly on Line Y.
• Room 2: [9] [ENTER] [3] [-] → 6 sits visibly on Line X.
• Combine: Both 12 and 6 are visible together! Press [/] → 12 / 6 = 2. It mirrors paper-and-pencil steps!

3. ANSWERING THE THREE BIGGEST TEACHER WORRIES

Worry 1: 'Will this confuse my students when they take state tests with a standard calculator?'

Answer: The exact opposite—it makes them dramatically better! Because the stack has no parentheses keys, students learn to decompose expressions from the inside out using true PEMDAS priority. When handed a basic state-test calculator, they already know which sub-expression to compute first.

Worry 2: 'What if a student gets stuck mid-problem and asks me for help?'

Answer: The 4-line screen (T, Z, Y, X) is a 2-second teacher X-ray. Instead of a mysterious single number on a screen, you can immediately see their first quantity sitting on Line Y and their second quantity on Line X.

Worry 3: 'How long will it take me to feel comfortable teaching with it?'

Answer: About 3 minutes! Turn to Page vii right now and try the 4 hands-on teacher exercises before your first class.

The 3-Minute Teacher Hands-On RPN Primer

Three golden rules, four 45-second try-it exercises, and a word-for-word Day 1 script to read to your class.

THE 3 GOLDEN RULES OF THE STACK (EVERYTHING YOU NEED TO KNOW)

1. [ENTER] = 'Put It on the Waiting Shelf': Press [ENTER] ONLY to separate two numbers you type back-to-back.

2. Action Keys Work Immediately: Press [+], [-], [x], or [/] AFTER both numbers are in place—never press [ENTER] after an operator!

3. Automatic Stack Lift: When you type a new number after an operation, your previous answer automatically lifts to Line Y!

TRY THESE 4 EXERCISES ON YOUR CALCULATOR RIGHT NOW (3 MINUTES TOTAL)

Try-It 1 (30 sec): Basic Addition — 8 + 5 = 13

Type [8] [ENTER] to lift 8 onto Shelf Y. Type [5] on Workbench X. Press [+] → 8 + 5 = 13!

Press Keys: 8 ENTER 5 +

Try-It 2 (45 sec): Two-Group PEMDAS Without Parentheses — (8 + 4) × (9 - 3) = 72

First group: [8] [ENTER] [4] [+] gives 12. Just start typing [9] [ENTER] [3] [-]! 12 lifts to Y while 6 forms on X. Press [x] → 72!

Press Keys: 8 ENTER 4 + 9 ENTER 3 - ×

Try-It 3 (45 sec): Two-Dot Fractions on the SC-6 — 1/3 + 1/4 = 7/12

Pressing [.] twice enters a fraction ([.] [1] [.] [3] = 1/3). Press [ENTER], type 1/4, and press [+] → displays 7/12 as a real fraction!

Press Keys: • 1 • 3 ENTER • 1 • 4 +

Try-It 4 (45 sec): Fixing Reversed Numbers Without Starting Over — 20 ÷ 4 = 5

Typed 4 then 20 when you meant 20 / 4? Never clear and start over! Press the Swap key to flip Y and X (Y: 20, X: 4), then press [/] → 5!

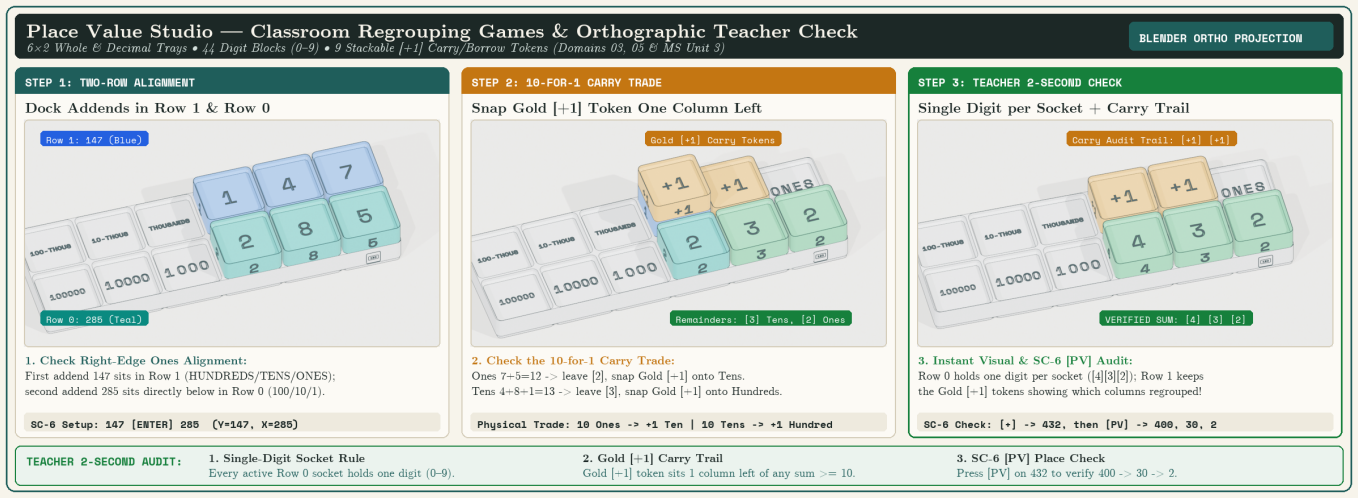
Press Keys: 4 ENTER 2 0 ↑↓ ÷

YOUR 60-SECOND DAY 1 SCRIPT (READ ALOUD TO INTRODUCE THE STACK TO STUDENTS)

"Class, look at your calculator—notice there is no equals sign and no parentheses! That's because this calculator works just like your two hands on your desk. The bottom line, Line X, is your Workbench where you type a number. The line right above it, Line Y, is your Waiting Shelf. When we want to add 8 + 5, we first type 8 and press [ENTER] to push 8 up onto the Waiting Shelf. Then we type 5 on our Workbench, and press [+] to combine them together! Put the two numbers on the shelves first, then press the action button!"

Place Value Studio: Games & Visual Teacher Check

Classroom games and 2-second orthographic visual checks for the 6×2 Whole/Decimal Trays, Digit Blocks (0–9), and [+1] Carry Tokens.



CLASSROOM GAMES & HANDS-ON ACTIVITIES (TIED TO CONCEPT DOMAINS & WORKBOOKS)

Game 1: 'Race to 1,000 Carry Relay' (Domains 03 & 05 • Grades 2–5 • 10-Min Partner Relay)

Partners start at 000 on the 6×2 Whole Number Tray. Take turns rolling two dice (or spinning 0–9) to form a 2-digit or 3-digit addend in Row 0 (100/10/1) and add it to the running total in Row 1. Whenever a column sum reaches 10+, the player calls 'Regroup!', leaves the ones digit in Row 0, and snaps a Gold [+1] Carry Token one column left. First pair to click [1] into THOUSANDS (1000) and verify on the SC-6 wins!

Check Keys (e.g., 147 + 285 = 432): `1` `4` `7` `ENTER` `2` `8` `5` `+` `123`

Game 2: 'The Decimal Point Shifter (×10 & ÷10 Slide)' (Domain 05 & MS Unit 03 • Grades 4–6 • 10 Min)

Dock the Whole Number Tray (100,000 to 1) and Decimal Tray (0.1 to 0.000001) side-by-side at the raised tactile decimal point (•). Call out a starting decimal (e.g., 4.75) and a power-of-10 action (× 10, × 100, or ÷ 10). Students physically slide every digit block one or two columns left (for ×10/×100) or right (for ÷10) across the stationary decimal point—seeing why digits shift columns rather than the point moving!

Check Keys (4.75 × 10 = 47.5): `4` `•` `7` `5` `ENTER` `1` `0` `×` `123`

Game 3: 'The Borrowing Bank Vault (Zero-Placeholder Heist)' (Domain 05 • Grades 3–6 • 12 Min)

Set a 'Vault Target' containing zeros in Row 1 (e.g., 502) and a subtraction code in Row 0 (e.g., 168). Because Ones (2 - 8) and Tens (0 - 6) cannot subtract directly, the 'Banker' partner must un-group 1 Hundred ([5] → [4]), trade across the zero Tens ([0] → [9]), and snap the Gold [+1] regrouping token onto the Ones column (12 - 8 = 4) to unlock 334.

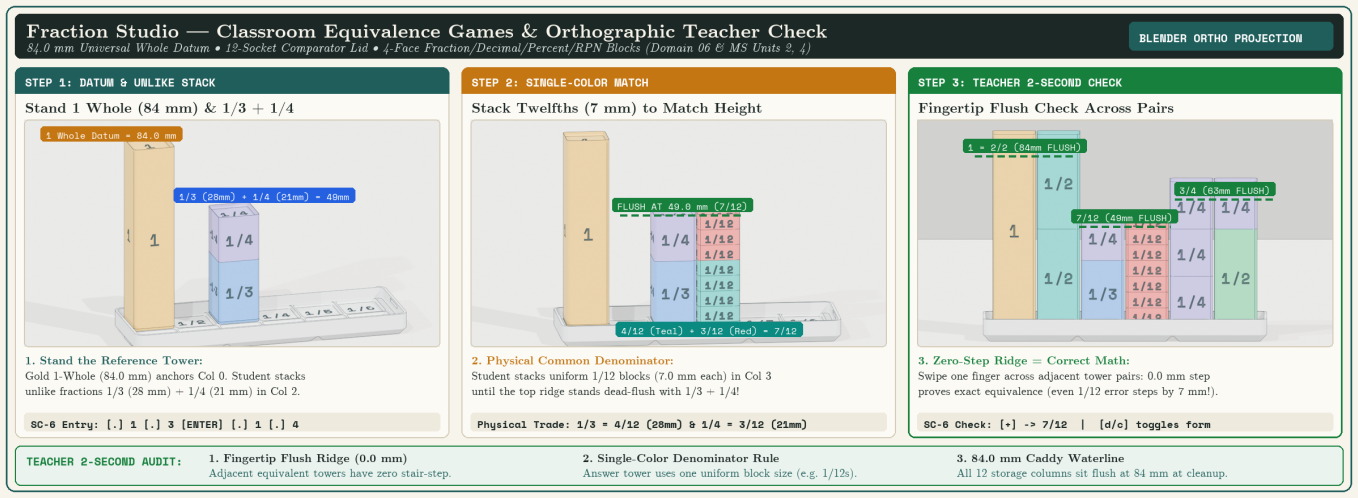
Check Keys (502 − 168 = 334): `5` `0` `2` `ENTER` `1` `6` `8` `-` `123`

HOW TO CHECK STUDENTS ARE DOING IT RIGHT (2-SECOND WALK-BY AUDIT)

- Check 1 — Single-Digit Socket Rule: Glance at Row 0—every active column must hold exactly ONE digit block (0–9). If a student stacked two digit blocks without trading, pause and ask: 'Did this column reach 10?'
- Check 2 — Gold [+1] Carry Trail: Every Gold [+1] token in Row 1 must sit one column to the LEFT of the column that crossed 10.
- Check 3 — SC-6 [PV] Place Decomposition: Have the student press [PV] on their SC-6 answer (432 → 400 → 30 → 2) while pointing to each tray column from left to right.

Fraction Studio: Games & Visual Teacher Check

Classroom games and 2-second orthographic flush checks for the 84.0 mm Proportional Fraction Towers & 12-Socket Comparator Lid.



CLASSROOM GAMES & HANDS-ON ACTIVITIES (TIED TO CONCEPT DOMAINS & WORKBOOKS)

Game 1: 'The 84 mm Flush Tower Challenge (Make 1 Whole)' (Domain 06 • Grades 3–5 • 10 Min)

Stand the Gold 1-Whole block (84.0 mm) in Col 0 of the 12-Socket Comparator Lid. Call out a constraint—e.g., 'Build 1 Whole in Col 1 using exactly 4 blocks from at least 3 different denominator colors!' (e.g., 1/2 + 1/4 + 1/6 + 1/12 = 42 + 21 + 14 + 7 = 84.0 mm). Students swipe a finger across the top ridge to prove 0.0 mm step, then verify with two-dot fraction entry on the SC-6.

Check Keys (1/2 + 1/4 + 1/6 + 1/12 = 1): [•] [1] [•] [2] [ENTER] [•] [1] [•] [4] [+] [•] [1] [•] [6] [+] [•] [1] [•] [1] [2] [+]

Game 2: 'Two-Dot Common Denominator Duel (1/3 + 1/4 = 7/12)' (Domain 06 & MS Unit 02 • Grades 4–6)

Partner A stacks two unlike fractions in Col 2 (e.g., 1/3 [28 mm] + 1/4 [21 mm] = 49.0 mm). Partner B must find a SINGLE-COLOR block family (twelfths, 7.0 mm each) and stack enough identical blocks in Col 3 (seven 1/12 blocks = 49.0 mm) to stand 100% flush with Partner A's tower. Both partners key the sum on the SC-6 and press [d/c] to inspect proper/improper forms.

Check Keys (1/3 + 1/4 = 7/12): [•] [1] [•] [3] [ENTER] [•] [1] [•] [4] [+] [f] [d/c]

Game 3: 'Mystery Tower Blind Fingertip Audit & 4-Face Spin' (Domain 06 & MS Unit 04 • Grades 3–6)

Student A builds a target fraction tower (e.g., three 1/4 blocks = 3/4 = 63.0 mm) and challenges Student B to build an equal-height tower using NO fourths (e.g., 1/2 + 1/4, or four 1/6 + one 1/12 = 63.0 mm). Student A closes their eyes and runs one fingertip across both tops to judge if they are flush, then both rotate a block 90° at a time to read Fraction (1/4), Decimal (0.25), Percent (25%), and RPN Key (.1.4).

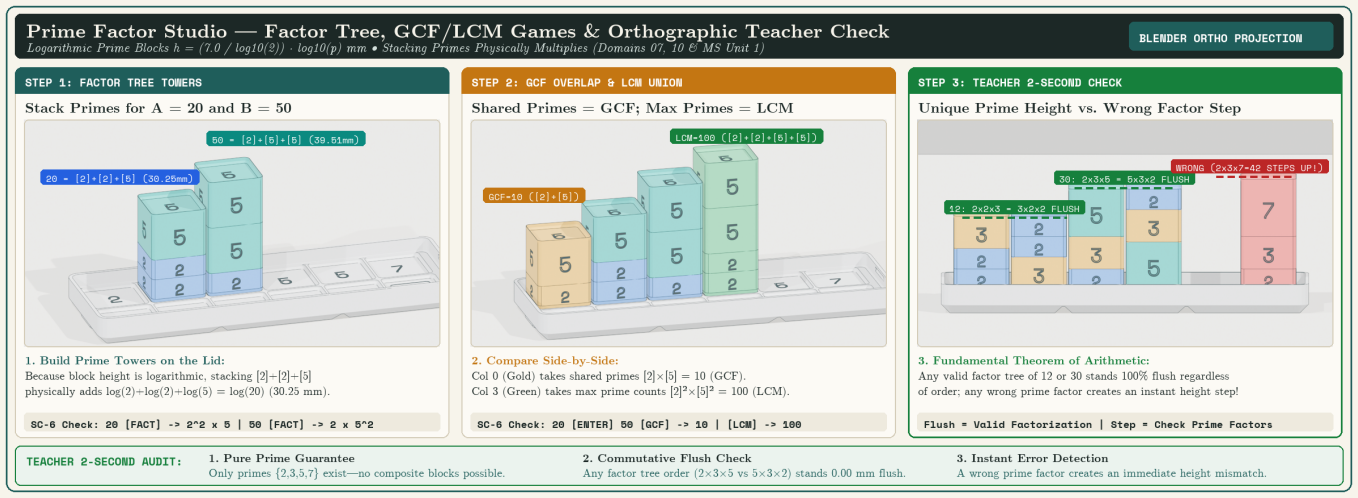
Check Keys (4/6 + 1/12 = 3/4): [•] [4] [•] [6] [ENTER] [•] [1] [•] [1] [2] [+]

HOW TO CHECK STUDENTS ARE DOING IT RIGHT (2-SECOND WALK-BY AUDIT)

- Check 1 — The 0.0 mm Fingertip Flush Test: Even the smallest fraction error (1/12) creates a bold 7.0 mm physical stair-step! If two adjacent towers are dead-flush across the top ridge, the student's equivalence or common denominator is 100% mathematically correct.
- Check 2 — Single-Color Answer Tower: When adding unlike fractions (1/3 + 1/4), check that the student's answer tower in the adjacent socket uses ONE uniform block color (seven 1/12 blocks) rather than mixed denominators.
- Check 3 — 3-Second Cleanup Waterline: When packed into the 12-column caddy, every column sits dead-flush at 84.0 mm—any missing block leaves an obvious gap.

Prime Factor Studio: Games & Visual Teacher Check

Classroom games and 2-second orthographic checks for the Logarithmic Prime Factor Blocks (h = k·log10(p)) & 12-Socket Comparator Lid.



CLASSROOM GAMES & HANDS-ON ACTIVITIES (TIED TO CONCEPT DOMAINS & WORKBOOKS)

Game 1: 'The Commutative Skyscraper Race (2 × 3 × 5 = 30)' (Domains 07 & 10 • Grades 3–6 • 10 Min)

Call out a composite number (e.g., 12, 30, or 42). Partner A factors it using one factor tree (e.g., 30 = 6 × 5 → [2] + [3] + [5]) and stacks the prime blocks in Col 2. Partner B uses a different factor tree (30 = 10 × 3 → [5] + [2] + [3]) in Col 3. Because block height is proportional to log10(p), both towers stand 100% flush at 34.35 mm—proving commutativity and unique factorization physically!

Check Keys (Factor 30 → 2 × 3 × 5): 3 0 ENTER g 人

Game 2: 'Impossible Impostor (Fundamental Theorem Challenge)' (Domain 10 & MS Unit 01 • Grades 5–8)

Student A builds a prime tower (e.g., 12 = [2] + [2] + [3] = 25.10 mm) and challenges Student B to match its exact height using DIFFERENT prime blocks (e.g., trying [5] + [2] = 10 = 23.25 mm or [2] + [7] = 14 = 26.65 mm). Students quickly discover that NO other combination of primes will ever stand flush—experiencing the Fundamental Theorem of Arithmetic in their hands!

Check Keys (Compare 12 vs 14): 1 2 g 人 1 4 g 人

Game 3: 'Side-by-Side GCF Overlap & LCM Union Duel' (Domain 10 & MS Unit 01 • Grades 5–8 • 12 Min)

Given A = 20 and B = 50, students stack Tower A ([2]+[2]+[5]) in Col 1 and Tower B ([2]+[5]+[5]) in Col 2. To build GCF(20,50) = 10 in Col 0, they copy only the SHARED prime blocks present in both towers ([2]+[5]). To build LCM(20,50) = 100 in Col 3, they stack the tallest count of each prime ([2]+[2]+[5]+[5]), then verify [FACT], [GCF], and [LCM] on the SC-6!

Check Keys (GCF & LCM of 20 and 50): 2 0 ENTER 5 0 GCF 2 0 ENTER 5 0 f LCM

HOW TO CHECK STUDENTS ARE DOING IT RIGHT (2-SECOND WALK-BY AUDIT)

- Check 1 — Pure Prime Guarantee: Because the kit contains ONLY prime blocks ({2, 3, 5, 7}), students can never stop a factor tree early at a composite number like 4, 6, 8, or 9.
- Check 2 — Logarithmic Flush Check: Any two valid factor trees for the same number stand dead-flush (0.000 mm step), while any wrong prime factor creates an immediate height step (see red tower in Step 3 above).
- Check 3 — GCF + LCM Height Symmetry: On the comparator lid, GCF is always the shortest tower (left), LCM is the tallest tower (right), and h(GCF) + h(LCM) = h(A) + h(B).

Master K–12 State Standards & District Crosswalk

Locate your state standard (CCSS / TEKS) or district basal textbook unit to find the matching Concept Domain and Student Workbook.

Concept Domain	State Standards (CCSS / TEKS)	Eureka / Illustrative / enVision	Big Ideas / Bridges	Guide & Lab Pages
Dom 01: Counting & Step	K.CC.B.4 / TEKS K.2A	Eureka GK M1 • enVision GK T1	Bridges GK U1 • Big Ideas GK Ch1	**pp. 4–5 • Lab U01**
Dom 02: Part-Part-Whole	1.OA.B.4, 1.OA.C.6 / TEKS 1.3B	Eureka G1 M1 • Illustrative G1 U1	Bridges G1 U2 • Big Ideas G1 Ch2	**pp. 6–7 • Lab U02**
Dom 03: Make-10 & Carry	1.NBT.B.2, 1.OA.C.6 / TEKS 1.3D	Eureka G1 M2 • enVision G1 T3	Bridges G1 U3 • Big Ideas G1 Ch3	**pp. 8–9 • Lab U03**
Dom 04: Doubles & Groups	2.OA.B.2, 2.OA.C.4 / TEKS 2.4A	Eureka G2 M6 • Illustrative G2 U8	Bridges G2 U1 • Big Ideas G2 Ch2	**pp. 10–11 • Lab U04**
Dom 05: Multi-Digit Stack	3.OA.A, 4.OA.A.3 / TEKS 4.4H	Eureka G4 M3 • Illustrative G4 U6	Bridges G4 U2 • Big Ideas G4 Ch2	**pp. 12–13 • Lab U05**
Dom 06: Fraction Equivalence	4.NF.A.1, 5.NF.A.1 / TEKS 5.3H	Eureka G5 M3 • Illustrative G5 U2	Bridges G5 U2 • Big Ideas G5 Ch8	**pp. 14–15 • Lab U06**
Dom 07: Prime Factors & RPN	6.NS.B.4, 6.EE.A.2 / TEKS 6.7A	Illustrative G6 U4 • Eureka G6 M4	Big Ideas G6 Ch1 (MS Unit 1)	**pp. 16–17, 29 • U07**
Dom 08: Quadratics & Poly	8.EE.A, HSA.APR.A / TEKS A.10E	Illustrative Alg1 U6 • Eureka M4	Big Ideas G8 & Alg1 Ch7	**pp. 18–19 • Lab U08**
Dom 09: Trig & 2D Vectors	HSG.SRT.C.8, HSN.VM / TEKS G.9A	Illustrative Geom U4 • enVision	Big Ideas Geometry Ch9 & Precalc	**pp. 20–21 • Lab U09**
Dom 10: Exponents & Logs	HSF.LE.A.4, HSF.BF / TEKS 2A.5D	Illustrative Alg2 U4 • Eureka M3	Big Ideas Algebra 2 Ch6	**pp. 22–23 • Lab U10**
Dom 11: Stats & Combinatorics	6.SP.B, HSS.ID, CP / AP Stats	Illustrative G6 U8 & Alg1 U1	Big Ideas G6 Ch9–10 & AP Stats	**pp. 24–25, 37 • U11**
Dom 12: Binary & Hex Logic	AP CSP Big Idea 2 / MP5	AP CS Principles • PLTW Digital	AP Computer Science A / CSP	**pp. 26–27 • Lab U12**

How District & Homeschool Teachers Use This Crosswalk

Find your state standard (CCSS / TEKS) or district curriculum unit above, open to the listed Teacher's Guide pages for the 10-minute Concept Launch and Misconception Remediations, and assign the matching Student Lab Unit (Book 2) or Middle School Section (Book 3).

Elementary District & Standards Concordance (Grades K–5)

Lesson-by-lesson mapping across Common Core (CCSS), Eureka Math, enVision, Bridges in Mathematics, and Dimensions K–5.

Grade	Standard / Unit	Mathematical Concept Topic	Manipulative / Model	Teacher Guide	Student Lab
Kindergarten	K.CC.B.4 • Unit 1, Mod 1, Session 1	**Welcome to Math: Counting Objects**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.CC.B.4 • Unit 1, Mod 1, Session 2	**Counting & Five-Benchmarks**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.CC.B.5 • Unit 1, Mod 2, Session 1	**Place Value Tray & Counting On**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.CC.A.3 • Unit 1, Mod 2, Session 3	**Zero & The [0] Digit Block**	Place Value Studio ([0] Digit Block)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.OA.A.1 • Unit 1, Mod 3, Session 2	**The Step Counter: Adding One More**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.CC.C.6 • Unit 2, Mod 1, Session 2	**Comparing Sets: Which Has More?**	Place Value Studio (Two-Row Tray)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.OA.A.3 • Unit 2, Mod 2, Session 3	**Number Bonds to 5**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Kindergarten	K.OA.A.2 • Unit 4, Mod 2, Session 1	**Part-Part-Whole Addition**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.OA.C.6 • Unit 1, Mod 2, Session 3	**Place Value & Complements to 10**	Place Value Studio (Digit Blocks & [+1] Carry)	**Dom 03 (pp.8–9)**	**Lab Unit 03**
Grade 1	1.OA.C.6 • Unit 2, Mod 1, Session 2	**Combinations to Ten**	Place Value Studio (Digit Blocks & [+1] Carry)	**Dom 03 (pp.8–9)**	**Lab Unit 03**
Grade 1	1.OA.B.4 • Unit 2, Mod 1, Session 4	**Finding the Missing Addend**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.OA.C.6 • Unit 2, Mod 3, Session 3	**Making 10 Addition Strategies**	Place Value Studio (Digit Blocks & [+1] Carry)	**Dom 03 (pp.8–9)**	**Lab Unit 03**
Grade 1	1.OA.C.6 • Unit 3, Mod 1, Session 1	**Introducing Twin-Row Doubles**	Place Value Studio (Two-Row Tray)	**Dom 04 (pp.10–11)**	**Lab Unit 04**
Grade 1	1.OA.C.6 • Unit 3, Mod 1, Session 2	**Double Clones with [ENTER] [+]**	Place Value Studio (Two-Row Tray)	**Dom 04 (pp.10–11)**	**Lab Unit 04**
Grade 1	1.OA.C.6 • Unit 3, Mod 1, Session 3	**Doubles Plus or Minus One**	Place Value Studio (Two-Row Tray)	**Dom 04 (pp.10–11)**	**Lab Unit 04**
Grade 1	1.OA.C.6 • Unit 3, Mod 2, Session 1	**Equal Groups on the Tray**	Place Value Studio (Two-Row Tray)	**Dom 04 (pp.10–11)**	**Lab Unit 04**
Grade 1	1.OA.A.1 • Unit 6, Mod 1, Session 2	**Two-Row Tray Comparisons**	Place Value Studio (Two-Row Tray)	**Dom 04 (pp.10–11)**	**Lab Unit 04**
Grade 1	1.NBT.B.2 • Unit 7, Mod 1, Session 3	**Place Value: Tens & Ones**	Place Value Studio (Digit Blocks & [+1] Carry)	**Dom 03 (pp.8–9)**	**Lab Unit 03**
Grade 2	2.OA.B.2 • Unit 1, Mod 1, Session 2	**Doubles Fact Fluency**	Place Value Studio (Two-Row Tray)	**Dom 04 (pp.10–11)**	**Lab Unit 04**
Grade 2	2.OA.B.2 • Unit 1, Mod 1, Session 4	**Near-Doubles Fact Strategies**	Place Value Studio (Two-Row Tray)	**Dom 04 (pp.10–11)**	**Lab Unit 04**
Grade 2	2.OA.B.2 • Unit 1, Mod 2, Session 1	**Combinations to 20**	Place Value Studio (Two-Row Tray)	**Dom 04 (pp.10–11)**	**Lab Unit 04**
Grade 2	2.NBT.B.5 • Unit 3, Mod 2, Session 4	**Two-Digit Subtraction on Stack**	Place Value Studio (Whole & Decimal Trays)	**Dom 05 (pp.12–13)**	**Lab Unit 05**
Grade 3	3.OA.D.8 • Unit 1, Mod 2, Session 3	**Multi-Step Addition & Subtraction**	Place Value Studio (Whole & Decimal Trays)	**Dom 05 (pp.12–13)**	**Lab Unit 05**
Grade 3	3.OA.A.1 • Unit 2, Mod 1, Session 2	**Multiplication Arrays & Groups**	Place Value Studio & Prime Factor Studio	**Dom 05 (pp.12–13)**	**Lab Unit 05**

Plugging Into Your State or District Pacing Guide

Match by your state standard code (e.g., 1.OA.C.6, 5.NF.A.1, 6.EE.A.2) or unit topic above, then turn to the listed Concept Domain in Part I (or Part II on pp. 28–40 for Middle School Grades 6–8).

Middle School, High School & AP Concordance (Grades 6–12)

Section-by-section mapping across Common Core (CCSS), Big Ideas Math, Illustrative Math, and AP Mathematics / CS.

Grade	Standard / Unit	Mathematical Concept Topic	Manipulative / Model	Teacher Guide	Student Lab
Kindergarten	K.CC.B.4 • Chapter 1, Section 1.2	**Count and Write 1 to 5**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.CC.B.5 • Chapter 3, Section 3.1	**Model and Count 6 to 10**	Place Value Studio (Digit Blocks & [+1] Tokens)	**Dom 01 (pp.4–5)**	**Lab Unit 01**
Kindergarten	K.OA.A.3 • Chapter 5, Section 5.1	**Number Bonds to 10 on the Tray**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Kindergarten	K.OA.A.1 • Chapter 7, Section 7.2	**Represent Addition as Putting Together**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.OA.C.6 • Chapter 1, Section 1.4	**Part-Part-Whole Addition**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.OA.B.4 • Chapter 2, Section 2.3	**Think Addition to Subtract**	Place Value Studio (Two-Row Tray)	**Dom 02 (pp.6–7)**	**Lab Unit 02**
Grade 1	1.OA.C.6 • Chapter 3, Section 3.1	**Doubles Strategies with ENTER**	Place Value Studio (Two-Row Tray)	**Dom 04 (pp.10–11)**	**Lab Unit 04**
Grade 1	1.OA.C.6 • Chapter 3, Section 3.2	**Doubles Plus 1 & Doubles Minus 1**	Place Value Studio (Two-Row Tray)	**Dom 04 (pp.10–11)**	**Lab Unit 04**
Grade 1	1.OA.C.6 • Chapter 3, Section 3.4	**Make a 10 to Add with Carry Token**	Place Value Studio (Digit Blocks & [+1] Carry)	**Dom 03 (pp.8–9)**	**Lab Unit 03**
Grade 1	1.OA.C.6 • Chapter 4, Section 4.2	**Use 10 to Subtract**	Place Value Studio (Digit Blocks & [+1] Carry)	**Dom 03 (pp.8–9)**	**Lab Unit 03**
Grade 2	2.OA.B.2 • Chapter 1, Section 1.2	**Mental Math: Doubles & Near-Doubles**	Place Value Studio (Two-Row Tray)	**Dom 04 (pp.10–11)**	**Lab Unit 04**
Grade 2	2.OA.B.2 • Chapter 1, Section 1.4	**Make a Ten Addition Strategies**	Place Value Studio (Digit Blocks & [+1] Carry)	**Dom 03 (pp.8–9)**	**Lab Unit 03**
Grade 2	2.NBT.B.5 • Chapter 3, Section 3.3	**Two-Digit Addition with Lift**	Place Value Studio (Whole & Decimal Trays)	**Dom 05 (pp.12–13)**	**Lab Unit 05**
Grade 3	3.OA.A.1 • Chapter 1, Section 1.3	**Multiplication as Repeated Addition**	Place Value Studio (Whole & Decimal Trays)	**Dom 05 (pp.12–13)**	**Lab Unit 05**
Grade 3	3.NF.A.1 • Chapter 8, Section 8.2	**Equal Parts & Unit Fractions**	Fraction Studio (84 mm Towers)	**Dom 06 (pp.14–15)**	**Lab Unit 06**
Grade 3	3.NF.A.3 • Chapter 8, Section 8.5	**Equivalent Fractions on Blocks**	Fraction Studio (84 mm Towers)	**Dom 06 (pp.14–15)**	**Lab Unit 06**
Grade 4	4.OA.A.3 • Chapter 3, Section 3.4	**Solve Multi-Step Word Problems**	Place Value Studio & 4-Register Stack	**Dom 05 (pp.12–13)**	**Lab Unit 05**
Grade 4	4.OA.A.3 • Chapter 3, Section 3.5	**Order of Operations Without Parens**	Place Value Studio & 4-Register Stack	**Dom 05 (pp.12–13)**	**Lab Unit 05**
Grade 4	4.NF.B.3 • Chapter 8, Section 8.3	**Add & Subtract Common Fractions**	Fraction Studio (84 mm Towers)	**Dom 06 (pp.14–15)**	**Lab Unit 06**
Grade 4	4.NF.B.3 • Chapter 8, Section 8.6	**Mixed Numbers: Two-Dot Entry**	Fraction Studio (84 mm Towers)	**Dom 06 (pp.14–15)**	**Lab Unit 06**
Grade 5	5.NF.A.1 • Chapter 5, Section 5.2	**Add Fractions with Unlike Denominators**	Fraction Studio (84 mm Towers)	**Dom 06 (pp.14–15)**	**Lab Unit 06**
Grade 5	5.NF.B.4 • Chapter 6, Section 6.2	**Multiply Fractions & Height Scaling**	Fraction Studio (84 mm Towers)	**Dom 06 (pp.14–15)**	**Lab Unit 06**
Grade 5	5.NF.B.7 • Chapter 6, Section 6.5	**Divide Fractions with Reciprocals**	Fraction Studio (84 mm Towers)	**Dom 06 (pp.14–15)**	**Lab Unit 06**
Grade 5	5.G.A.1 • Chapter 12, Section 12.1	**Graphing Points on Coordinate Plane**	Place Value Studio (Two-Row Tray)	**Dom 08 (pp.18–19)**	**Lab Unit 08**

Plugging Into Your State or District Pacing Guide

Match by your state standard code (e.g., 1.OA.C.6, 5.NF.A.1, 6.EE.A.2) or unit topic above, then turn to the listed Concept Domain in Part I (or Part II on pp. 28–40 for Middle School Grades 6–8).

PART I • DOMAIN 01 • COUNTING & CARDINALITY • PRE-K – KINDERGARTEN (SC-6 ELEMENTARY)

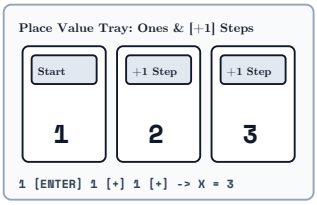
1-to-1 Correspondence: The Concrete Step Counter

Core Concept Lesson • Bridges in Mathematics (Numbers to 5 & 10) • Standard: CCSS.MATH.K.CC.B.4

1. MATHEMATICAL CONCEPT & STACK APPROACH: Establish one-to-one correspondence between physical digit blocks and step tokens on the Place Value Tray and calculator screen increments.

Counting by ones is repeated addition of the unit 1 (n → n + 1). In a stack-based approach, the current total waits on Shelf Y while the unit step (1) enters Workbench X; pressing [+] combines them into the new total.

Classroom Hook: Close your eyes and clap slowly 4 times. Before opening your eyes, which Digit Block belongs in the Ones socket of your Place Value Tray and what number waits on Line X?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Digit Blocks 0–9 & Gold [+1] Step Tokens)

Hands-On Setup: Place the [1] Digit Block into the Ones socket of the Place Value Tray. Keep Gold [+1] step tokens ready beside the tray.

Predict First (Before Calculator): When we place 1 Gold [+1] step token and step the Ones block up by 1, what digit block replaces [1]? (Prediction: [2])

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Place 1 Gold [+1] step token on the tray first, then press [1] [+] to match the count you just touched."

2B. AVOID SAYING (RULE WITHOUT REASON):

"Just press 1 + 1 + 1 as fast as you can until the screen says 5."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [1] [ENTER] [1] [+] [1] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[1]	1	First count placed: 1 enters register X
[ENTER]	1	1 copied to register Y: holds the current count
[1]	1	Gold [+1] step token added: 1 enters register X
[+]	2	Press [+]: 1 + 1 = 2! Swap Ones block to [2]
[1]	1	Next Gold [+1] step token added: 1 enters X
[+]	3	Press [+]: 2 + 1 = 3! Tray [3] and screen match perfectly

4. STORY CONTEXT (How Ten Fingers & an Empty Shell Created Place Value & The 1-to-10 Staircase)

Long before paper or keyboards existed, ancient shepherds counted flocks with clay pebbles ('calculi' in Latin—the root of 'calculator'!) and Maya astronomers drew a shell for zero so an empty socket had its own symbol. Stepping 1 token at a time on the Place Value Tray while pressing [1] [+] builds the exact 1-to-10 Staircase from Chapter 3 of From Blocks to the Stack.

Core Investigation (Student Warm-Up A): Place the [4] Digit Block in the Ones column of your Place Value Tray. Add 1 Gold [+1] step token. Write the keystrokes to show 4 + 1 on the calculator.

Expected Answer: 5 ([5] Digit Block).

Verification Keys: [4] [ENTER] [1] [+]

More, Less & Zero: Finding Differences

Follow-Up Extension (Common Core / TEKS (Eureka, enVision, Bridges) • CCSS.MATH.K.CC.C.6) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Comparison subtraction finds the difference ('how many more?') between two quantities. In stack order, the larger reference quantity sits on Waiting Shelf Y while the smaller quantity enters Workbench X before pressing [-].

Blocks & Prediction: Place the [5] Digit Block in Row 1 (top row) and the [2] Digit Block in Row 0 (bottom row) of the Ones column on the Place Value Tray. Prediction: Look at the two rows on the tray. Which row has the larger quantity? How many more is 5 than 2?

Extension Key Sequence: [5] [ENTER] [2] [-]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[5]	5	Row 1 quantity (5) enters register X
[ENTER]	5	5 lifted to register Y (first set)
[2]	2	Row 0 quantity (2) enters register X
[-]	3	Minus subtracts X from Y: Row 1 has 3 more!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Reciting number names faster than placing physical step tokens (losing 1-to-1 correspondence).

Hands-On Block & Stack Fix: Have the student place 1 Gold [+1] token on the Place Value Tray and step the Ones Digit Block first, say the number aloud, and only then press [1] [+] so the tray and Register X stay locked in step.

Misconception 2: Confusing the action of adding 1 more with the total quantity on the tray (cardinality).

Hands-On Block & Stack Fix: Ask the student to point to the Digit Block in the Ones socket of the Place Value Tray and verify that its numeral matches Register X on the screen.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 01 INVESTIGATIONS 1–4)

Investigation 1: Place [3] in Row 1 and [2] in Row 0 of the Ones column. Write the keystrokes to find the total. **Ans: 5**

Check Keys: [3] [ENTER] [2] [+]

Investigation 2: The Ones column holds [5]. If you subtract [2], what digit block remains? **Ans: 3**

Check Keys: [5] [ENTER] [2] [-]

Investigation 3: Clear the calculator by pressing [C]. What Digit Block represents an empty Ones count? **Ans: 0 (Zero)**

Check Keys: [C]

Investigation 4: Step count by ones starting at 1: [1] [ENTER] [1] [+]. Press [1] [+] two more times. What is X? **Ans: 4**

Check Keys: [1] [+] [1] [+]

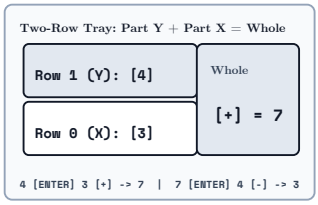
Composing Numbers: Part + Part = Whole

Core Concept Lesson • Common Core / TEKS (Eureka, enVision, Bridges) (Number Bonds & Part-Part-Whole) • Standard: CCSS.MATH.1.OA.C.6

1. MATHEMATICAL CONCEPT & STACK APPROACH: Physically compose two parts on the two rows of the Place Value Tray into a whole and connect to RPN binary addition.

Every addition problem composes two independent parts into a single whole (Part 1 + Part 2 = Whole). The stack makes this structure visible: Part 1 rests on Waiting Shelf Y, Part 2 is built on Workbench X, and [+] joins them.

Classroom Hook: Place [7] as the Whole and cover Row 0 on your Place Value Tray so only Row 1 ([4]) shows. How does the stack uncover the hidden part?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Two-Row Tray & Digit Blocks 0–9)

Hands-On Setup: Place the [4] Digit Block in Row 1 (Part 1) and the [3] Digit Block in Row 0 (Part 2) of the Ones column on your Place Value Tray.

Predict First (Before Calculator): Combine both parts in the Ones column: 4 + 3 = 7 (swap to the [7] Digit Block).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Row 1 ([4]) sits on Waiting Shelf Y and Row 0 ([3]) sits on Workbench X—press [+] to combine both parts into the Whole!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Move the 4 across the equals sign and change its sign to minus."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [4] [ENTER] [3] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[4]	4	Part 1 (4) enters register X
[ENTER]	4	4 copied up to register Y; represents Row 1 (Part 1)
[3]	3	Part 2 (3) enters register X; represents Row 0 (Part 2)
[+]	7	Plus combines Part 1 (Y) + Part 2 (X) = Whole (7) in X!

4. STORY CONTEXT (Number Partners on the Workbench: Part-Part-Whole Bonds & Friends of Ten)

Your brain loves structure: when you place two parts on the top and bottom rows of the Place Value Tray, you can see how they compose a whole! In From Blocks to the Stack, Row 1 of the Place Value Tray sits on Waiting Shelf Y (Part 1), Row 0 sits on Workbench X (Part 2), and pressing [+] snaps both rows together into the Whole.

Core Investigation (Student Warm-Up A): Place [5] in Row 1 (Part 1) and [4] in Row 0 (Part 2) of your Place Value Tray to find the whole. Write the keystrokes.

Expected Answer: 9 ([9] Digit Block).

Verification Keys: [5] [ENTER] [4] [+]

Decomposing the Whole: Finding the Missing Part

Follow-Up Extension (Bridges in Mathematics • CCSS.MATH.1.OA.B.4) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Finding a missing addend (6 + ? = 10) is the inverse of composing a sum. By placing the known Whole (10) on Shelf Y and the known Part (6) on Workbench X, pressing [-] isolates the unknown part.

Blocks & Prediction: Place [1][0] in Row 1 of the Place Value Tray (Whole = 10) and [6] in Row 0 (Known Part 1 = 6). Prediction: What missing part combines with 6 to make 10? (10 - 6 = 4).

Extension Key Sequence: [1] [0] [ENTER] [6] [-]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	Enter Whole (10)
[0]	10	10 in register X
[ENTER]	10	Whole (10) lifted to register Y
[6]	6	Known Part 1 (6) enters register X
[-]	4	Minus subtracts known Part from Whole: Missing Part is 4!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Pressing [+] immediately after the first number without pressing [ENTER] to separate Part 1 and Part 2.

Hands-On Block & Stack Fix: Point to the two rows of the Place Value Tray: pressing [ENTER] is the physical divider that lifts Row 1 (Part 1) up to Register Y before Row 0 (Part 2) is typed into Register X.

Misconception 2: Adding the whole and the known part when solving for a missing addend (e.g., computing 10 + 6 instead of 10 − 6).

Hands-On Block & Stack Fix: Place [1][0] in Row 1 (Whole) and [6] in Row 0 (Known Part) of the Place Value Tray; show that the Whole (10) goes into Register Y, the known part (6) enters X, and [−] reveals the missing part.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 02 INVESTIGATIONS 1–4)

Investigation 1: Make 6: Part 1 is 2. What is Part 2? Write the keystrokes. **Ans: 4**

Check Keys: [6] [ENTER] [2] [-]

Investigation 2: Part 1 is 7, Part 2 is 2. What is the Whole? **Ans: 9**

Check Keys: [7] [ENTER] [2] [+]

Investigation 3: Three parts make a whole: 2, 3, and 4. Add them on the stack. **Ans: 9**

Check Keys: [2] [ENTER] [3] [+] [4] [+]

Investigation 4: The Whole is 10. Part 1 is 7. What is Part 2? **Ans: 3**

Check Keys: [1] [0] [ENTER] [7] [-]

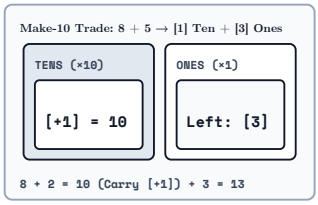
Complements to 10: Making Ten & Carry Trade

Core Concept Lesson • Big Ideas Math (Addition Strategies to 10) • Standard: CCSS.MATH.1.OA.C.6

1. MATHEMATICAL CONCEPT & STACK APPROACH: Determine complements to 10 using Place Value Digit Blocks, Gold [+1] Carry Tokens, and RPN subtraction.

Our base-10 system regroups every collection of 10 ones into 1 ten. Finding a number's complement to 10 (10 - 7 = 3) places the target base (10) on Shelf Y and the current ones digit (7) on Workbench X.

Classroom Hook: Place [8] in the Ones socket of your Place Value Tray. How many more ones do you need to trigger a Gold [+1] Carry Token in the Tens column?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Whole-Number Tray, Digit Blocks 0–9 & Gold [+1] Carry Tokens)

Hands-On Setup: Place the [7] Digit Block into the Ones socket of the Place Value Tray and hold a Gold [+1] Carry Token beside the Tens socket.

Predict First (Before Calculator): How many more ones must be added to [7] to reach 10 and trigger the Gold [+1] Carry Token? (7 + 3 = 10).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"How many more to reach 10? Let's borrow 2 from 5 to complete 10 first and crown the Tens column with a Gold [+1] Carry Token!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Memorize 8 + 5 = 13 from your flash cards."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [1] [0] [ENTER] [7] [-]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[1]	1	Enter 1
[0]	10	Target ten (10) in X
[ENTER]	10	10 lifted to Y
[7]	7	Current Ones digit (7) in X
[-]	3	10 - 7 = 3: Adding 3 completes 10 and triggers the [+1] Carry!

4. STORY CONTEXT (Making a Ten First (8 + 5 = 10 + 3 = 13) & The Gold [+1] Carry Token Trade)

Why is 10 + 3 so much easier for our brains than 8 + 5? Because base-10 place value groups everything into full tens! In Chapter 3 of From Blocks to the Stack, students bridge across 10 on the Place Value Tray and crown the Tens column with a Gold [+1] Carry Token.

Core Investigation (Student Warm-Up A): The Ones socket holds [6]. How many more ones are needed to reach 10 and earn a Gold [+1] Carry Token?

Expected Answer: 4 ones.

Verification Keys: [1] [0] [ENTER] [6] [-]

Bridging Through 10: The Make-Ten & Carry Strategy

Follow-Up Extension (Common Core / TEKS (Eureka, enVision, Bridges) • CCSS.MATH.1.OA.C.6) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Bridging across 10 decomposes an addend so the first part completes a full ten (8 + 5 = (8 + 2) + 3 = 10 + 3). On the stack, the intermediate sum (10) stays in X and automatically lifts to Y when the remaining part (3) is entered.

Blocks & Prediction: Place [8] in Row 1 and [5] in Row 0 of the Ones column on the Place Value Tray. Prediction: Take 2 from the 5 to bring 8 up to 10. Place a Gold [+1] Carry Token in the Tens socket and [3] in the Ones socket: 10 + 3 = 13.

Extension Key Sequence: [8] [ENTER] [2] [+] [3] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[8]	8	8 in Ones socket
[ENTER]	8	8 lifted to Y
[2]	2	2 borrowed from 5 to make 10
[+]	10	8 + 2 = 10 (Gold [+1] Carry Token placed in Tens column!)
[3]	3	Remaining 3 ones
[+]	13	10 + 3 = 13 total ([1] Ten, [3] Ones)!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Counting on by ones across 10 (e.g., 8 + 5 as 9, 10, 11, 12, 13) instead of decomposing 5 into 2 + 3 to complete 10 first.

Hands-On Block & Stack Fix: Have the student bridge [8] to 10 on the Place Value Tray, place a Gold [+1] Carry Token in the Tens socket with [3] in the Ones socket, and key [8] [ENTER] [2] [+] [3] [+].

Misconception 2: Reading teen numbers as two separate single digits (seeing 14 as 1 and 4 rather than 1 Ten plus 4 Ones).

Hands-On Block & Stack Fix: Place [1] in the Tens socket and [4] in the Ones socket of the Place Value Tray and key [10] [ENTER] [4] [+] so students see 14 formed from 10 in Register Y and 4 in Register X.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 03 INVESTIGATIONS 1–4)

Investigation 1: What is the complement to 10 for 4? Write the keystrokes. **Ans: 6**

Check Keys: [1] [0] [ENTER] [4] [-]

Investigation 2: Solve 7 + 5 using Make-Ten: first make 10, then add 2. **Ans: 12**

Check Keys: [7] [ENTER] [3] [+] [2] [+]

Investigation 3: The Ones socket holds [9]. How many more to trigger a Gold [+1] Carry Token? **Ans: 1**

Check Keys: [1] [0] [ENTER] [9] [-]

Investigation 4: Solve 8 + 6 using Make-Ten on the calculator. **Ans: 14**

Check Keys: [8] [ENTER] [2] [+] [4] [+]

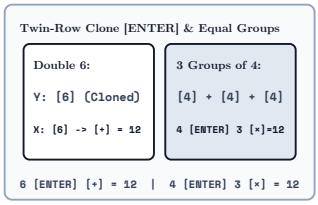
Doubles Strategy: The ENTER Duplicate Key

Core Concept Lesson • Bridges in Mathematics (Doubles & Fact Strategies) • Standard: CCSS.MATH.1.OA.C.6

1. MATHEMATICAL CONCEPT & STACK APPROACH: Model doubles physically using matching digit blocks in Row 1 and Row 0 of the Place Value Tray and computationally using ENTER duplication.

Doubling a quantity means adding two identical copies of the same number (n + n = 2n). On a stack calculator, pressing [ENTER] duplicates the active number in X onto Waiting Shelf Y so pressing [+] immediately doubles it.

Classroom Hook: Place a [6] Digit Block in Row 1 and a matching [6] Digit Block in Row 0 of your Place Value Tray. Which single key clones a number on the calculator?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Two-Row Tray & Digit Blocks 0–9)

Hands-On Setup: Place a [6] Digit Block in Row 1 (top) and a matching [6] Digit Block in Row 0 (bottom) of the Ones column on the Place Value Tray.

Predict First (Before Calculator): Observe the matching rows: 6 + 6 = 12 (1 Ten and 2 Ones on the tray).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Press [6] [ENTER]—look how [ENTER] clones your top row of 6 onto Shelf Y so both rows are ready to add!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Press 6, then plus, then 6, then equals."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [6] [ENTER] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[6]	6	Top row quantity (6) enters register X
[ENTER]	6	ENTER duplicates 6 into register Y! Both rows now on stack
[+]	12	Plus combines top and bottom rows: 6 + 6 = 12!

4. STORY CONTEXT (Twin Rows on the Workbench: How Matching Tray Rows Mirror Instant [ENTER] [+] Doubles)

When you place matching Digit Blocks in the top and bottom rows of your Place Value Tray, your eyes immediately recognize a double! In Chapters 1 & 2 of From Blocks to the Stack, pressing [ENTER] clones Workbench X onto Waiting Shelf Y just like matching Row 1 and Row 0 on the tray for instant [ENTER] [+] doubles.

Core Investigation (Student Warm-Up A): Place [7] in both rows of your Place Value Tray. Use [ENTER] [+] to double it on the calculator.

Expected Answer: 14 ([1][4] on the tray).

Verification Keys: [7] [ENTER] [+]

Near-Doubles: Doubles Plus One

Follow-Up Extension (Bridges in Mathematics • CCSS.MATH.2.OA.B.2) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: A near-double (such as 7 + 8) is a double plus or minus 1 ((7 + 7) + 1). Evaluating the known double first leaves 14 on the stack, ready to combine with the +1 adjustment.

Blocks & Prediction: Place [7] in Row 1 (top) and [8] in Row 0 (bottom) of the Place Value Tray. Prediction: 7 and 8 is double 7 plus 1 more: 14 + 1 = 15.

Extension Key Sequence: [7] [ENTER] [+] [1] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[7]	7	Enter base number 7
[ENTER]	7	Duplicate 7
[+]	14	Double: 7 + 7 = 14
[1]	1	Near-double difference (1)
[+]	15	14 + 1 = 15!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Retyping the number to double it (e.g., [7] [ENTER] [7] [+]) instead of recognizing that [ENTER] duplicates Register X into Register Y.

Hands-On Block & Stack Fix: Place matching [7] Digit Blocks in Row 1 and Row 0 of the Place Value Tray, then press [7] [ENTER] and point to the screen: both Y and X already hold 7, so pressing [+] doubles it immediately.

Misconception 2: Confusing '3 groups of 4' with '3 + 4' when transitioning from repeated addition to multiplication.

Hands-On Block & Stack Fix: Model three groups of [4] on the Place Value Tray; have the student key [4] [ENTER] [4] [+] [4] [+] = 12 first, then compare directly with [4] [ENTER] [3] [×] = 12.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 04 INVESTIGATIONS 1–4)

Investigation 1: Calculate double 8 using [ENTER] [+]. **Ans: 16**

Check Keys: [8] [ENTER] [+]

Investigation 2: Solve near-double 8 + 9: double 8 then add 1. **Ans: 17**

Check Keys: [8] [ENTER] [+] [1] [+]

Investigation 3: Calculate double 9 using [ENTER] [+]. **Ans: 18**

Check Keys: [9] [ENTER] [+]

Investigation 4: Solve 5 + 6 using doubles plus one. **Ans: 11**

Check Keys: [5] [ENTER] [+] [1] [+]

Multi-Step Word Problems Without Parentheses

Core Concept Lesson • Common Core / TEKS (Eureka, Illustrative, enVision) (Multi-Step Word Problems) • Standard: CCSS.MATH.4.OA.A.3

1. MATHEMATICAL CONCEPT & STACK APPROACH: Evaluate chained arithmetic expressions without parentheses by managing stack register flow.

In a multi-step expression like $(3 \times 4) + (2 \times 5)$, each product is a self-contained quantity that must be evaluated before the two groups can be added. On the 4-level stack, the first product (12) waits safely on Shelf Y while the second product (10) is formed on X, requiring zero parentheses.

Classroom Hook: Beat the Smartphone Challenge: Ask a volunteer to type 8 + 4 × 5 left-to-right on a basic phone calculator (it says 60!). Why is the true answer 28?



Place Value Studio (p. vi) • 10-for-1 Carry Trade

2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Place Value Blocks & 4-Register Stack

Hands-On Setup: Use Place Value unit blocks to represent quantities alongside the 4-line stack display.

Predict First (Before Calculator): Problem: (8 - 3) × 2. Predict: 8 - 3 = 5, then 5 × 2 = 10.

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Solve the inner bundle first and let it wait on Shelf Y; when you type the next number, watch Automatic Stack Lift push it up!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Type the expression from left to right and hope the calculator knows PEMDAS."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [8] [ENTER] [3] [-] [2] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[8]	8	8 in X
[ENTER]	8	8 copied to Y
[3]	3	3 entered into X
[-]	5	8 - 3 = 5 in X
[2]	2	Automatic lift! 5 lifts to Y; 2 enters X
[×]	10	5 × 2 = 10 in X!

4. STORY CONTEXT (Inside the 4-Floor Number Elevator (T, Z, Y, X) & Young Gauss's 10-Second Shortcut)

In 1786, 9-year-old Carl Friedrich Gauss stunned his teacher by adding 1 through 100 in seconds—folding the list into 50 pairs of 101! In Chapters 2 & 4 of From Blocks to the Stack, students discover how the 4-Floor Number Elevator (Shelves X, Y, Z, T) automatically lifts finished answers up to Shelf Y when you type the next number, solving multi-step stories with zero parentheses.

Core Investigation (Student Warm-Up A): A baker has 15 cookies, sells 6, and packs the rest into 3 bags. How many per bag? Keystrokes for (15 - 6) / 3.

Expected Answer: 3 cookies per bag.

Verification Keys: [1] [5] [ENTER] [6] [-] [3] [÷]

Automatic Stack Lift & Register Retention

Follow-Up Extension (Big Ideas Math • CCSS.MATH.4.OA.A.3) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: When an operation finishes on the stack, its result remains in Register X and automatically lifts to Waiting Shelf Y as soon as the next digit is typed—allowing chained calculations like (12 ÷ 4) + 7 to flow continuously from the inside out.

Blocks & Prediction: Observe the 4-line stack screen as values automatically lift into register Y: (12 / 4) + 7. Prediction: 12 / 4 = 3. Entering 7 will lift 3 to Y. 3 + 7 = 10.

Extension Key Sequence: [1] [2] [ENTER] [4] [÷] [7] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	First digit of 12 enters X
[2]	12	12 in register X
[ENTER]	12	12 copied to Y
[4]	4	4 enters X
[÷]	3	12 / 4 = 3 in X
[7]	7	7 entered: 3 lifts to Y automatically!
[+]	10	3 + 7 = 10!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Pressing [ENTER] after an operator (+, −, ×, ÷) before typing the next number in a chained calculation, wasting a stack register.

Hands-On Block & Stack Fix: Demonstrate Automatic Stack Lift: after [18] [ENTER] [6] [−] produces 12, typing [3] automatically lifts 12 to Register Y—no [ENTER] needed before [3] [×]!

Misconception 2: Reversing operand order in subtraction or division (computing X − Y instead of Y − X).

Hands-On Block & Stack Fix: Remind students that Register Y is always the starting quantity on top (Row 1) and Register X is the operator's action (Row 0); if entered backward, press [x swap y] before [−] or [÷].

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 05 INVESTIGATIONS 1–4)

Investigation 1: Calculate (14 - 5) × 3 on the stack. **Ans: 27**

Check Keys: [1] [4] [ENTER] [5] [-] [3] [×]

Investigation 2: Calculate 30 / (2 + 3): add denominator first! **Ans: 6**

Check Keys: [2] [ENTER] [3] [+] [3] [0] [↑↓] [÷]

Investigation 3: Calculate (9 + 7) / (8 - 4) using 4 registers. **Ans: 4**

Check Keys: [9] [ENTER] [7] [+] [8] [ENTER] [4] [-] [÷]

Investigation 4: A farmer collects 20 eggs, breaks 2, and divides among 6 cartons. **Ans: 3 eggs**

Check Keys: [2] [0] [ENTER] [2] [-] [6] [÷]

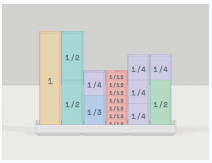
Mixed Numbers & Height Equivalence: 3/4 + 2 1/2

Core Concept Lesson • Common Core / TEKS (Eureka, Illustrative, Big Ideas) (Fraction Addition & Mixed Numbers) • Standard: CCSS.MATH.5.NF.A.1

1. MATHEMATICAL CONCEPT & STACK APPROACH: Represent proper and mixed fractions physically with block towers and computationally with two-dot notation.

Fractions can only be added or subtracted when they share a common unit (denominator), and mixed numbers combine whole units with fractional parts. On the stack, two-dot notation ([.] n [.] d) treats each fraction or mixed number as a single unified value on Y and X.

Classroom Hook: Hold up the 1-Whole tower and a millimeter ruler: Why did engineers make the Whole block exactly 84 mm tall instead of 100 mm?



Fraction Studio (p. vii) • 84 mm Flush Towers

2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: 84 mm Fraction Towers & Twin Comparator Well

Hands-On Setup: Place three 1/4 blocks (3 × 21 = 63 mm). Beside them, stack two Whole blocks and one 1/2 block (84 + 84 + 42 = 210 mm).

Predict First (Before Calculator): Combine heights: 63 mm + 210 mm = 273 mm. In wholes: 3 1/4 (273 / 84 = 3.25).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Stand your 1/2 (42 mm) and 1/4 (21 mm) towers in the comparator lid—do they match three 1/4 towers (63 mm) in height?"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Just cross-multiply the denominators and don't ask why."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [•] [3] [•] [4] [ENTER] [2] [•] [1] [•] [2] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[•]	0.	Leading dot indicates 0 wholes
[3]	0.3	Numerator = 3
[•]	0.3.	Second dot sets fraction entry mode
[4]	3/4	Denominator = 4: displays 3/4 (or 0.75)
[ENTER]	3/4	3/4 lifted to register Y
[2]	2	Whole = 2
[•]	2.	Dot
[1]	2.1	Numerator = 1
[•]	2.1.	Dot
[2]	2 1/2	Denominator = 2: displays 2 1/2
[+]	3 1/4	Plus combines: 3 1/4 (decimal 3.25)!

4. STORY CONTEXT (The Pharaoh's Bakery, Pythagoras's Guitar Strings & Why Your Fraction Towers Stand 84.0 mm Tall)

Four thousand years ago, Egyptian scribes had to divide 3 loaves of bread fairly among 4 workers—and instead of cutting every loaf into quarters, they gave each worker 1/2 + 1/4! In Chapter 5 of From Blocks to the Stack, students discover why the Whole block is engineered to 84.0 mm (LCM of 1,2,3,4,6,8,12 → 84 mm), letting 42 mm (1/2) + 21 mm (1/4) stack flush with three 21 mm (3/4) blocks.

Core Investigation (Student Warm-Up A): Build 1/2 and 1/4 with blocks. Verify their combined height against three 1/4 blocks, then check on the calculator.

Expected Answer: 3/4 (height 63 mm).

Verification Keys: [•] [1] [•] [2] [ENTER] [•] [1] [•] [4] [+]

Fraction Multiplication & Reciprocals

Follow-Up Extension (Bridges in Mathematics • CCSS.MATH.5.NF.B.4) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Multiplying fractions ((3/4) × (2/3)) scales one rational quantity by another, while dividing by a fraction multiplies by its reciprocal. On the stack, placing the dividend in Y and divisor in X lets you either divide directly or invert X ([1/x]) and multiply.

Blocks & Prediction: Set up 3/4 of a tower (63 mm) and divide into 3 equal parts (21 mm each), taking 2 parts: (3/4) × (2/3). Prediction: 3/4 × 2/3 = 6/12 = 1/2. The resulting block height matches one 1/2 block (42 mm).

Extension Key Sequence: [.] [3] [.] [4] [ENTER] [.] [2] [.] [3] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[.]	0.	Start proper fraction 3/4
[3]	0.3	Numerator 3
[.]	0.3.	Fraction separator
[4]	3/4	3/4 in register X
[ENTER]	3/4	3/4 lifted to register Y
[.]	0.	Start proper fraction 2/3
[2]	0.2	Numerator 2
[.]	0.2.	Fraction separator
[3]	2/3	2/3 in register X
[×]	1/2	Product 6/12 = 1/2 (matches 42 mm block)!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Adding numerators and denominators directly (e.g., 1/2 + 1/4 = 2/6) because 6 looks larger.

Hands-On Block & Stack Fix: Place the 1/2 tower (42 mm) + 1/4 tower (21 mm) beside two 1/6 towers (28 mm) in the Fraction Studio comparator lid; the physical height difference proves 1/2 + 1/4 = 3/4 (63 mm).

Misconception 2: Entering a proper fraction like 3/4 as [3] [.] [4] (which is decimal 3.4) instead of using Two-Dot fraction notation [.][3][.][4].

Hands-On Block & Stack Fix: Teach the rule: 'First dot = whole number zero, second dot = fraction bar.' Typing [.][3][.][4] displays exact 3/4 on the stack.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 06 INVESTIGATIONS 1–4)

Investigation 1: Add 1/3 + 1/6 using fraction blocks and verify on the calculator. **Ans: 1/2 (42 mm)**

Check Keys: [.] [1] [.] [3] [ENTER] [.] [1] [.] [6] [+]

Investigation 2: Calculate 1 1/2 - 3/4. What block height corresponds to this? **Ans: 3/4 (63 mm)**

Check Keys: [1] [.] [1] [.] [2] [ENTER] [.] [3] [.] [4] [-]

Investigation 3: Compare 3/8 (31.5 mm) to 1/2 (42 mm). What is the decimal equivalent of 3/8? **Ans: 0.375**

Check Keys: [.] [3] [.] [8] [FDISP]

Investigation 4: Multiply 2/3 × 3/4. Check against your fraction blocks. **Ans: 1/2 (42 mm)**

Check Keys: [.] [2] [.] [3] [ENTER] [.] [3] [.] [4] [×]

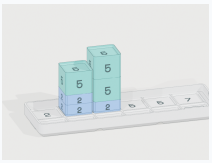
Chained Operations: (a + b) × (c - d) on the Stack

Core Concept Lesson • Big Ideas Math (Numerical & Algebraic Expressions) • Standard: CCSS.MATH.6.EE.A.2

1. MATHEMATICAL CONCEPT & STACK APPROACH: Map an algebraic expression to a postfix sequence and execute on the 4-register stack without parentheses.

Written algebra uses parentheses to show which inner operations must happen first. Postfix (stack) notation evaluates expressions in true mathematical order from the inside out: compute the first branch (6 + 1 = 7), let it wait on Y while computing the second branch (9 - 5 = 4), and multiply.

Classroom Hook: Why does 24 factor into the Prime Factor tower [2]×[2]×[2]×[3] (giving 8 factor pairs), while 23 is a single prime number?



Prime Factor Studio (p. viii) • Log Prime Towers

2. MODELING WITH BLOCKS & STUDENT PREDICTION

Published Block Kit: Prime Factor Studio (Logarithmic Prime Blocks) & 4-Register Stack

Hands-On Setup: Use Prime Factor Studio blocks in the 12-socket comparator lid to model the two factor branches: (6 + 1) = [7] prime block and (9 - 5) = [2] + [2] = 4 tower, then combine their heights for 7 × 4 = 28.

Predict First (Before Calculator): Left sub-expression: 6 + 1 = 7. Right sub-expression: 9 - 5 = 4. Combined product: 7 × 4 = 28 (prime tower 2 × 2 × 7).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Evaluate the left branch (6 + 1 = 7) onto Shelf Y and the right branch (9 − 5 = 4) on Shelf X, then press [×] to combine!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Count open and close parentheses before typing anything."

3. CALCULATOR EXECUTION: SC-6 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [6] [ENTER] [1] [+] [9] [ENTER] [5] [-] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[6]	6	Value a (6) placed into Register X
[ENTER]	6	ENTER lifts 6 into Register Y
[1]	1	Value b (1) placed into Register X
[+]	7	Left operator (+): Combines Y(6) + X(1) = 7 in X
[9]	9	Value c (9) entered into X; 7 lifted to Y
[ENTER]	9	ENTER lifts 9 into Z and 7 into T
[5]	5	Value d (5) placed into Register X
[-]	4	Right operator (-): Combines Z/Y(9) - X(5) = 4 in X; 7 drops to Y
[×]	28	Root operator (×): Combines Y(7) × X(4) = 28 in X!

4. STORY CONTEXT (Why 24 Is the Champion Packing Number, The Typesetter's Bargain & Dijkstra's 1961 Railway Siding)

In the 1500s, typesetters invented parentheses only to squash 2D fraction bars onto flat lines of lead type—then in 1924 Jan Łukasiewicz proved that placing operators after their numbers eliminates every parenthesis, and in 1961 Edsger Dijkstra modeled the 4-register stack as a railway shunting yard! Alongside the Prime Factor Studio (where 24 = 2 × 2 × 2 × 3 decomposes into prime factor towers), students evaluate nested

Core Investigation (Student Warm-Up A): Evaluate (8 + 2) × (14 - 9) by solving the inner quantities first, then checking on the stack.

Expected Answer: 50.

Verification Keys: [8] [ENTER] [2] [+] [1] [4] [ENTER] [9] [-] [×]

Deep Stack Evaluation & Register Limits

Follow-Up Extension (Common Core / TEKS (Illustrative, Eureka, Big Ideas) • CCSS.MATH.6.EE.A.2) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Deeply nested fractions such as (12 - 4) ÷ (2 × (1 + 1)) require holding completed sub-expressions aside while inner terms are resolved. The 4-register stack (T, Z, Y, X) holds up to four nested levels visibly without memory variables or parentheses.

Blocks & Prediction: Track the 4 stack registers (T, Z, Y, X) alongside Prime Factor towers for (12 - 4) / (2 × (1 + 1)). Prediction: Numerator: 12 - 4 = 8 (three [2] blocks). Denominator: 2 × 2 = 4 (two [2] blocks). Result: 8 / 4 = 2.

Extension Key Sequence: [1] [2] [ENTER] [4] [-] [2] [ENTER] [1] [ENTER] [1] [+] [×] [÷]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	First digit of 12
[2]	12	12 in register X
[ENTER]	12	12 lifted to Y
[4]	4	4 in register X
[-]	8	Numerator 12 - 4 = 8 held in X
[2]	2	Factor 2 enters X; 8 lifts to Y
[ENTER]	2	2 lifted to Y; 8 lifts to Z
[1]	1	Inner 1 enters X
[ENTER]	1	1 lifted to Y; 2 in Z; 8 in T
[1]	1	Second 1 enters X
[+]	2	Innermost sum: 1 + 1 = 2 in X
[×]	4	Denominator combined: 2 × 2 = 4 in X; 8 in Y
[÷]	2	8 / 4 = 2!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Evaluating expressions strictly left-to-right (computing 10 + 5 × 2 as 15 × 2 = 30) instead of respecting operator structure.

Hands-On Block & Stack Fix: Model the product branch 5 × 2 with Prime Factor [5] and [2] blocks first: push 10 to Register Y ([10] [ENTER]), evaluate 5 × 2 in X ([5] [ENTER] [2] [×] → 10), then press [+] → 20.

Misconception 2: Losing track of intermediate results when evaluating two parenthesized branches like (8 + 2) × (9 − 4).

Hands-On Block & Stack Fix: Show how Register Y holds the first branch (8 + 2 = 10) automatically while the second branch (9 − 4 = 5) is computed in X, ready for [×] = 50.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 07 INVESTIGATIONS 1–4)

Investigation 1: Evaluate (4 + 3) × (10 - 6) and check on the stack. **Ans: 28**

Check Keys: [4] [ENTER] [3] [+] [1] [0] [ENTER] [6] [-] [×]

Investigation 2: Solve (20 - 4) / (2 × 4) using inner-first evaluation. **Ans: 2**

Check Keys: [2] [0] [ENTER] [4] [-] [2] [ENTER] [4] [×] [÷]

Investigation 3: Evaluate (18 - 6) / (1 + 2) on the 4-level stack. **Ans: 4**

Check Keys: [1] [8] [ENTER] [6] [-] [1] [ENTER] [2] [+] [÷]

Investigation 4: Evaluate 5 + (3 × 4) by computing the inner product first. **Ans: 17**

Check Keys: [3] [ENTER] [4] [×] [5] [+]

Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ with Prime Factor Blocks

Core Concept Lesson • Big Ideas Math (Polynomial Equations & Factoring) • Standard: CCSS.MATH.HSA.APR.A.1

1. MATHEMATICAL CONCEPT & STACK APPROACH: Factor the constant term with Prime Factor blocks, sketch the $(x + p)(x + q)$ area grid, and verify with the SC-32 [POLY] solver.

A polynomial $ax^2 + bx + c$ mirrors base-10 place value with variable powers (x^2, x^1, x^0) in place of 10^2, 10^1, 10^0. Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ decomposes the constant term into prime factors ($2 \times 3 = 6$) whose sum equals the linear coefficient ($2 + 3 = 5$), which [POLY] verifies when [a, b, c] are pushed into Z, Y, X.

Classroom Hook: How is the polynomial $x^2 + 5x + 6 = (x + 2)(x + 3)$ just like the base-10 number $156 = 12 \times 13$ when $x = 10$?

Area Grid (x+2)(x+3) & Prime Pair 2×3=6

x²	+3x
+2x	2×3=6

b = 2+3 = 5, c = 2×3 = 6 -> Roots -2, -3

2. MODELING WITH BLOCKS & STUDENT PREDICTION

Visual / Block Model: Prime Factor Studio (PRIME-01) & Algebraic Area Diagram

Hands-On Setup: Stack Prime Factor blocks [2] and [3] ($2 \times 3 = 6$) in the comparator lid and sketch the 2×2 area grid for $(x + 2)(x + 3) = x^2 + 3x + 2x + 6$.

Predict First (Before Calculator): Prime factors of 6 are 2 and 3, and their sum is $2 + 3 = 5$. The factored form is $(x + 2)(x + 3)$ with roots -2 and -3.

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Factor the constant 6 with Prime Factor blocks [2] and [3]—their sum $2 + 3 = 5$ matches the middle x^1 coefficient, giving factors $(x + 2)(x + 3)$!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Just FOIL the letters and memorize the quadratic formula."

3. CALCULATOR EXECUTION: SC-32 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: 1 ENTER 5 ENTER 6 ↗ POLY

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
1	1	Coefficient a = 1 entered into X
ENTER	1	1 lifted to Y
5	5	Coefficient b = 5 entered into X
ENTER	5	b in Y, a in Z
6	6	Coefficient c = 6 entered into X
↗ POLY	-2	Press [g:] [POLY]: Root 1 = -2 in Y (factor: x + 2), Root 2 = -3 in X (factor: x + 3)!

4. STORY CONTEXT (From Al-Khwarizmi's Squares to the 1535 Duel in Milan & Kahan's [POLY] Solver)

In 820 AD in Baghdad, Al-Khwarizmi solved $x^2 + 10x = 39$ by completing the square—and in 1535 in Milan, Niccolò Tartaglia defeated Antonio Fiore in a 30-problem public algebra duel overnight! By mapping polynomial coefficients onto the Place Value Tray and factoring constant terms with Prime Factor Studio blocks, students discover how $(x + a)(x + b)$ connects directly to the SC-32 [POLY] solver.

Core Investigation (Student Warm-Up A): Place coefficients [1], [7], [12] for $x^2 + 7x + 12$ on your tray, factor 12 with Prime Factor blocks to find the pair summing to 7, then verify with [POLY].

Expected Answer: $(x + 3)(x + 4)$, roots -3 and -4.

Verification Keys: 1 ENTER 7 ENTER 1 2 ↗ POLY

Signed Factor Pairs: Factoring Trinomials with Negative Terms

Follow-Up Extension (Singapore Math • CCSS.MATH.HSA.REI.B.4) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: When a quadratic has a negative constant or linear term (such as $x^2 - x - 6$), the factor pair must have opposite signs (-3 and +2) so their product is -6 and their sum is -1. Entering signed coefficients with [+/-] onto the stack verifies both real roots.

Blocks & Prediction: Set up coefficients for $x^2 - x - 6$ on the Place Value Tray and factor $|c| = 6$ into Prime Factor blocks [2] and [3]. Prediction: Assign signs to the factor pair (2, -3) so their product is -6 and their sum is $2 + (-3) = -1$: factors are $(x - 3)(x + 2)$.

Extension Key Sequence: 1 ENTER 1 +/- ENTER 6 +/- g POLY

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
1	1	Coefficient a = 1
ENTER	1	1 lifted to Y
1	1	Enter magnitude of b
+/-	-1	Coefficient b = -1 in X
ENTER	-1	b in Y, a in Z
6	6	Enter magnitude of c
+/-	-6	Coefficient c = -6 in X
g POLY	3	Press [g:] [POLY]: Root 1 = 3 in Y (factor: x - 3), Root 2 = -2 in X (factor: x + 2)!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Expanding $(x + 2)(x + 3)$ as $x^2 + 6$ by forgetting the two linear cross-terms (2x and 3x).

Hands-On Block & Stack Fix: Place coefficients [1], [5], [6] in the x^2, x^1, x^0 sockets of the Place Value Tray and factor 6 into Prime Factor blocks [2] and [3] so students see that $b = 2 + 3 = 5$ and $c = 2 \times 3 = 6$.

Misconception 2: Using the subtraction operator [−] instead of the change-sign key [+/−] when entering negative polynomial coefficients into [POLY].

Hands-On Block & Stack Fix: Point out that [−] immediately subtracts X from Y, whereas [+/−] negates the number currently in Register X without collapsing the stack.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 08 INVESTIGATIONS 1–4)

Investigation 1: Factor $x^2 + 6x + 8$ by factoring 8 with Prime Factor blocks, then verify with [POLY]. **Ans: $(x + 2)(x + 4)$; roots -2, -4**

Check Keys: 1 ENTER 6 ENTER 8 g POLY

Investigation 2: Factor $x^2 - 4$ (difference of two squares) where b = 0. **Ans: $(x - 2)(x + 2)$; roots 2, -2**

Check Keys: 1 ENTER 0 ENTER 4 +/- g POLY

Investigation 3: Factor $x^2 + 5x + 4$ using Prime Factor blocks and verify. **Ans: $(x + 1)(x + 4)$; roots -1, -4**

Check Keys: 1 ENTER 5 ENTER 4 g POLY

Investigation 4: Factor $x^2 - 5x + 6$ using signed factor pairs. **Ans: $(x - 2)(x - 3)$; roots 2, 3**

Check Keys: 1 ENTER 5 +/- ENTER 6 g POLY

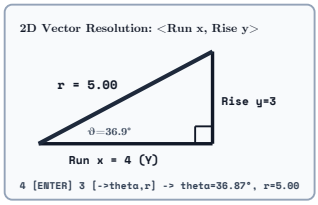
Vector Magnitude & Direction: Cartesian to Polar

Core Concept Lesson • Big Ideas Math (Right Triangle Trigonometry & Vectors) • Standard: CCSS.MATH.HSN.VM.A.1

1. MATHEMATICAL CONCEPT & STACK APPROACH: Decompose 2D vectors into horizontal and vertical components on the stack (Y = Δx, X = Δy) and convert to polar coordinates (r, ϑ).

Every 2D vector or right triangle relates two perpendicular legs (Δx, Δy) to a hypotenuse r and angle ϑ via $r = \sqrt{(x^2 + y^2)}$ and $\tan(\vartheta) = y/x$. By placing Δy in Register Y and Δx in Register X, the stack keeps both components paired to compute magnitude and angle simultaneously.

Classroom Hook: If a surveyor walks 4 blocks East and 3 blocks North, why is the straight-line shortcut 5 blocks instead of 4 + 3 = 7?



2. VISUAL MATHEMATICAL MODEL & STUDENT PREDICTION

Visual / Block Model: Right-Triangle & 2D Vector Diagram (Optional Ratio Check: Fraction Studio)

Hands-On Setup: Draw a right triangle with horizontal run Δx = 4 (Register Y) and vertical rise Δy = 3 (Register X) to model the 3-4-5 vector.

Predict First (Before Calculator): Run = 4, Rise = 3. Predict magnitude: r = sqrt(4^2 + 3^2) = 5. Predict angle: arctan(3/4) ~ 36.87°.

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Put your horizontal Run (Δx) on Waiting Shelf Y and vertical Rise (Δy) on Workbench X, then press [→ϑ,r] to see angle ϑ and hypotenuse r!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Type arctan(y/x) and guess which quadrant you're in."

3. CALCULATOR EXECUTION: SC-32 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [4] [ENTER] [3] [↶] [▸θ,r] [x⇄y]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[4]	4	Enter horizontal component x = 4
[ENTER]	4	4 lifted to register Y
[3]	3	Enter vertical component y = 3 in X
[↶] [▸θ,r]	36.87	Rectangular to Polar! Screen displays angle ϑ = 36.87°
[x⇄y]	5.00	Swap registers: Y displays vector magnitude r = 5.00!

4. STORY CONTEXT (A Stick in Alexandria: How Eratosthenes Measured the Earth with a Shadow & Crosswind Vectors)

In 240 BC, Eratosthenes measured a 7.2° shadow angle in Alexandria at noon on the summer solstice while a well in Syene 800 km away had zero shadow—calculating Earth's circumference within 1% using pure geometry! Using 84.0 mm Fraction Studio ratio towers and [→ϑ,r] polar conversion, students resolve right triangles and aircraft crosswind vectors in a single keystroke.

Core Investigation (Student Warm-Up A): A surveyor walks 60 m East and 80 m North. Find the straight-line displacement magnitude and heading.

Expected Answer: Displacement: 100 m, Heading: 53.13°.

Verification Keys: [6] [0] [ENTER] [8] [0] [↶] [▸θ,r] [x⇄y]

2D Vector Addition: Resultant Forces

Follow-Up Extension (Big Ideas Math • CCSS.MATH.HSN.VM.A.3) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Non-collinear vectors cannot be added by simply summing their lengths; their horizontal (x) and vertical (y) components must be summed separately first. The 4-level stack accumulates ☒x and ☒y from the inside out before converting the resultant back to polar form.

Blocks & Prediction: Sketch component vectors F1 = <3, 2> and F2 = <1, 5> tip-to-tail on a coordinate grid. Prediction: Add components: x_net = 3 + 1 = 4; y_net = 2 + 5 = 7. Magnitude r = sqrt(4^2 + 7^2) = sqrt(65) ~ 8.06.

Extension Key Sequence: [3] [ENTER] [1] [+] [2] [ENTER] [5] [+] [↶] [▸θ,r] [x⇄y]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[3]	3	Horizontal component x1 = 3
[ENTER]	3	Lifted to Y
[1]	1	Horizontal component x2 = 1
[+]	4	Net horizontal force x_net = 4 in X
[2]	2	Vertical component y1 = 2 (4 lifts to Y)
[ENTER]	2	2 lifted to Y; 4 in Z
[5]	5	Vertical component y2 = 5 in X
[+]	7	Net vertical force y_net = 7 in X; x_net = 4 in Y
[↶] [▸θ,r]	60.26	Resultant angle ϑ = 60.26°
[x⇄y]	8.06	Resultant force magnitude r = 8.06!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Evaluating trigonometric functions in Radian mode when angles are given in Degrees (or vice versa).

Hands-On Block & Stack Fix: Have students check the DEG/RAD indicator on the SC-32 status bar and verify with the 1/2 (42 mm) and 1 Whole (84 mm) Fraction Studio towers that sin(30°) = 42/84 = 0.5000 before starting vector problems.

Misconception 2: Swapping x and y components when converting rectangular coordinates to polar via [→ϑ,r].

Hands-On Block & Stack Fix: Place horizontal Run (x) in Row 1 (Register Y) and vertical Rise (y) in Row 0 (Register X) of the Place Value Tray before pressing [→ϑ,r].

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 09 INVESTIGATIONS 1–4)

Investigation 1: Find magnitude and angle for vector <12, 5>. **Ans: ϑ = 22.62°, r = 13**

Check Keys: [1] [2] [ENTER] [5] [↶] [▸θ,r]

Investigation 2: Find the hypotenuse of a right triangle with legs 7 and 24. **Ans: 25**

Check Keys: [7] [ENTER] [2] [4] [↶] [▸θ,r] [x⇄y]

Investigation 3: Compare the 1/2 tower (42 mm) to the 1 Whole tower (84 mm) and calculate sin(30°). **Ans: 0.5**

Check Keys: [3] [0] [SIN]

Investigation 4: A vector has magnitude 10 at angle 45°. What are its x and y components? **Ans: x = 7.07, y = 7.07**

Check Keys: [1] [0] [ENTER] [4] [5] [↷] [▸y,x]

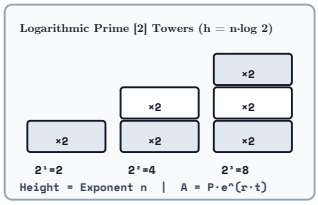
Continuous Compounding & Euler's Constant e

Core Concept Lesson • Big Ideas Math (Exponential & Logarithmic Functions) • Standard: CCSS.MATH.HSF.LE.A.4

1. MATHEMATICAL CONCEPT & STACK APPROACH: Model continuous exponential accumulation and compound interest using the natural exponential function [e^x].

Exponential growth P·e^(rt) compounds continuously by scaling a base quantity according to an inner rate-time product (r × t). Evaluating from the inside out on the stack—computing r × t first, applying [e^x], and multiplying by P—follows the exact mathematical hierarchy without parentheses.

Classroom Hook: Stack three [2] Prime Factor blocks (15 mm each) beside the tower for 8 (45 mm). How did adding physical block heights multiply 2 × 2 × 2 = 8?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Visual / Block Model: Prime Factor Studio (PRIME-01 Logarithmic Blocks) & Growth Curve

Hands-On Setup: Stack Prime Factor [2] blocks (15 mm each) to show linear height addition representing exponential doubling ($2^1 = 2$, $2^2 = 4$, $2^3 = 8$).

Predict First (Before Calculator): Invest $1,000 at 7% continuous interest for 10 years: A = 1000 × e^(0.07 × 10) = 1000 × e^0.7. Will it double? (e^0.7 ~ 2.01).

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Compute the inner growth exponent r × t first, press [e^x] to get the growth factor, and then multiply by principal P!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Multiply the base by the exponent or round intermediate decimals to two places."

3. CALCULATOR EXECUTION: SC-32 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: `0` `.` `0` `7` `ENTER` `1` `0` `×` `e^x` `1` `0` `0` `0` `×`

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
`0` `.` `0` `7`	0.07	Rate r = 0.07 entered in X
`ENTER`	0.07	Rate lifted to Y
`1` `0`	10	Time t = 10 entered in X
`×`	0.7	Exponent r × t = 0.70
`e^x`	2.0138	e^0.70 = 2.01375 (Growth factor: more than doubled!)
`1` `0` `0` `0`	1000	Principal P = 1000 (growth factor lifts to Y)
`×`	2013.75	Total accumulated amount A = $2,013.75!

4. STORY CONTEXT (The Scottish Baron Who Doubled the Lifespan of Astronomers (John Napier), Euler's e & The Rule of 72)

In 1614, John Napier at Merchiston Castle realized that logarithms turn tedious multiplication into simple height addition—the exact engineering principle of our Prime Factor blocks (where h = 15 × log2(p) mm)! Paired with Jakob Bernoulli's continuous compounding limit e and Fibonacci's Liber Abaci (1202), students model exponential doubling and half-life decay.

Core Investigation (Student Warm-Up A): A bacteria culture starts at 500 and triples every hour. Use [y^x] to calculate population after 4 hours.

Expected Answer: 40,500 bacteria.

Verification Keys: `3` `ENTER` `4` `y^x` `5` `0` `0` `×`

Solving for Time: Doubling Time & Natural Log [LN]

Follow-Up Extension (Big Ideas Math • CCSS.MATH.HSF.BF.B.5) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: A logarithm is the inverse of exponentiation: it converts multiplicative scaling into additive height (ln(a·b) = ln(a) + ln(b)). To solve for time t in P·e^(rt) = A on the stack, divide A by P first, take [LN] to unwrap the exponent, and divide by r.

Blocks & Prediction: Observe how Prime Factor block heights scale with ln(p) on the comparator lid while configuring target multiplier in Y and growth rate in X. Prediction: At 5% growth (r = 0.05), doubling time is t = ln(2) / 0.05 ~ 0.69315 / 0.05 ~ 13.86 years.

Extension Key Sequence: [2] [LN] [0] [•] [0] [5] [÷]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[2]	2	Doubling target factor 2
[LN]	0.6931	Natural log ln(2) = 0.693147... in X
[0] [•] [0] [5]	0.05	Rate r = 0.05 (ln(2) lifts to Y)
[÷]	13.86	t = ln(2) / r = 13.86 years!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Multiplying the base by the exponent (computing 2^5 as 2 × 5 = 10) or reversing Base and Exponent on [y^x].

Hands-On Block & Stack Fix: Stack five [2] Prime Factor logarithmic blocks in the comparator lid to show that adding 5 block heights multiplies 2×2×2×2×2 = 32, and verify that Base y goes in Register Y and Exponent x goes in X before [y^x].

Misconception 2: Applying the power [y^x] after multiplying by principal P in P(1 + r)^t (computing (P·(1+r))^t instead of P·((1+r)^t)).

Hands-On Block & Stack Fix: Use the Inside-Out RPN rule: compute (1 + r)^t first ([1.05] [ENTER] [2] [y^x]), then multiply by P ([1000] [×]).

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 10 INVESTIGATIONS 1–4)

Investigation 1: Calculate e^2 on the calculator to 4 decimal places. **Ans: 7.3891**

Check Keys: [2] [e^x]

Investigation 2: Calculate doubling time for an investment growing at 8% continuous interest. **Ans: 8.66 years**

Check Keys: [2] [LN] [0] [•] [0] [8] [÷]

Investigation 3: Compute 10^3.5 using [f:] [10^x]. **Ans: 3162.28**

Check Keys: [3] [•] [5] [↶] [10^x]

Investigation 4: Evaluate log10(500) using [f:] [LOG]. **Ans: 2.6990**

Check Keys: [5] [0] [0] [↶] [LOG]

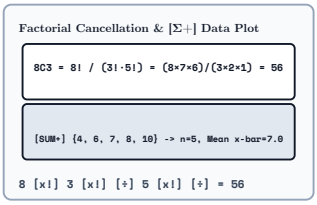
Combinations vs. Permutations: Subsets & Prime Cancellation

Core Concept Lesson • AP Statistics (Probability, Random Variables & Combinatorics) • Standard: CCSS.MATH.HSS.CP.B.9

1. MATHEMATICAL CONCEPT & STACK APPROACH: Calculate permutations and combinations by evaluating factorials across the 4-register stack and canceling prime factors.

Combinations nCr = n! / (r!(n-r)!) count unordered selections by dividing ordered permutations by duplicate arrangements. On the stack, you can either evaluate and cancel the factorial branches step-by-step across Y and X or verify directly with [nCr].

Classroom Hook: Compare the 1/6 tower (14 mm) and 1/2 tower (42 mm) in the Fraction Studio comparator lid: how many 1/6 outcomes equal a 50% probability?



2. VISUAL MATHEMATICAL MODEL & STUDENT PREDICTION

Visual / Block Model: Prime Factor Studio (PRIME-01 Factorial Cancellation) & [Σ+] Data Plot

Hands-On Setup: Use Prime Factor blocks to model prime factor cancellation in 8! / (3! × 5!) = (8 × 7 × 6) / 6 = 8 × 7 = 56.

Predict First (Before Calculator): How many ways to choose a committee of 3 from 8 candidates? 8C3 = 8! / (3! × 5!) = 40320 / (6 × 120) = 56.

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Press [CLΣ] first, then feed each sample value in with [Σ+]—watch Register X count n while the stack tracks the running mean!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Re-type all 12 numbers from scratch if you want both the mean and standard deviation."

3. CALCULATOR EXECUTION: SC-32 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: 8 ↶ x! 3 ↶ x! ÷ 5 ↶ x! ÷

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
8	8	Total candidates n = 8
↶ x!	40320	8! = 40,320 in register X
3	3	Subset r = 3 (40320 lifts to Y)
↶ x!	6	3! = 6 in X
÷	6720	40320 / 6 = 6720 in X
5	5	Remaining n - r = 5 (6720 lifts to Y)
↶ x!	120	5! = 120 in X
÷	56	6720 / 120 = 56 combinations!

4. STORY CONTEXT (Sir Francis Galton's 1873 Bean Machine, The Asteroid Ceres & The Guinness Brewery Chemist)

In 1873, Sir Francis Galton built a board studded with pins (the Quincunx) to show how independent random bounces settle into a bell-shaped curve—while in 1908 at the Guinness Brewery in Dublin, chemist William Sealy Gosset ('Student') invented small-sample statistics with n - 1 degrees of freedom! Students model exact probability ratios on 84 mm Fraction Studio towers, cancel factorial primes, and accumulate trials

Core Investigation (Student Warm-Up A): A lottery draws 4 winning numbers from a pool of 10. How many combinations? Calculate 10C4.

Expected Answer: 210 combinations.

Verification Keys: 1 0 ↶ x! 4 ↶ x! ÷ 6 ↶ x! ÷

Statistical Accumulation: Mean & Variance with [Σ+]

Follow-Up Extension (AP Statistics • CCSS.MATH.HSS.ID.A.2) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Statistical measures of center (Mean x-bar) and spread (Standard Deviation s) summarize an entire distribution of values. Pressing [Σ+] on the stack calculator accumulates n, ☒x, and ☒x^2 simultaneously in dedicated statistical registers so x-bar and s can be read without re-entering data.

Blocks & Prediction: Lay out the 5 sample values (4, 6, 7, 8, 10) with Digit Blocks on the Place Value Tray and accumulate them with [Σ+]. Prediction: Sum = 4 + 6 + 7 + 8 + 10 = 35. Sample size n = 5. Mean x-bar = 35/5 = 7.

Extension Key Sequence: [4] [Σ+] [6] [Σ+] [7] [Σ+] [8] [Σ+] [1] [0] [Σ+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[4] [Σ+]	1	First score entered: sample size n = 1
[6] [Σ+]	2	Second score: n = 2
[7] [Σ+]	3	Third score: n = 3
[8] [Σ+]	4	Fourth score: n = 4
[1] [0] [Σ+]	5	Fifth score: n = 5 total data points

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Forgetting to clear the statistical registers ([CLΣ]) before entering a new dataset with [Σ+].

Hands-On Block & Stack Fix: Notice that pressing [Σ+] displays the sample count n in Register X; if the first data point shows n > 1, press [CLΣ] first so prior data does not corrupt the mean.

Misconception 2: Dividing by r! and then multiplying by (n − r)! when computing C(n, r) = n! / (r!(n − r)!).

Hands-On Block & Stack Fix: Compute n! into Y, compute r! and (n − r)! and multiply them together in X ([×]), then press [÷] once to divide n! by the combined denominator.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 11 INVESTIGATIONS 1–4)

Investigation 1: Calculate 6! (6 factorial) on the calculator. **Ans: 720**

Check Keys: [6] [↶] [x!]

Investigation 2: How many ways can 5 runners finish 1st, 2nd, and 3rd? (5P3 = 5! / 2!). **Ans: 60**

Check Keys: [5] [↶] [x!] [2] [↶] [x!] [÷]

Investigation 3: Calculate 7C2 (combinations of 2 from 7). **Ans: 21**

Check Keys: [7] [↶] [x!] [2] [↶] [x!] [÷] [5] [↶] [x!] [÷]

Investigation 4: Accumulate 5 and 15 with [Σ+]. What is the mean? **Ans: Mean = 10**

Check Keys: [5] [Σ+] [1] [5] [Σ+] [↷] [$\bar{x}, \bar{y}$]

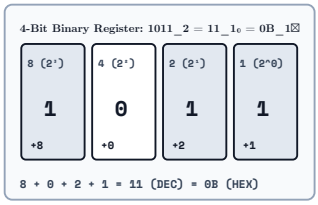
Number Representation: Decimal to Binary and Hexadecimal

Core Concept Lesson • AP Computer Science Principles (Data Representation & Binary Number Systems) • Standard: CCSS.MATH.PRACTICE.MP5

1. MATHEMATICAL CONCEPT & STACK APPROACH: Represent integers physically as [0] and [1] bit registers on the Place Value Tray and computationally convert between bases with [f:] [BASE].

Binary (base-2) and hexadecimal (base-16) follow the exact same positional place-value principle as base-10, using powers of 2 (8, 4, 2, 1) instead of powers of 10. Switching base modes on the stack changes how the active value in Register X is grouped and displayed without changing the underlying number.

Classroom Hook: How can you represent every number from 0 to 15 using only [0] and [1] Digit Blocks across 4 sockets of your Place Value Tray?



2. MODELING WITH BLOCKS & STUDENT PREDICTION

Visual / Block Model: Place Value Studio (Binary [0]/[1] Register Trays & Base-N Columns)

Hands-On Setup: Use the sockets of the Place Value Tray as binary bit positions (8, 4, 2, 1). Place [1] Digit Blocks for active '1' bits and [0] Digit Blocks for '0' bits.

Predict First (Before Calculator): Place [1], [0], [1], [1] in the 8, 4, 2, 1 sockets: binary $1011_2 = 8 + 0 + 2 + 1 = 11_10$. Predict hex value: $0B_16$.

2A. TEACHER TALK — WHAT TO SAY & ASK:

"Place [1] and [0] Digit Blocks in the 8-4-2-1 sockets of your Place Value Tray, then toggle [BIN] and [HEX] on the SC-32 to see the exact same bit pattern!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Convert binary to decimal only by long-division remainders on scratch paper."

3. CALCULATOR EXECUTION: SC-32 KEYSTROKES & 4-REGISTER STACK TRACE

Complete Key Sequence: [1] [1] [↶] [BASE]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[1] [1]	11	Enter decimal integer 11 in DEC mode
[↶] [BASE]	HEX 0B	Press [f:] [BASE]: screen displays Hexadecimal 0B!
[↶] [BASE]	BIN 1011	Press [f:] [BASE]: screen displays Binary 1011!
[↶] [BASE]	OCT 13	Press [f:] [BASE]: screen displays Octal 13!
[↶] [BASE]	DEC 11	Press [f:] [BASE]: returns to Decimal 11

4. STORY CONTEXT (Leibniz's Binary Medallion, Claude Shannon's 1937 Relay Basement & NASA's 36-Bit IBM 7090 Word)

In 1697, Gottfried Wilhelm Leibniz designed a medallion showing that every number can be built from 1 and 0 alone; in 1937, 21-year-old MIT student Claude Shannon proved electrical switches execute Boolean algebra; and in 1961, NASA's IBM 7090 computed orbital trajectories in 36-bit binary words! Using [0] and [1] Digit Blocks on the Place Value Tray and [BASE], students decode bytes, hex nibbles, and bitwise masks.

Core Investigation (Student Warm-Up A): Enter decimal 255. Use [f:] [BASE] to convert to HEX and BIN. What is the binary pattern?

Expected Answer: HEX: FF, BIN: 11111111 (8 ones).

Verification Keys: [2] [5] [5] [↶] [BASE]

Bitwise Logic: AND Masking & Network Subnets

Follow-Up Extension (AP Computer Science Principles • CCSS.MATH.PRACTICE.MP5) + Misconception Remediations & Lab Workbook Key

1. EXTENSION LESSON (STUDENT WARM-UP B): CONCEPT → BLOCKS → CALCULATOR

Concept & Stack Approach: Bitwise logic operations (AND, OR, XOR, NOT) compare corresponding binary place-value columns of two integers in parallel. Placing value A on Waiting Shelf Y and mask B on Workbench X lets [AND] filter bit columns in a single stack operation.

Blocks & Prediction: Row 1 of the Place Value Tray holds nibble A ([1][1][0][0] = 12). Row 0 holds mask B ([1][0][1][0] = 10). Prediction: Bitwise AND: bit 3 (1 & 1 = 1), bit 2 (1 & 0 = 0), bit 1 (0 & 1 = 0), bit 0 (0 & 0 = 0). Result: 1000_2 = 8_10.

Extension Key Sequence: [1] [2] [ENTER] [1] [0] [AND]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1] [2]	12	Enter operand A = 12 (binary 1100)
[ENTER]	12	12 lifted to register Y
[1] [0]	10	Enter operand B = 10 (binary 1010) in X
[AND]	8	Bitwise AND evaluates 1100 & 1010 = 1000 (8 decimal)!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Confusing bitwise [AND] / [OR] with decimal addition or multiplication (e.g., thinking 12 AND 10 is 22).

Hands-On Block & Stack Fix: Model 12 ([1][1][0][0]_2) in Row 1 and 10 ([1][0][1][0]_2) in Row 0 of the Place Value Tray using [1] and [0] Digit Blocks; only columns where BOTH rows hold [1] remain 1 in 12 AND 10 = 8 ([1][0][0][0]_2).

Misconception 2: Reading hexadecimal [10] as ten instead of decimal 16 ($1 \times 16^1 + 0 \times 16^0$).

Hands-On Block & Stack Fix: Place [1] in the Sixteens column and [0] in the Ones column on the Place Value Tray, then switch from [HEX] [10] to [DEC] on the SC-32 to see 16.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 12 INVESTIGATIONS 1–4)

Investigation 1: Convert decimal 42 to binary using [f:] [BASE]. **Ans: 101010 (32 + 8 + 2)**

Check Keys: [4] [2] [↰] [BASE]

Investigation 2: Convert hexadecimal 2A to decimal. **Ans: 42**

Check Keys: [In] [HEX] [mode:] [2] [A] [->] [switch] [to] [DEC]

Investigation 3: Evaluate bitwise AND: 15 (1111) AND 7 (0111). **Ans: 7 (0111)**

Check Keys: [1] [5] [ENTER] [7] [AND]

Investigation 4: Evaluate bitwise XOR: 12 (1100) XOR 10 (1010). **Ans: 6 (0110)**

Check Keys: [1] [2] [ENTER] [1] [0] [XOR]

Middle School Section Coaching & 900 Keys

Complete teacher coaching notes, misconception alerts, and worked Practice A & B setups for Book 3 (Middle School Math Lab).

HOW PART II CONNECTS CONCEPT DOMAINS TO DAILY MIDDLE SCHOOL SECTIONS

While Part I organizes K–12 instruction by macro Concept Domain, middle school teachers using Book 3 (Middle School Math Lab — Aligned to Big Ideas Math Grades 6–8) also need rapid section-by-section coaching notes and complete worked solutions for all 45 sections and 900 dual-method problems (450 Practice A + 450 Practice B). Pages 29–40 provide the complete Middle School Teacher & Coach Guide right inside this master volume.

Middle School Unit (Book 3)	Sections & Core Concepts	Related Part I Domains	Coach Page
Unit 01: Numerical Expressions & Factors	Sec 1.1–1.6 (Powers, Order of Ops, Prime Factor, GCF/LCM)	**Domains 05 & 07**	**p. 29**
Unit 02: Fractions & Decimals	Sec 2.1–2.6 (Multiplying/Dividing Fractions & Decimals)	**Domains 05 & 06**	**p. 30**
Unit 03: Ratios, Rates & Percents	Sec 3.1–3.6 (Ratios, Tape Diagrams, Unit Rates & Percents)	**Domains 04 & 06**	**p. 31**
Unit 04: Algebraic Expressions & Properties	Sec 4.1–4.4 (Evaluating, Writing, Properties & Factoring)	**Domains 07 & 08**	**p. 32**
Unit 05: Areas of Polygons & Elapsed Time	Sec 5.1–5.5 (Parallelograms, Triangles, Trapezoids & Time)	**Domains 05 & 09**	**p. 33**
Unit 06: Integers, Rationals & Coord. Plane	Sec 6.1–6.5 (Signed Numbers, Absolute Value & 4 Quadrants)	**Domains 03, 06 & 08**	**p. 34**
Unit 07: Equations & Inequalities	Sec 7.1–7.4 (1-Step Equations, Tables/Graphs & Inequalities)	**Domains 07 & 08**	**p. 35**
Unit 08: Surface Area & Volume of 3D Solids	Sec 8.1–8.4 (3D Nets, Prisms, Pyramids & Fractional Edge Vol)	**Domains 06 & 09**	**p. 36**
Unit 09: Statistical Measures & Data Displays	Sec 9.1–9.5 (Mean, Median, Mode, IQR, MAD & Box Plots)	**Domain 11**	**p. 37**
Unit 10: Grade 7 Bridge — Signed, 2-Step & Prob.	Sec 10.1–10.4 (Signed Rationals, 2-Step Eq, Circles & Prob.)	**Domains 08, 09 & 11**	**p. 38**
Middle School Misconception Matrix	10 High-Frequency Errors & Block Remediations	**All Domains**	**p. 39**
Section Mastery & Token Roster	45-Section Pacing & Classroom Check-Off Grid	**All Domains**	**p. 40**

Dual-Method Verification Rule for Middle School Classrooms

Every problem in Book 3 requires (1) written mathematical setup and mental/paper estimation first, followed by (2) RPN stack verification on the SC-6 calculator. Pages 29–38 list both the written setup and exact SC-6 keystrokes for all 900 problems.

Unit 1: Numerical Expressions & Factors — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 1 • Concept → Blocks (When Applicable: Place Value & Prime Factor Blocks) → Calculator

1. **Concept (Stack Approach): Multi-step expressions evaluate from the inside out—holding completed groups on shelf Y while building the next group on X. Every whole number factors uniquely into primes.**

2. Blocks (Place Value & Prime): Model division quotients/remainders on the Place Value tray and stack Prime Factor blocks to see exponents (2^3), GCF (overlap), and LCM (union).

3. **Calculator (SC-6): Use |g||÷R| to split Quotient (Y) and Remainder (X), |g||FACT] for prime powers, and |GCF]/|f||LCM] to verify Venn diagram results.**



SEC 1.1 (pp.4-5): WHOLE NUMBER OPERATIONS & DIVISION WITH REMAINDERS

Methods: W1: Standard Long Division | W2: Partial Quotients | W3: Equal-Groups Array Model

PRACTICE A (p.4): KEY IDEAS (#1-#10) — ExA: 5 R 3		PRACTICE B (p.5): REAL LIFE (#1-#10) — ExB: 17 vans (16 full vans + 1 van with 7 students)	
A1: 5 R 1 |31 ÷ 6 (Q & R)| SC-6: 31 [ENTER] 6 [g][÷R]	A2: 9 R 2 |47 ÷ 5 (Q & R)| SC-6: 47 [ENTER] 5 [g][÷R]	B1: 8 full packs, 5 cells left |53 ÷ 6 (Q & R)| SC-6: 53 [ENTER] 6 [g][÷R]	B2: 11 figures, 6 tickets left |94 ÷ 8 (Q & R)| SC-6: 94 [ENTER] 8 [g][÷R]
A3: 8 R 2 |58 ÷ 7 (Q & R)| SC-6: 58 [ENTER] 7 [g][÷R]	A4: 9 R 4 |85 ÷ 9 (Q & R)| SC-6: 85 [ENTER] 9 [g][÷R]	B3: 20 canoes (19 full + 1 with 2 campers) |78 ÷ 4 (Q & R)| SC-6: 78 [ENTER] 4 [g][÷R]	B4: 9 full racks, 2 on last |110 ÷ 12 (Q & R)| SC-6: 110 [ENTER] 12 [g][÷R]
A5: 12 R 2 |146 ÷ 12 (Q & R)| SC-6: 146 [ENTER] 12 [g][÷R]	A6: 7 full lines, 5 cm left |250 ÷ 35 (Q & R)| SC-6: 250 [ENTER] 35 [g][÷R]	B5: 28 knights, 2 m left |450 ÷ 16 (Q & R)| SC-6: 450 [ENTER] 16 [g][÷R]	B6: 18 full, 19 caches total |165 ÷ 9 (Q & R)| SC-6: 165 [ENTER] 9 [g][÷R]
A7: 117 |(14 × 8) + 5| SC-6: 14 [ENTER] 8 [×] 5 [+]	A8: 9 R 3 (Rem < 8) |75 ÷ 8 (Q & R)| SC-6: 75 [ENTER] 8 [g][÷R]	B7: 187 marbles |(12 × 15) + 7| SC-6: 12 [ENTER] 15 [×] 7 [+]	B8: 8 full trains, 8 on final |200 ÷ 24 (Q & R)| SC-6: 200 [ENTER] 24 [g][÷R]
A9: 19 rovers, 2 tires left |78 ÷ 4 (Q & R)| SC-6: 78 [ENTER] 4 [g][÷R]	A10: 8 tents (7 full + 1 for the 29th explorer) |29 ÷ 4 (Q & R)| SC-6: 29 [ENTER] 4 [g][÷R]	B9: 9 pizzas (5 slices left) |67 ÷ 8 (Q & R)| SC-6: 67 [ENTER] 8 [g][÷R]	B10: 11 full crates, 17 apples left |325 ÷ 28 (Q & R)| SC-6: 325 [ENTER] 28 [g][÷R]

SEC 1.2 (pp.6-7): POWERS AND EXPONENTS

Methods: W1: Multiply Step-by-Step | W2: Group Factors in Pairs | W3: Square & Cube Array Model

PRACTICE A (p.6): KEY IDEAS (#1-#10) — ExA: $4^3 = 16$, $4^3 = 64$		PRACTICE B (p.7): REAL LIFE (#1-#10) — ExB: 81 shares	
A1: $5 \times 5 = 25$ |5^2| SC-6: 5 [x²]	A2: 49 (vs. 14) |7^2| SC-6: 7 [x²]	B1: $8^2 = 64$ pixels |8^2| SC-6: 8 [x²]	B2: $6^3 = 216$ blocks |6^3| SC-6: 6 [x³]
A3: $3 \times 3 \times 3 = 27$ |3^3| SC-6: 3 [x³]	A4: 125 |5^3| SC-6: 5 [x³]	B3: 2^6 = 64 bacteria |2^6| SC-6: 2 [ENTER] 6 [g][y^x]	B4: 4 × 4 × 4 × 4 = 256 players |4^4| SC-6: 4 [ENTER] 4 [g][y^x]
A5: 32 |2^5| SC-6: 2 [ENTER] 5 [g][y^x]	A6: 1,296 |6^4| SC-6: 6 [ENTER] 4 [g][y^x]	B5: 144 + 25 = 169 ft² |$12^2 + 5^2$| SC-6: 12 [x²] 5 [x²] [+]	B6: 5^4 = 625 starlings |5^4| SC-6: 5 [ENTER] 4 [g][y^x]
A7: 10,000 (4 zeros) and 1 |10^4| SC-6: 10 [ENTER] 4 [g][y^x]	A8: 16 (2×2×2×2 = 16) |2^4| SC-6: 2 [ENTER] 4 [g][y^x]	B7: 14 meters |√(196)| SC-6: 196 [f]|√x|	B8: 2^5 = 32 > 5^2 = 25 (by 7) |2^5 - 5^2| SC-6: 2 [ENTER] 5 [g][y^x] 5 [x²] [-]
A9: 96 kN (3 × 32) |2^5 × 3| SC-6: 2 [ENTER] 5 [g][y^x] 3 [×]	A10: 8 m^3 per room; 64 m^3 total |$2^3 \times 8$| SC-6: 2 [x³] 8 [×]	B9: 1,024 bytes |2^10| SC-6: 2 [ENTER] 10 [g][y^x]	B10: 27 + 343 = 370 in^3 |$3^3 + 7^3$| SC-6: 3 [x³] 7 [x³] [+]

SEC 1.3 (pp.8-9): ORDER OF OPERATIONS

Methods: W1: PEMDAS Step-by-Step Funnel | W2: Distributive Property / Fraction-Bar Grouping

PRACTICE A (p.8): KEY IDEAS (#1-#10) — ExA: 72		PRACTICE B (p.9): REAL LIFE (#1-#10) — ExB: 25	
A1: 11 |5 + (2 × 3)| SC-6: 5 [ENTER] 2 [ENTER] 3 [×] [+]	A2: 4 |(14 + 6) ÷ (7 - 2)| SC-6: 14 [ENTER] 6 [+] 7 [ENTER] 2 [-] [÷]	B1: 4(9 + 5) = $56 |(9 + 5) × 4| SC-6: 9 [ENTER] 5 [+] 4 [×]	B2: 45 + 4(7) = $73 |45 + (4 × 7)| SC-6: 45 [ENTER] 4 [ENTER] 7 [×] [+]
A3: 6 |30 - (4 × 6)| SC-6: 30 [ENTER] 4 [ENTER] 6 [×] [-]	A4: 84 |(9 + 3) × (15 - 8)| SC-6: 9 [ENTER] 3 [+] 15 [ENTER] 8 [-] [×]	B3: 101 points |(100 - (3 × 8)) + 25| SC-6: 100 [ENTER] 3 [ENTER] 8 [×] [-] 25 [+]	B4: $10^2 - 6^2 = 64$ m² |$10^2 - 6^2$| SC-6: 10 [x²] 6 [x²] [-]
A5: 1 |50 - (3 + 4)²| SC-6: 50 [ENTER] 3 [ENTER] 4 [+] [x²] [-]	A6: 20 |120 - (4 × (3 + 2)²)| SC-6: 120 [ENTER] 4 [ENTER] 3 [ENTER] 2 [+] [x²] [×] [-]	B5: $73 |(2 × 14) + (5 × 9)| SC-6: 2 [ENTER] 14 [×] 5 [ENTER] 9 [×] [+]	B6: 70 points |((3² + 5) × 6) - 14| SC-6: 3 [x²] 5 [+] 6 [×] 14 [-]
A7: 2 (24 ÷ 12) |(48 - (4 × 6)) ÷ (3 + (2 × 4.5))| SC-6: 48 [ENTER] 4 [ENTER] 6 [×] [-] 3 [ENTER] 2 [ENTER] 4.5 [×] [+] [÷]	A8: 24 |(3^3 + (15 ÷ 3)) - (2 × 4)| SC-6: 3 [x³] 15 [ENTER] 3 [÷] [+] 2 [ENTER] 4 [×] [-]	B7: 10 bracelets per box |(72 - 12) ÷ 6| SC-6: 72 [ENTER] 12 [-] 6 [÷]	B8: 42 (÷ and × go left to right) |((48 ÷ 6) × 4) + 10| SC-6: 48 [ENTER] 6 [÷] 4 [×] 10 [+]
A9: $92 (60 + 32) |(5 × 12) + (4 × 8)| SC-6: 5 [ENTER] 12 [×] 4 [ENTER] 8 [×] [+]	A10: $62 (90 - 28) |(45 × 2) - 28| SC-6: 45 [ENTER] 2 [×] 28 [-]	B9: 12 muffins per basket |((5 × 12) + (3 × 8)) ÷ 7| SC-6: 5 [ENTER] 12 [×] 3 [ENTER] 8 [×] [+] 7 [÷]	B10: 74 |((4³ - 14) ÷ (2 + 3)) + 8²| SC-6: 4 [x³] 14 [-] 2 [ENTER] 3 [+] [÷] 8 [x²] [+]

SEC 1.4 (pp.10-11): PRIME FACTORIZATION

Methods: W1: Factor Tree (Pair A) | W2: Factor Tree (Pair B) | W3: Division Ladder

PRACTICE A (p.10): KEY IDEAS (#1-#10) — ExA: 60 = 2² × 3 × 5		PRACTICE B (p.11): REAL LIFE (#1-#10) — ExB: 2² × 3 × 5 (e.g., 8 × 15 or 10 × 12)	
A1: $2^2 \times 3$ |Factors(12)| SC-6: 12 [g][FACT]	A2: $2^2 \times 3^2$ |Factors(36)| SC-6: 36 [g][FACT]	B1: 1×48, 2×24, 3×16, 4×12, 6×8; $2^4 \times 3$ |Factors(48)| SC-6: 48 [g][FACT]	B2: 2×3^3 |Factors(54)| SC-6: 54 [g][FACT]
A3: 29 is prime; 91 is composite (7 × 13) |Factors(91)| SC-6: 29 [PRIME], 91 [g][FACT]	A4: $3^2 \times 5$ |Factors(45)| SC-6: 45 [g][FACT]	B3: 51 is composite (3 × 17) |Factors(51)| SC-6: 51 [g][FACT]	B4: $2^4 \times 5$ |Factors(80)| SC-6: 80 [g][FACT]
A5: $2^3 \times 3^2$ |Factors(72)| SC-6: 72 [g][FACT]	A6: $2^2 \times 3 \times 7$ |Factors(84)| SC-6: 84 [g][FACT]	B5: 126 |$(2 \times 3^2) \times 7$| SC-6: 2 [ENTER] 3 [x²] [×] 7 [×]	B6: $2^2 \times 5^2$ |Factors(100)| SC-6: 100 [g][FACT]
A7: 2 × 3 × 5 × 7 |Factors(210)| SC-6: 210 [g][FACT]	A8: $2^3 \times 3^2 \times 7$ |Factors(504)| SC-6: 504 [g][FACT]	B7: $2^2 \times 5 \times 7$ |Factors(140)| SC-6: 140 [g][FACT]	B8: $2^4 \times 3^2$ |Factors(144)| SC-6: 144 [g][FACT]
A9: 7 × 11 × 13 |Factors(1001)| SC-6: 1001 [g][FACT]	A10: $2^4 \times 3^2 \times 5^2$ |Factors(3600)| SC-6: 3600 [g][FACT]	B9: Both give $2 \times 3^2 \times 5$ |Factors(90)| SC-6: 90 [g][FACT]	B10: 2 × 3 × 5 × 7 = 210 |((2 × 3) × 5) × 7| SC-6: 2 [ENTER] 3 [×] 5 [×] 7 [×]

SEC 1.5 (pp.12-13): GREATEST COMMON FACTOR (GCF) & LEAST COMMON MULTIPLE (LCM)

Methods: W1: List Factors & Multiples | W2: Prime Factorization | W3: Prime Factor Venn Diagram

PRACTICE A (p.12): KEY IDEAS (#1-#10) — ExA: GCF = 10, LCM = 100		PRACTICE B (p.13): REAL LIFE (#1-#10) — ExB: 6 packs (3 pretzels and 4 fruit bars per pack)	
A1: GCF = 6, LCM = 36 |GCF(12, 18); LCM(12, 18)| SC-6: 12 [ENTER] 18 [GCF] / [f][LCM]	A2: GCF = 4, LCM = 24 |GCF(8, 12); LCM(8, 12)| SC-6: 8 [ENTER] 12 [GCF] / [f][LCM]	B1: 14 kits (2 bandages, 3 wipes each) |GCF(28, 42)| SC-6: 28 [ENTER] 42 [GCF]	B2: 60 years |LCM(12, 15)| SC-6: 12 [ENTER] 15 [f][LCM]
A3: GCF = 5, LCM = 75 |GCF(15, 25); LCM(15, 25)| SC-6: 15 [ENTER] 25 [GCF] / [f][LCM]	A4: GCF = 7, LCM = 70 |GCF(14, 35); LCM(14, 35)| SC-6: 14 [ENTER] 35 [GCF] / [f][LCM]	B3: 24 (3 packs of hot dogs, 2 packs of buns) |LCM(8, 12)| SC-6: 8 [ENTER] 12 [f][LCM]	B4: 16 singers per row |GCF(32, 48)| SC-6: 32 [ENTER] 48 [GCF]
A5: 24 |GCF(48, 72)| SC-6: 48 [ENTER] 72 [GCF]	A6: 28 |GCF(84, 140)| SC-6: 84 [ENTER] 140 [GCF]	B5: 10:12 AM (72 minutes later) |LCM(18, 24)| SC-6: 18 [ENTER] 24 [f][LCM]	B6: 12 inches |GCF(36, 60)| SC-6: 36 [ENTER] 60 [GCF]
A7: GCF = 15, LCM = 225 |GCF(45, 75); LCM(45, 75)| SC-6: 45 [ENTER] 75 [GCF] / [f][LCM]	A8: 60 minutes |LCM(15, 20)| SC-6: 15 [ENTER] 20 [f][LCM]	B7: GCF = 7, LCM = 42 |LCM(14, 21)| SC-6: 14 [ENTER] 21 [GCF] 14 [ENTER] 21 [f][LCM]	B8: 60 minutes |LCM(LCM(4, 6), 10)| SC-6: 4 [ENTER] 6 [f][LCM] 10 [f][LCM]
A9: 180 seconds (3 minutes) |LCM(45, 60)| SC-6: 45 [ENTER] 60 [f][LCM]	A10: 12 bags (3 red, 4 blue per bag) |GCF(36, 48)| SC-6: 36 [ENTER] 48 [GCF]	B9: 15 bouquets (3 gold, 5 silver each) |GCF(45, 75)| SC-6: 45 [ENTER] 75 [GCF]	B10: GCF=5, LCM=60; 5×60 = 300 = 15×20 |GCF(15, 20) × LCM(15, 20)| SC-6: 15 [ENTER] 20 [GCF] 15 [ENTER] 20 [f][LCM] [×]

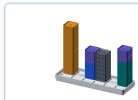
Unit 2: Fractions & Decimals — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 2 • Concept → Blocks (When Applicable: Fraction Strips & Place Value Blocks) → Calculator

1. Concept (Stack Approach): Rational operations require common units before adding/subtracting and multiplicative inverses (reciprocals) when dividing, holding the first operand on Y while transforming X.

2. Blocks (Fraction & Place Value): Compare 84 mm Fraction Strips side-by-side to see common denominators and 'how many fit' division; align tenths and hundredths on the Decimal Tray.

3. Calculator (SC-6): Enter fractions with Two-Dot notation (|.| n |.| d), flip only the divisor in X with |f||1/x| before |×|, and split decimals with |f||IP| and |g||FP|.



SEC 2.1 (pp.15-16): EQUIVALENT FRACTIONS, MIXED NUMBERS & LCD ADDITION

Methods: W1: Divide Once by GCF / Use LCD | W2: Step-by-Step Common Factors | W3: Fraction Strips Model

PRACTICE A (p.15): KEY IDEAS (#1-#10) — ExA: 2/3

A1: 2/3 |Simp(8/12)×2|
SC-6: [.] 8 [.] 12 [ENTER] [g][SIMP]×2

A2: 8/3 = 2 2/3 trays |Simp(48/18)|
SC-6: [.] 48 [.] 18 [ENTER] [g][SIMP]

A3: 13/5 and 2.6 |Improper(2 3/5)|
SC-6: 2 [.] 3 [.] 5 [ENTER] [f][d/c]

A4: 3 5/6 pizzas (3 whole, 5 sixths) |23/6|
SC-6: [.] 23 [.] 6 [ENTER]

A5: 5/6 |1/2 + 1/3|
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 3 [+]

A6: 1 3/20 gallons (23/20) |3/4 + 2/5|
SC-6: [.] 3 [.] 4 [ENTER] [.] 2 [.] 5 [+]

A7: 3 5/6 feet |1 1/2 + 2 1/3|
SC-6: 1 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 3 [+]

A8: 2 7/12 feet |5 1/3 - 2 3/4|
SC-6: 5 [.] 1 [.] 3 [ENTER] 2 [.] 3 [.] 4 [-]

A9: 7/8 climbed; 1/8 remains |(1/2 + 1/4) + 1/8|
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 4 [+] [.] 1 [.] 8 [+]

A10: 5 1/4 pints |3 1/2 + 1 3/4|
SC-6: 3 [.] 1 [.] 2 [ENTER] 1 [.] 3 [.] 4 [+]

PRACTICE B (p.16): REAL LIFE (#1-#10) — ExB: 2 11/12 miles (35/12 miles)

B1: 5/8 cup |3/8 + 1/4|
SC-6: [.] 3 [.] 8 [ENTER] [.] 1 [.] 4 [+]

B2: 3/4 complete; 1/4 remains |Simp(18/24)|
SC-6: [.] 18 [.] 24 [g][SIMP]

B3: 13/8 and 1.625 in |Improper(1 5/8)|
SC-6: 1 [.] 5 [.] 8 [ENTER] [f][d/c] [FDISP]

B4: 11/24 foot |5/6 - 3/8|
SC-6: [.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]

B5: 3 5/6 pizzas (23/6) |2 1/3 + 1 1/2|
SC-6: 2 [.] 1 [.] 3 [ENTER] 1 [.] 1 [.] 2 [+]

B6: 2 3/4 feet (11/4) |4 1/2 - 1 3/4|
SC-6: 4 [.] 1 [.] 2 [ENTER] 1 [.] 3 [.] 4 [-]

B7: 3/5 = 0.6 |Simp(15/25)|
SC-6: [.] 15 [.] 25 [g][SIMP] [FDISP]

B8: 1/5 liter (0.2 L) |(2/5 + 3/10) - 1/2|
SC-6: [.] 2 [.] 5 [ENTER] [.] 3 [.] 10 [+] [.] 1 [.] 2 [-]

B9: 4 5/12 yd used; 7/12 yd left |2 2/3 + 1 3/4|
SC-6: 2 [.] 2 [.] 3 [ENTER] 1 [.] 3 [.] 4 [+]

B10: 4 5/8 miles |(3/8 + 2 1/2) + 1 3/4|
SC-6: [.] 3 [.] 8 [ENTER] 2 [.] 1 [.] 2 [+] 1 [.] 3 [.] 4 [+]

SEC 2.2 (pp.17-18): MULTIPLYING FRACTIONS AND MIXED NUMBERS

Methods: W1: Multiply Straight Across | W2: Cross-Cancel Common Factors First | W3: Area Model / Distributive Split

PRACTICE A (p.17): KEY IDEAS (#1-#10) — ExA: 3/8

A1: 1/3 |1/2 × 2/3|
SC-6: [.] 1 [.] 2 [ENTER] [.] 2 [.] 3 [×]

A2: 1/2 |2/3 × 3/4|
SC-6: [.] 2 [.] 3 [ENTER] [.] 3 [.] 4 [×]

A3: 1/3 |3/5 × 5/9|
SC-6: [.] 3 [.] 5 [ENTER] [.] 5 [.] 9 [×]

A4: 6 cups |16 × 3/8|
SC-6: 16 [ENTER] [.] 3 [.] 8 [×]

A5: 4 |1 3/4 × 2 2/7|
SC-6: 1 [.] 3 [.] 4 [ENTER] 2 [.] 2 [.] 7 [×]

A6: 3 |2 1/4 × 1 1/3|
SC-6: 2 [.] 1 [.] 4 [ENTER] 1 [.] 1 [.] 3 [×]

A7: 9 cups |2 1/4 × 4|
SC-6: 2 [.] 1 [.] 4 [ENTER] 4 [×]

A8: 3 1/3 (Less than 5 because 2/3 < 1) |5 × 2/3|
SC-6: 5 [ENTER] [.] 2 [.] 3 [×]

A9: 14 7/8 miles (119/8) |3 1/2 × 4 1/4|
SC-6: 3 [.] 1 [.] 2 [ENTER] 4 [.] 1 [.] 4 [×]

A10: 1/4 of the input speed |(1/2 × 2/3) × 3/4|
SC-6: [.] 1 [.] 2 [ENTER] [.] 2 [.] 3 [×] [.] 3 [.] 4 [×]

PRACTICE B (p.18): REAL LIFE (#1-#10) — ExB: 4 cups

B1: 2/5 of the garden |2/3 × 3/5|
SC-6: [.] 2 [.] 3 [ENTER] [.] 3 [.] 5 [×]

B2: 2/5 of a full battery |4/5 × 1/2|
SC-6: [.] 4 [.] 5 [ENTER] [.] 1 [.] 2 [×]

B3: 4 3/8 cups (35/8) |1 3/4 × 2 1/2|
SC-6: 1 [.] 3 [.] 4 [ENTER] 2 [.] 1 [.] 2 [×]

B4: 3 in² |2 1/4 × 1 1/3|
SC-6: 2 [.] 1 [.] 4 [ENTER] 1 [.] 1 [.] 3 [×]

B5: 4 miles |3 1/5 × 1 1/4|
SC-6: 3 [.] 1 [.] 5 [ENTER] 1 [.] 1 [.] 4 [×]

B6: 2/3 yd² |3/4 × 8/9|
SC-6: [.] 3 [.] 4 [ENTER] [.] 8 [.] 9 [×]

B7: 10 gallons |(24 × 5/8) × 2/3|
SC-6: 24 [ENTER] [.] 5 [.] 8 [×] [.] 2 [.] 3 [×]

B8: 1/8 in² |1/2²|
SC-6: [.] 1 [.] 2 [x²]

B9: 8 ft² |3 1/2 × 2 2/7|
SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 2 [.] 7 [×]

B10: 1/12 of the day (2 hours) |1/3 × 1/4|
SC-6: [.] 1 [.] 3 [ENTER] [.] 1 [.] 4 [×]

SEC 2.3 (pp.19-20): DIVIDING FRACTIONS AND MIXED NUMBERS

Methods: W1: Multiply by the Reciprocal | W2: Common-Denominator Division | W3: Fraction Strip / Number-Line Model

PRACTICE A (p.19): KEY IDEAS (#1-#10) — ExA: 6

A1: 2 |1/2 ÷ 1/4|
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 4 [÷]

A2: 4 |2/3 ÷ 1/6|
SC-6: [.] 2 [.] 3 [ENTER] [.] 1 [.] 6 [÷]

A3: 2 |5/6 × 1/(5/12)|
SC-6: [.] 5 [.] 6 [ENTER] [.] 5 [.] 12 [f][1/x] [×]

A4: 8 |6 ÷ 3/4|
SC-6: 6 [ENTER] [.] 3 [.] 4 [÷]

A5: 1 7/8 (15/8) |3/4 ÷ 2/5|
SC-6: [.] 3 [.] 4 [ENTER] [.] 2 [.] 5 [÷]

A6: 7/8 cup |3 1/2 ÷ 4|
SC-6: 3 [.] 1 [.] 2 [ENTER] 4 [÷]

A7: 2 2/3 bike frames |2/3 ÷ 1/4|
SC-6: [.] 2 [.] 3 [ENTER] [.] 1 [.] 4 [÷]

A8: 3 |5 1/4 ÷ 1 3/4|
SC-6: 5 [.] 1 [.] 4 [ENTER] 1 [.] 3 [.] 4 [÷]

A9: 5 full slats (5 7/37 → 3 1/2 in left) |96 ÷ 18 1/2|
SC-6: 96 [ENTER] 18 [.] 1 [.] 2 [÷]

A10: 6 glasses |3 3/4 ÷ 5/8|
SC-6: 3 [.] 3 [.] 4 [ENTER] [.] 5 [.] 8 [÷]

PRACTICE B (p.20): REAL LIFE (#1-#10) — ExB: 6 shelves

B1: 6 slices |3/4 ÷ 1/8|
SC-6: [.] 3 [.] 4 [ENTER] [.] 1 [.] 8 [÷]

B2: 12 pebbles |4 1/2 ÷ 3/8|
SC-6: 4 [.] 1 [.] 2 [ENTER] [.] 3 [.] 8 [÷]

B3: 6 runners |3 3/4 ÷ 5/8|
SC-6: 3 [.] 3 [.] 4 [ENTER] [.] 5 [.] 8 [÷]

B4: 7 segments |5 1/4 ÷ 3/4|
SC-6: 5 [.] 1 [.] 4 [ENTER] [.] 3 [.] 4 [÷]

B5: 40 layers |6 ÷ 3/20|
SC-6: 6 [ENTER] [.] 3 [.] 20 [÷]

B6: 6 sections |7 1/2 ÷ 1 1/4|
SC-6: 7 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 4 [÷]

B7: 2 meters |4 1/2 ÷ 2 1/4|
SC-6: 4 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [÷]

B8: 8 cups |2 2/3 ÷ 1/3|
SC-6: 2 [.] 2 [.] 3 [ENTER] [.] 1 [.] 3 [÷]

B9: 5/12 (12/5 × 5/12 = 1) |1/(2 2/5)|
SC-6: 2 [.] 2 [.] 5 [f][1/x]

B10: 12 pieces |(10 1/2 - 1 1/2) ÷ 3/4|
SC-6: 10 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 2 [-] [.] 3 [.] 4 [÷]

SEC 2.4 (pp.21-22): ADDING, SUBTRACTING, AND MULTIPLYING DECIMALS

Methods: W1: Vertical Place-Value Algorithm | W2: Convert to Fractions/Hundredths | W3: Place-Value Split / Area Model

PRACTICE A (p.21): KEY IDEAS (#1-#10) — ExA: 18.45

A1: 345.67 mm |(345 + 0.6) + 0.07|
SC-6: 345 [ENTER] 0.6 [+] 0.07 [+]

A2: 17.55 |12.8 + 4.75|
SC-6: 12.8 [ENTER] 4.75 [+]

A3: 11.35 |20 - 8.65|
SC-6: 20 [ENTER] 8.65 [-]

A4: 2.7 |4.5 × 0.6|
SC-6: 4.5 [ENTER] 0.6 [×]

A5: 18.74 |(25.4 - 9.86) + 3.2|
SC-6: 25.4 [ENTER] 9.86 [-] 3.2 [+]

A6: 8.1 (Estimate: 4 × 2 = 8) |3.6 × 2.25|
SC-6: 3.6 [ENTER] 2.25 [×]

A7: 0.06 (3/50) |0.15 × 0.4|
SC-6: 0.15 [ENTER] 0.4 [×] [FDISP]

A8: 35.79 g (35.7864) |28.86 × 1.24|
SC-6: 28.86 [ENTER] 1.24 [×]

A9: $6.65 |20 - (8.75 + 4.60)|
SC-6: 20 [ENTER] 8.75 [ENTER] 4.60 [+] [-]

A10: 161 meters |((30 × 1.20) + (40 × 1.35)) + (50 × 1.42)|
SC-6: 30 [ENTER] 1.20 [×] 40 [ENTER] 1.35 [×] [+] 50 [ENTER] 1.42 [×] [+]

PRACTICE B (p.22): REAL LIFE (#1-#10) — ExB: $8.10

B1: $41.24 |18.75 + 22.49|
SC-6: 18.75 [ENTER] 22.49 [+]

B2: $6.35 |20 - 13.65|
SC-6: 20 [ENTER] 13.65 [-]

B3: 0.55 seconds faster |13.42 - 12.87|
SC-6: 13.42 [ENTER] 12.87 [-]

B4: $7.80 |2.40 × 3.25|
SC-6: 2.40 [ENTER] 3.25 [×]

B5: 1.19 cm² |1.4 × 0.85|
SC-6: 1.4 [ENTER] 0.85 [×]

B6: $41.40 |11.5 × 3.60|
SC-6: 11.5 [ENTER] 3.60 [×]

B7: 30.35 meters |(24.8 - 6.35) + 11.9|
SC-6: 24.8 [ENTER] 6.35 [-] 11.9 [+]

B8: $35.75 |(2 × 14.50) + 6.75|
SC-6: 2 [ENTER] 14.50 [×] 6.75 [+]

B9: 2.5 mm gap |170.5 - (4 × 42)|
SC-6: 170.5 [ENTER] 4 [ENTER] 42 [×] [-]

B10: $35.00 profit |(3.20 - 1.80) × 25|
SC-6: 3.20 [ENTER] 1.80 [-] 25 [×]

SEC 2.5 (pp.23-24): DIVIDING WHOLE NUMBERS AND DECIMALS

Methods: W1: Power-of-10 Decimal Shift & Long Division | W2: Equivalent Fraction Scaling | W3: Reciprocal Fraction Division

PRACTICE A (p.23): KEY IDEAS (#1-#10) — ExA: 5 R 3 = 5 3/4 = 5.75

A1: 4 R 7, 4 7/8, and 4.875 |39 ÷ 8|
SC-6: 39 [ENTER] 8 [÷] [FDISP]

A2: 2.3 |18.4 ÷ 8|
SC-6: 18.4 [ENTER] 8 [÷]

A3: 45 quarters |11.25 ÷ 0.25|
SC-6: 11.25 [ENTER] 0.25 [÷]

A4: 8 |7.2 ÷ 0.9|
SC-6: 7.2 [ENTER] 0.9 [÷]

A5: 130 |15.6 ÷ 0.12|
SC-6: 15.6 [ENTER] 0.12 [÷]

A6: 6.3 (6 3/10) |9.45 ÷ 1.5|
SC-6: 9.45 [ENTER] 1.5 [÷] [FDISP]

A7: $20.80 |124.80 ÷ 6|
SC-6: 124.80 [ENTER] 6 [÷]

A8: 24 (24 × 0.2 = 4.8) |4.8 ÷ 0.2|
SC-6: 4.8 [ENTER] 0.2 [÷]

A9: 6 laps |4.5 ÷ 0.75|
SC-6: 4.5 [ENTER] 0.75 [÷]

A10: $1.25 per bottle; 16 bottles |7.50 ÷ 6; 20 ÷ 1.25|
SC-6: 7.50 [ENTER] 6 [÷] → 1.25; 20 [ENTER] 1.25 [÷]

PRACTICE B (p.24): REAL LIFE (#1-#10) — ExB: 27 quarters

B1: 4 R 3 = 4 3/8 = 35/8 = 4.375 |35 ÷ 8|
SC-6: 35 [ENTER] 8 [÷] [FDISP]

B2: 16 quarters |28 ÷ 1.75|
SC-6: 28 [ENTER] 1.75 [÷]

B3: $8.65 each |34.60 ÷ 4|
SC-6: 34.60 [ENTER] 4 [÷]

B4: 72 layers |14.4 ÷ 0.20|
SC-6: 14.4 [ENTER] 0.20 [÷]

B5: 15 flowers |5.25 ÷ 0.35|
SC-6: 5.25 [ENTER] 0.35 [÷]

B6: 6 miles per hour |4.5 ÷ 0.75|
SC-6: 4.5 [ENTER] 0.75 [÷]

B7: 18 sections |22.5 ÷ 1.25|
SC-6: 22.5 [ENTER] 1.25 [÷]

B8: 30 cups |13.5 ÷ 0.45|
SC-6: 13.5 [ENTER] 0.45 [÷]

B9: 7 tickets |87.50 ÷ 12.50|
SC-6: 87.50 [ENTER] 12.50 [÷]

B10: 6 |10.5 ÷ 1.75|
SC-6: 10.5 [ENTER] 1.75 [÷]

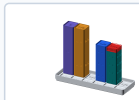
Unit 3: Ratios, Rates & Percents — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapters 3, 4 & 5 (Ratios, Rates & Percents) • Concept → Blocks (When Applicable: Fraction Strips & Place Value Blocks) → Calculator

1. Concept (Stack Approach): Ratios, unit rates, and percents are multiplicative comparisons. Keeping Y and X paired preserves the two quantities being compared across scaling steps.

2. Blocks (Fraction & Place Value): Lay out equal-length Fraction Strips as physical Tape Diagram bars to distinguish Part-to-Part (3:2) from Part-to-Whole (3/5) and model benchmark percents.

3. Calculator (SC-6): Simplify ratios on Y:X with [g][a:b], step through ratio tables with [g][REP], swap upside-down unit rates with [x swap y], and toggle forms with [FDISP].



SEC 3.1 (pp.26-27): RATIOS AND TAPE DIAGRAMS

Methods: W1: Tape Diagram (Equal Boxes) | W2: Equivalent Ratio Fraction | W3: Divide by GCF

PRACTICE A (p.26): KEY IDEAS (#1-#10) — ExA: Part-to-Part = 3:2; Part-to-Whole = 3:5		PRACTICE B (p.27): REAL LIFE (#1-#10) — ExB: 20 wins (and 12 losses)	
A1: 2 : 3 [Ratio(12:18)] SC-6: 12 [ENTER] 18 [g][a:b]	A2: 3 : 2 [Ratio(24:16)] SC-6: 24 [ENTER] 16 [g][a:b]	B1: Part-to-Part = 3:2; Part-to-Whole = 3:5 [Ratio(15:10)] SC-6: 15 [ENTER] 10 [g][a:b]	B2: 16 strawberries, 12 bananas [(28 ÷ 7) × 4] SC-6: 28 [ENTER] 7 [÷] 4 [×]
A3: 2 : 5 [Ratio(4:10)] SC-6: 4 [ENTER] 10 [g][a:b]	A4: 18 and 24 (Total = 42) [4 × 6] SC-6: 3 [ENTER] 6 [×], 4 [ENTER] 6 [×]	B3: 6 cups yellow (21 cups total) [(15 ÷ 5) × 2] SC-6: 15 [ENTER] 5 [÷] 2 [×]	B4: 35 secret and 25 open [(60 ÷ 12) × 7] SC-6: 60 [ENTER] 12 [÷] 7 [×]
A5: 4 L concentrate, 30 L water [34 ÷ 17] SC-6: 34 [ENTER] 17 [÷] → 2/part	A6: 20 dogs [(36 ÷ 9) × 5] SC-6: 36 [ENTER] 9 [÷] 5 [×]	B5: 42 gold coins, 18 silver coins [(24 ÷ 4) × 7] SC-6: 24 [ENTER] 4 [÷] 7 [×]	B6: 3 : 1 [Ratio(48:16)] SC-6: 48 [ENTER] 16 [g][a:b]
A7: 25 seventh graders [(15 ÷ 3) × 5] SC-6: 15 [ENTER] 3 [÷] 5 [×]	A8: 30 electric guitars (12 acoustic) [(18 ÷ 3) × 5] SC-6: 18 [ENTER] 3 [÷] 5 [×]	B7: 4 : 9 (16 out of 36) [Ratio(16:36)] SC-6: 16 [ENTER] 36 [g][a:b]	B8: 16 : 9 [Ratio(1920:1080)] SC-6: 1920 [ENTER] 1080 [g][a:b]
A9: 48 gears and 36 gears [84 ÷ 7] SC-6: 84 [ENTER] 7 [÷] → 12/box	A10: 77 coins total (49 gold, 28 silver) [(21 ÷ 3) × 11] SC-6: 21 [ENTER] 3 [÷] 11 [×]	B9: 150g almonds, 100g cashews, 250g raisins [500 ÷ 10] SC-6: 500 [ENTER] 10 [÷]	B10: 3 : 4 [Ratio(24:32)] SC-6: 24 [ENTER] 32 [g][a:b]

SEC 3.2 (pp.28-29): USING RATIO TABLES AND DOUBLE NUMBER LINES

Methods: W1: Multiplicative Scale Factor | W2: Double Number Line / Repeated Addition | W3: Scale Down Then Up

PRACTICE A (p.28): KEY IDEAS (#1-#10) — ExA: 40 meters		PRACTICE B (p.29): REAL LIFE (#1-#10) — ExB: $75	
A1: 4:9, 8:18, 12:27, 16:36 [Repeat(+4), Repeat(+9)] SC-6: 4 [g][REP] [+], 9 [g][REP] [+]	A2: 63 [(28 ÷ 4) × 9] SC-6: 28 [ENTER] 4 [÷] 9 [×]	B1: 45 grams [15 × 3] SC-6: 15 [ENTER] 3 [×]	B2: 56 scoops [(16 ÷ 2) × 7] SC-6: 16 [ENTER] 2 [÷] 7 [×]
A3: 25 [(60 ÷ 12) × 5] SC-6: 60 [ENTER] 12 [÷] 5 [×]	A4: 60 pages [(8 ÷ 2) × 15] SC-6: 8 [ENTER] 2 [÷] 15 [×]	B3: $15 for 10 notebooks [(12 ÷ 4) × 5] SC-6: 12 [ENTER] 4 [÷] 5 [×]	B4: 112 beats per minute [28 × 4] SC-6: 28 [ENTER] 4 [×]
A5: 48 ounces [(18 ÷ 3) × 8] SC-6: 18 [ENTER] 3 [÷] 8 [×]	A6: $15 [(12 ÷ 4) × 5] SC-6: 12 [ENTER] 4 [÷] 5 [×]	B5: 260 gallons of sap [40 × 6.5] SC-6: 40 [ENTER] 6.5 [×]	B6: 25 pounds [150 ÷ 6] SC-6: 150 [ENTER] 6 [÷]
A7: 84 beats [(60 ÷ 15) × 21] SC-6: 60 [ENTER] 15 [÷] 21 [×]	A8: Ratios scale by multiplication (e.g., ×3 gives 9:15), not addition [Ratio(7:9)] SC-6: 7 [ENTER] 9 [g][a:b]	B7: 15 cups of flour [(6 ÷ 2) × 5] SC-6: 6 [ENTER] 2 [÷] 5 [×]	B8: 75 minutes (1 hr 15 min) [(90 ÷ 18) × 15] SC-6: 90 [ENTER] 18 [÷] 15 [×]
A9: 20 cups [(12 ÷ 3) × 5] SC-6: 12 [ENTER] 3 [÷] 5 [×]	A10: 40 miles [(14 ÷ 7) × 20] SC-6: 14 [ENTER] 7 [÷] 20 [×]	B9: 8 inches [(60 ÷ 15) × 2] SC-6: 60 [ENTER] 15 [÷] 2 [×]	B10: 30 lemons [(9 ÷ 3) × 10] SC-6: 9 [ENTER] 3 [÷] 10 [×]

SEC 3.3 (pp.30-31): RATES, UNIT RATES, AND COMPARING RATES

Methods: W1: Divide to Find the Unit Rate (1 Unit) | W2: Common Multiple Comparison | W3: Double Number Line

PRACTICE A (p.30): KEY IDEAS (#1-#10) — ExA: 63 words per minute		PRACTICE B (p.31): REAL LIFE (#1-#10) — ExB: Pack B ($1.40 per battery vs. $1.45 per battery)	
A1: 65 mph [195 ÷ 3] SC-6: 195 [ENTER] 3 [÷]	A2: $9 per hour [36 ÷ 4] SC-6: 36 [ENTER] 4 [÷]	B1: 150 miles per hour [450 ÷ 3] SC-6: 450 [ENTER] 3 [÷]	B2: Box B ($0.70/bar vs. $0.75/bar) [7.00 ÷ 10] SC-6: 4.50 [ENTER] 6 [÷] 7.00 [ENTER] 10 [÷]
A3: $1.30 per pound [7.80 ÷ 6] SC-6: 7.80 [ENTER] 6 [÷]	A4: 15 grams per hour [90 ÷ 6] SC-6: 90 [ENTER] 6 [÷]	B3: Spool B ($1.95 vs. $2.20 per 100g) [19.50 ÷ 10] SC-6: 11 [ENTER] 5 [÷] 19.50 [ENTER] 10 [÷]	B4: 15 gal/min; 375 gallons [(180 ÷ 12) × 25] SC-6: 180 [ENTER] 12 [÷] 25 [×]
A5: Supplier B ($0.17/oz vs. $0.18/oz) [3.24 ÷ 18; 4.08 ÷ 24] SC-6: 3.24 [ENTER] 18 [÷]; 4.08 [ENTER] 24 [÷]	A6: 10-pack ($0.82/pen vs. $0.85/pen) [5.10 ÷ 6; 8.20 ÷ 10] SC-6: 5.10 [ENTER] 6 [÷]; 8.20 [ENTER] 10 [÷]	B5: B (57 taps/min vs. A's 55 taps/min) [285 ÷ 5] SC-6: 220 [ENTER] 4 [÷] 285 [ENTER] 5 [÷]	B6: Car A (28 mpg vs. 27 mpg) [405 ÷ 15] SC-6: 336 [ENTER] 12 [÷] 405 [ENTER] 15 [÷]
A7: 300 miles [(210 ÷ 3.5) × 5] SC-6: 210 [ENTER] 3.5 [÷] 5 [×]	A8: Pump A (9 gal/min vs. 8 gal/min) [45 ÷ 5; 64 ÷ 8] SC-6: 45 [ENTER] 5 [÷]; 64 [ENTER] 8 [÷]	B7: $9 per dog → 11 dogs [99 ÷ (54 ÷ 6)] SC-6: 54 [ENTER] 6 [÷] 99 [x swap y] [÷]	B8: 12-pack ($0.42 vs. $0.45 each) [5.04 ÷ 12] SC-6: 3.60 [ENTER] 8 [÷] 5.04 [ENTER] 12 [÷]
A9: 180 mm [(45 ÷ 15) × 60] SC-6: 45 [ENTER] 15 [÷] 60 [×]	A10: Car B by 1 mpg (29 mpg vs. 28 mpg) [168 ÷ 6; 232 ÷ 8] SC-6: 168 [ENTER] 6 [÷]; 232 [ENTER] 8 [÷]	B9: 4 inches per hour [14 ÷ 3.5] SC-6: 14 [ENTER] 3.5 [÷]	B10: 4.5 m/s → 90 meters [(27 ÷ 6) × 20] SC-6: 27 [ENTER] 6 [÷] 20 [×]

SEC 3.4 (pp.32-33): PERCENTS, FRACTIONS, AND DECIMALS

Methods: W1: Equivalent Fraction over 100 | W2: Divide Numerator ÷ Denominator | W3: Percent Bar / Benchmark Model

PRACTICE A (p.32): KEY IDEAS (#1-#10) — ExA: 0.35 and 35%		PRACTICE B (p.33): REAL LIFE (#1-#10) — ExB: 75% (3/4 = 0.75)	
A1: 0.6 and 60% [3/5 → Dec/Frac] SC-6: [.] 3 [.] 5 [ENTER] [FDISP]	A2: 0.36 and 36% [9/25 → Dec/Frac] SC-6: [.] 9 [.] 25 [ENTER] [FDISP]	B1: 0.85 = 85% [17/20 × 100] SC-6: [.] 17 [.] 20 [FDISP] 100 [×]	B2: 0.36 = 36% [9/25 × 100] SC-6: [.] 9 [.] 25 [FDISP] 100 [×]
A3: 0.45 and 9/20 [45 ÷ 100] SC-6: 45 [ENTER] 100 [÷] [FDISP]	A4: 12.5% and 1/8 [0.125 → Dec/Frac] SC-6: 0.125 [ENTER] [FDISP]	B3: 20 + 40 = 60 points [40% of 50 + 80% of 50] SC-6: 50 [ENTER] 40 [%] 50 [ENTER] 80 [%] [+]	B4: 104 students [65% of 160] SC-6: 160 [ENTER] 65 [%]
A5: 270 seats [60% of 450] SC-6: 450 [ENTER] 60 [%]	A6: 3,510 grams [65% of 5400] SC-6: 5400 [ENTER] 65 [%]	B5: 0.15 and 3/20 [Simp(15/100)] SC-6: [.] 15 [.] 100 [g][SIMP]	B6: 45 students total [18 ÷ 0.40] SC-6: 18 [ENTER] 0.40 [÷]
A7: 84% [42 ÷ 50] SC-6: 42 [ENTER] 50 [÷]	A8: 1.5 and 1 1/2 [150 ÷ 100] SC-6: 150 [ENTER] 100 [÷] [FDISP]	B7: $9.00 [15% of 60] SC-6: 60 [ENTER] 15 [%]	B8: 7/8 = 0.875 = 87.5% [21 ÷ 24] SC-6: 21 [ENTER] 24 [÷] [FDISP]
A9: 60%, 0.61, 5/8 [5/8 → Dec/Frac] SC-6: [.] 5 [.] 8 [FDISP] → 0.625	A10: 1,400 mAh [35% of 4000] SC-6: 4000 [ENTER] 35 [%]	B9: $480 goal [360 ÷ 0.75] SC-6: 360 [ENTER] 0.75 [÷]	B10: 58% < 3/5 (60%) < 0.62 (62%) [3/5 → Dec/Frac] SC-6: [.] 3 [.] 5 [FDISP]

SEC 3.5 (pp.34-35): SOLVING PERCENT PROBLEMS, DISCOUNTS & CONVERSIONS

Methods: W1: Find Discount & Subtract | W2: Multiply by Remaining % | W3: Percent Proportion / Ratio Bar

PRACTICE A (p.34): KEY IDEAS (#1-#10) — ExA: Discount = $15; Sale Price = $45		PRACTICE B (p.35): REAL LIFE (#1-#10) — ExB: 30% increase; 81.28 cm	
A1: Discount $17; Sale price $68 [20% of 85] SC-6: 85 [ENTER] 20 [%] [-]	A2: Tax $10.20; Total $130.20 [8.5% of 120] SC-6: 120 [ENTER] 8.5 [%] [+]	B1: Discount = $15; Sale Price = $60 [20% of 75] SC-6: 75 [ENTER] 20 [%] [-]	B2: Tip = $9.60; Total = $73.60 [15% of 64] SC-6: 64 [ENTER] 15 [%] [+]
A3: +400% ((75-15)÷15 = 400%) [%Chg(15→75)] SC-6: 15 [ENTER] 75 [g][%CHG]	A4: 25% decrease [%Chg(80→60)] SC-6: 80 [ENTER] 60 [g][%CHG]	B3: Tax = $3.60; Total = $51.60 [7.5% of 48] SC-6: 48 [ENTER] 7.5 [%] [+]	B4: 30% increase [%Chg(40→52)] SC-6: 40 [ENTER] 52 [g][%CHG]
A5: 25.4 cm [10 (to cm)] SC-6: 10 [f][→cm]	A6: 20 in [50.8 (to in)] SC-6: 50.8 [g][→in]	B5: 30% decrease [%Chg(90→63)] SC-6: 90 [ENTER] 63 [g][%CHG]	B6: 16.09 km [10 × 1.609344] SC-6: 10 [ENTER] 1.609344 [×]
A7: $39.20 [30% of 56] SC-6: 56 [ENTER] 30 [%] [-]	A8: 72 [18 ÷ 0.25] SC-6: 18 [ENTER] 0.25 [÷]	B7: 44.09 lb [20 (to lb)] SC-6: 20 [g][→lb]	B8: 3.79 liters; 128 fl oz [1 (to L)] SC-6: 1 [f][→L]
A9: $144 ($200 → $160 → $144) [10% of 20% of 200] SC-6: 200 [ENTER] 20 [%] [-] 10 [%] [-]	A10: 81.28 cm and 2.27 kg [32 (to cm); 5 (to kg)] SC-6: 32 [f][→cm]; 5 [f][→kg]	B9: $72 (10% is taken off $80, not $100) [10% of 20% of 100] SC-6: 100 [ENTER] 20 [%] [-] 10 [%] [-]	B10: 68°F [20 (to °F)] SC-6: 20 [g][→°F]

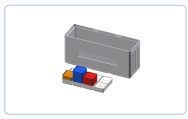
Unit 4: Algebraic Expressions & Properties — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 3 / Chapter 5 (Algebraic Expressions) • Concept → Blocks (When Applicable: Place Value & Prime Factor Blocks) → Calculator

1. **Concept (Stack Approach): Algebraic expressions are recipes of operations waiting for an input value. Evaluating an expression means pushing the substituted value onto the stack and executing inner terms first.**

2. Blocks (Place Value & Prime): Treat 1, x, and x^2 as distinct Place Value columns so unlike terms are never combined, and use Prime Factor blocks to factor out the GCF.

3. **Calculator (SC-6): Substitute a test value (e.g., x = 5) into both original and simplified expressions on Y and X, then press [!=] (flashes NO when equivalent).**



SEC 4.1 (pp.37-38): WRITING AND EVALUATING ALGEBRAIC EXPRESSIONS

Methods: W1: Substitute in Parentheses & Follow PEMDAS | W2: Equal-Groups Bar / Place Value Model

PRACTICE A (p.37): KEY IDEAS (#1-#10) — ExA: 27		PRACTICE B (p.38): REAL LIFE (#1-#10) — ExB: 99 points	
A1: 33 [(6 × 7) - 9] SC-6: 6 [ENTER] 7 [×] 9 [-]	A2: 54 [(5 × 8) + 14] SC-6: 5 [ENTER] 8 [×] 14 [+]	B1: 5g + 6; $26 [(5 × 4) + 6] SC-6: 5 [ENTER] 4 [×] 6 [+]	B2: 50 - 4r; 14 tickets [50 - (4 × 9)] SC-6: 50 [ENTER] 4 [ENTER] 9 [×] [-]
A3: 13 [40 - (3 × 9)] SC-6: 40 [ENTER] 3 [ENTER] 9 [×] [-]	A4: 47 [6^2 + 11] SC-6: 6 [x²] 11 [+]	B3: $225 [($5^2$ × 8) + 25] SC-6: 5 [x²] 8 [×] 25 [+]	B4: 82 points [(10 × 7) + (3 × 4)] SC-6: 10 [ENTER] 7 [×] 3 [ENTER] 4 [×] [+]
A5: 27 [(4^2 × 2) - 5] SC-6: 4 [x²] 2 [×] 5 [-]	A6: 55 [(7 × 4) + (3 × 9)] SC-6: 7 [ENTER] 4 [×] 3 [ENTER] 9 [×] [+]	B5: 144 feet [3^2 × 16] SC-6: 3 [x²] 16 [×]	B6: 96 in² [4^2 × 6] SC-6: 4 [x²] 6 [×]
A7: 8 [(19 + 13) ÷ 4] SC-6: 19 [ENTER] 13 [+] 4 [÷]	A8: 15.1 [(5 × 3.5) - 2.4] SC-6: 5 [ENTER] 3.5 [×] 2.4 [-]	B7: 7p + 4d; $41 [(7 × 3) + (4 × 5)] SC-6: 7 [ENTER] 3 [×] 4 [ENTER] 5 [×] [+]	B8: 78 [A = ½((11 + 15) × 6)] SC-6: 11 [ENTER] 15 [+] 6 [g][TRI]
A9: 8 + 4.50h; $26 [(4.50 × 4) + 8] SC-6: 4.50 [ENTER] 4 [×] 8 [+]	A10: 150 cm² [5^2 × 6] SC-6: 5 [x²] 6 [×]	B9: 111 minutes (1 hr 51 min) [(12 × 8) + 15] SC-6: 12 [ENTER] 8 [×] 15 [+]	B10: 128 - 75 = 53 [(4^3 × 2) - (5^2 × 3)] SC-6: 4 [x²] 2 [×] 5 [x²] 3 [×] [-]

SEC 4.2 (pp.39-40): SIMPLIFYING EXPRESSIONS BY COMBINING LIKE TERMS

Methods: W1: Reorder & Add Coefficients (Properties) | W2: Sort by Variable Power Columns

PRACTICE A (p.39): KEY IDEAS (#1-#10) — ExA: 6x + 8		PRACTICE B (p.40): REAL LIFE (#1-#10) — ExB: 5x² + 4x + 6	
A1: 9x + 13 [7 + 2; 4 + 9] SC-6: 7 [ENTER] 2 [+]; 4 [ENTER] 9 [+]	A2: 5x + 7 [8 - 3; 11 - 4] SC-6: 8 [ENTER] 3 [-]; 11 [ENTER] 4 [-]	B1: 9x + 9 [(3 + 1) + 5] SC-6: 2 [ENTER] 3 [+] 4 [+] 3 [ENTER] 1 [+] 5 [+]	B2: 8x + 12 [(4 + 2) × 2] SC-6: 3 [ENTER] 1 [+] 2 [×] 4 [ENTER] 2 [+] 2 [×]
A3: 8y + 6 [9 - 1] SC-6: 9 [ENTER] 1 [-]	A4: 8a + 13 [12 - 4; 5 + 8] SC-6: 12 [ENTER] 4 [-]; 5 [ENTER] 8 [+]	B3: 4x + 14 [9 + 5] SC-6: 6 [ENTER] 2 [-] 9 [ENTER] 5 [+]	B4: 6x² + 4x + 11 [3 + 8] SC-6: 4 [ENTER] 2 [+] 7 [ENTER] 3 [-] 3 [ENTER] 8 [+]
A5: 6x² + 8x + 7 [4+2=6; 3+5=8] SC-6: Mental / Table Check	A6: 4n² + 5n + 10 [6-2=4; 9-4=5] SC-6: Mental / Table Check	B5: 13c + 31 [12 + 19] SC-6: 8 [ENTER] 5 [+] 12 [ENTER] 19 [+]	B6: 7x + 19 (coeff = 7, constant = 19) [14 + 5] SC-6: 9 [ENTER] 2 [-] 14 [ENTER] 5 [+]
A7: 5.3x + 2.7 [3.5 + 1.8; 4.2 - 1.5] SC-6: 3.5 [ENTER] 1.8 [+]; 4.2 [ENTER] 1.5 [-]	A8: 53 != 80; 5x + 3 is already in simplest form [5(10)+3=53 vs 8(10)=80] SC-6: Mental / Table Check	B7: (1/2)n + 13 [4 + 9] SC-6: [.] 3 [.] 8 [ENTER] [.] 1 [.] 8 [+] 4 [ENTER] 9 [+]	B8: 4y + 3.5 [5.8 - 2.3] SC-6: 2.4 [ENTER] 1.6 [+] 5.8 [ENTER] 2.3 [-]
A9: 6x + 16; 76 meters [2+2+1+1=6x; 3+3+5+5=16] SC-6: Mental / Table Check	A10: 9x + 8; 53 cm [3+2+4=9x; 2+7-1=8] SC-6: Mental / Table Check	B9: 6m + 9 (both equal 39 at m=5) [((10 - 4) × 5) + (15 - 6)] SC-6: 10 [ENTER] 4 [-] 5 [×] 15 [ENTER] 6 [-] [+]	B10: 3x² + 7 [10 - 3] SC-6: 5 [ENTER] 2 [-] 10 [ENTER] 3 [-]

SEC 4.3 (pp.41-42): THE DISTRIBUTIVE PROPERTY

Methods: W1: Multiply Each Term Inside a(b + c) = ab + ac | W2: Split-Rectangle Area Model | W3: Repeated Addition of Rows

PRACTICE A (p.41): KEY IDEAS (#1-#10) — ExA: 84		PRACTICE B (p.42): REAL LIFE (#1-#10) — ExB: 14x + 15	
A1: 140 + 21 = 161 [(7 × 20) + (7 × 3)] SC-6: 7 [ENTER] 20 [×] 7 [ENTER] 3 [×] [+]	A2: 4x + 20 [4 × 5] SC-6: 4 [ENTER] 5 [×]	B1: 8(50 + 3) = 400 + 24 = $424 [(8 × 50) + (8 × 3)] SC-6: 8 [ENTER] 50 [×] 8 [ENTER] 3 [×] [+]	B2: 6(10 - 0.10) = $60.00 - $0.60 = $59.40 [(6 × 10) - (6 × 0.10)] SC-6: 6 [ENTER] 10 [×] 6 [ENTER] 0.10 [×] [-]
A3: 18x + 24 [6×3=18; 6×4=24] SC-6: Mental / Table Check	A4: 16y - 40 [8×2=16; 8×5=40] SC-6: Mental / Table Check	B3: 7(x + 8) = 7x + 56 [7 × 8] SC-6: 7 [ENTER] 8 [×]	B4: 9(2x + 7) = 18x + 63 [9 × 7] SC-6: 9 [ENTER] 2 [×] 9 [ENTER] 7 [×]
A5: 387 [(9 × 40) + (9 × 3)] SC-6: 9 [ENTER] 40 [×] 9 [ENTER] 3 [×] [+]	A6: 17x + 6 [3×4+5=17; 3×2=6] SC-6: Mental / Table Check	B5: 4(t - 3) = 4t - 12 ($48 at t = $15) [(4 × 15) - (4 × 3)] SC-6: 4 [ENTER] 15 [×] 4 [ENTER] 3 [×] [-]	B6: 14x + 15 [3 × 5] SC-6: 3 [ENTER] 4 [×] 2 [+] 3 [ENTER] 5 [×]
A7: 12n + 13 [4×3=12; 20-7=13] SC-6: Mental / Table Check	A8: 5x + 30 (She forgot to multiply 6 by 5) [5 × 6] SC-6: 5 [ENTER] 6 [×]	B7: 13x + 22 [(4 × 3) + (5 × 2)] SC-6: 4 [ENTER] 2 [×] 5 [+] 4 [ENTER] 3 [×] 5 [ENTER] 2 [×] [+]	B8: 3x + 5 [10 ÷ 2] SC-6: 6 [ENTER] 2 [÷] 10 [ENTER] 2 [÷]
A9: 14x + 15 (Both equal 43 when x = 2) [5×2+4=14; 5×3=15] SC-6: Mental / Table Check	A10: 6(x + 3.50) = 6x + 21; $60 total [6 × 3.50] SC-6: 6 [ENTER] 3.50 [×] → 6x+21	B9: 2(2x + 5) + 2(x + 3) = 6x + 16 [(2 × 5) + (2 × 3)] SC-6: 2 [ENTER] 2 [×] 2 [+] 2 [ENTER] 5 [×] 2 [ENTER] 3 [×] [+]	B10: 12(15) = 180 and 120 + 60 = 180 [12 × 15] SC-6: 12 [ENTER] 15 [×]

SEC 4.4 (pp.43-44): FACTORING EXPRESSIONS USING THE GCF

Methods: W1: Factor Out the GCF Using Prime Factors | W2: Equal-Rows Area Model / Division Ladder

PRACTICE A (p.43): KEY IDEAS (#1-#10) — ExA: 6(3x + 4)		PRACTICE B (p.44): REAL LIFE (#1-#10) — ExB: 15(3x - 2)	
A1: 4(3 + 5) [GCF(12, 20)] SC-6: 12 [ENTER] 20 [GCF] → 4	A2: 4(2x + 3) [GCF(8, 12)] SC-6: 8 [ENTER] 12 [GCF] → 4	B1: 8(2x + 3) → 8 bags [GCF(16, 24)] SC-6: 16 [ENTER] 24 [GCF]	B2: 7(3x + 5) → length = 3x + 5 m [GCF(21, 35)] SC-6: 21 [ENTER] 35 [GCF]
A3: 5(3y - 5) [GCF(15, 25)] SC-6: 15 [ENTER] 25 [GCF] → 5	A4: 12(2x + 3) [GCF(24, 36)] SC-6: 24 [ENTER] 36 [GCF] → 12	B3: 4(5x + 9) → side = 5x + 9 in [GCF(20, 36)] SC-6: 20 [ENTER] 36 [GCF]	B4: 3(6y - 9) → side = 6y - 9 cm [GCF(18, 27)] SC-6: 18 [ENTER] 27 [GCF]
A5: 16(2n - 3) [GCF(32, 48)] SC-6: 32 [ENTER] 48 [GCF] → 16	A6: 14(3a + 1) [GCF(42, 14)] SC-6: 42 [ENTER] 14 [GCF] → 14	B5: 8(5x + 8) [GCF(40, 64)] SC-6: 40 [ENTER] 64 [GCF]	B6: 18(2s + 3) → 18 player kits [GCF(36, 54)] SC-6: 36 [ENTER] 54 [GCF]
A7: 12(3m - 5) [GCF(36, 60)] SC-6: 36 [ENTER] 60 [GCF] → 12	A8: 12(2x + 3) (4 and 6 still share a factor of 2) [GCF(4, 6)] SC-6: 4 [ENTER] 6 [GCF] → 2 left!	B7: 6(2x + 3y + 5) [GCF(GCF(12, 18), 30)] SC-6: 12 [ENTER] 18 [GCF] 30 [GCF]	B8: 7(5d - 2) → 5d - 2 per row [GCF(35, 14)] SC-6: 35 [ENTER] 14 [GCF]
A9: Height = 14 in.; Width = (2x + 3) in. [GCF(28, 42)] SC-6: 28 [ENTER] 42 [GCF] → 14	A10: (9x + 12) feet [36 ÷ 4; 48 ÷ 4] SC-6: 36 [ENTER] 4 [÷]; 48 [ENTER] 4 [÷]	B9: 24x + 36 = 12(2x + 3) [GCF((10 + 14), (18 + 18))] SC-6: 10 [ENTER] 14 [+] 18 [ENTER] 18 [+] [GCF]	B10: 9(3x - 2); at x=2 both equal 36 [GCF(27, 18)] SC-6: 27 [ENTER] 18 [GCF]

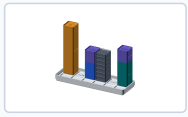
Unit 5: Areas of Polygons & Elapsed Time — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 4 (Areas of Polygons) • Concept → Blocks (When Applicable: Fraction Strips & Place Value Blocks) → Calculator

1. Concept (Stack Approach): Every polygon area formula decomposes into base-times-height products held on the stack, while elapsed time regroups in base-60 rather than base-10.

2. Blocks (When Applicable — Fraction & Place Value): Use Fraction Strips for fractional edge lengths and Place Value columns to contrast base-10 regrouping with 60-minute hours.

3. Calculator (SC-6): Compute rectangle |f|[RECT] and triangle |g|[TRI] areas, accumulate composite sub-areas with |Σ+| and |f|[Σx], and convert H.MMSS time via |f|[→HR] and |g|[→HMS].



SEC 5.1 (pp.46-47): AREAS OF PARALLELOGRAMS AND RECTANGLES

Methods: W1: Area Formula A = bh | W2: Cut-and-Slide Rectangle Decomposition

PRACTICE A (p.46): KEY IDEAS (#1-#10)

ExA: 96 cm'

A1: 135 m² |A = 15 × 9|
SC-6: 15 [ENTER] 9 [f][RECT]

A2: 36 m² |A = 8 × 4.5|
SC-6: 8 [ENTER] 4.5 [f][RECT]

A3: 54.4 m² |A = 8.5 × 6.4|
SC-6: 8.5 [ENTER] 6.4 [f][RECT]

A4: 169 cm² |13²|
SC-6: 13 [x²]

A5: 26 in.² |A = 6 1/2 × 4|
SC-6: 6 [.] 1 [.] 2 [ENTER] 4 [f][RECT]

A6: 9 cm |108 ÷ 12|
SC-6: 108 [ENTER] 12 [÷]

A7: 18 ft |135 ÷ 7.5|
SC-6: 135 [ENTER] 7.5 [÷]

A8: 60 cm² (He multiplied by the slanted side 7.5) |A = 10 × 6|
SC-6: 10 [ENTER] 6 [f][RECT]

A9: 666 ft² |(9×18.5) × 4|
SC-6: 9 [ENTER] 18.5 [f][RECT] 4 [×]

A10: 168 ft² |(15×12) - (4×3)|
SC-6: 15 [ENTER] 12 [f][RECT] 4 [ENTER] 3 [f][RECT] [-]

PRACTICE B (p.47): REAL LIFE (#1-#10)

ExB: 12 inches

B1: 180 ft² |A = 10 × 18|
SC-6: 10 [ENTER] 18 [f][RECT]

B2: 60 ft |(10 + 20) × 2|
SC-6: 10 [ENTER] 20 [+] 2 [×]

B3: 14.4 ft² |A = 4.5 × 3.2|
SC-6: 4.5 [ENTER] 3.2 [f][RECT]

B4: 15 inches |135 ÷ 9|
SC-6: 135 [ENTER] 9 [÷]

B5: 216 cm² |(6×4.5) × 8|
SC-6: 6 [ENTER] 4.5 [f][RECT] 8 [×]

B6: 228 ft² |A = 19 × 12|
SC-6: 19 [ENTER] 12 [f][RECT]

B7: 2,520 ft² |(65×40) - (10×8)|
SC-6: 65 [ENTER] 40 [f][RECT] 10 [ENTER] 8 [f][RECT] [-]

B8: 22 in² |5 1/2 × 4|
SC-6: 5 [.] 1 [.] 2 [ENTER] 4 [×]

B9: Both equal 144 cm² (same base and perpendicular height) |A = 16 × 9|
SC-6: 16 [ENTER] 9 [f][RECT]

B10: 28 ft² × $4 = $112.00 |(8×3.5) × 4|
SC-6: 8 [ENTER] 3.5 [f][RECT] 4 [×]

SEC 5.2 (pp.48-49): AREAS OF TRIANGLES

Methods: W1: Formula A = (b × h) ÷ 2 | W2: Take Half of Base or Height First (½b × h)

PRACTICE A (p.48): KEY IDEAS (#1-#10)

ExA: 30 cm'

A1: 750 cm² |A = ½(25 × 60)|
SC-6: 25 [ENTER] 60 [g][TRI]

A2: 18 ft² |A = ½(9 × 4)|
SC-6: 9 [ENTER] 4 [g][TRI]

A3: 88 cm² |A = ½(16 × 11)|
SC-6: 16 [ENTER] 11 [g][TRI]

A4: 52.5 m² |A = ½(12.5 × 8.4)|
SC-6: 12.5 [ENTER] 8.4 [g][TRI]

A5: 13 1/2 in.² (13.5) |A = ½(4 1/2 × 6)|
SC-6: 4 [.] 1 [.] 2 [ENTER] 6 [g][TRI]

A6: 9 cm |(54 × 2) ÷ 12|
SC-6: 54 [ENTER] 2 [×] 12 [÷]

A7: 12 m |(84 × 2) ÷ 14|
SC-6: 84 [ENTER] 2 [×] 14 [÷]

A8: 20 cm² (He forgot to divide by 2) |A = ½(8 × 5)|
SC-6: 8 [ENTER] 5 [g][TRI]

A9: 210 ft² |½(12×25) + ½(8×15)|
SC-6: 12 [ENTER] 25 [g][TRI] 8 [ENTER] 15 [g][TRI] [+]

A10: 1,400 cm² |A = ½(40 × 70)|
SC-6: 40 [ENTER] 70 [g][TRI]

PRACTICE B (p.49): REAL LIFE (#1-#10)

ExB: 8 inches

B1: 150 ft² |A = ½(12 × 25)|
SC-6: 12 [ENTER] 25 [g][TRI]

B2: 14 ft² (1 side); 28 ft² (both) |A = ½(8 × 3.5)|
SC-6: 8 [ENTER] 3.5 [g][TRI]

B3: 141 ft² |A = ½(30 × 9.4)|
SC-6: 30 [ENTER] 9.4 [g][TRI]

B4: 36 inches |(450 × 2) ÷ 25|
SC-6: 450 [ENTER] 2 [×] 25 [÷]

B5: 6 feet |(18 × 2) ÷ 6|
SC-6: 18 [ENTER] 2 [×] 6 [÷]

B6: 10.125 in² (10 1/8 in²) |A = ½(4.5 × 4.5)|
SC-6: 4.5 [ENTER] 4.5 [g][TRI]

B7: 54 ft² |A = ½(9 × 12)|
SC-6: 9 [ENTER] 12 [g][TRI]

B8: 630 cm² (315 cm² per wing) |½(18×35) × 2|
SC-6: 18 [ENTER] 35 [g][TRI] 2 [×]

B9: 13 in² |(6 1/2 × 4) ÷ 2|
SC-6: 6 [.] 1 [.] 2 [ENTER] 4 [×] 2 [÷]

B10: 64 ft² |½(5×6.4) × 4|
SC-6: 5 [ENTER] 6.4 [g][TRI] 4 [×]

SEC 5.3 (pp.50-51): AREAS OF TRAPEZOIDS AND COMPOSITE FIGURES

Methods: W1: Decompose & Add Sub-Areas | W2: Trapezoid Formula A = ½h(b1 + b2) | W3: Enclosing Box Minus Cutouts

PRACTICE A (p.50): KEY IDEAS (#1-#10)

ExA: 126 ft'

A1: 72 cm² |A = ½((10 + 14) × 6)|
SC-6: 10 [ENTER] 14 [+] 6 [g][TRI]

A2: 135 ft² |A = ½((18 + 12) × 9)|
SC-6: 18 [ENTER] 12 [+] 9 [g][TRI]

A3: 92 ft² |Stats(10, 6, 8, 4)|
SC-6: 10 [ENTER] 6 |f|[RECT] |Σ+| 8 [ENTER] 4 |f|[RECT] |Σ+| |f|[Σx]

A4: 120 in.² |Stats(15, 6, 10, 6)|
SC-6: 15 [ENTER] 6 |f|[RECT] |Σ+| 10 [ENTER] 6 |g|[TRI] |Σ+| |f|[Σx]

A5: 29.4 m² |A = ½((8.5 + 5.5) × 4.2)|
SC-6: 8.5 [ENTER] 5.5 [+] 4.2 [g][TRI]

A6: 56 in.² |Stats(10, 8, 6, 4)|
SC-6: 10 [ENTER] 8 |f|[RECT] |Σ+| 6 [ENTER] 4 |f|[RECT] |f|[Σ-] |f|[Σx]

A7: 10 cm |(90 × 2) ÷ (11 + 7)|
SC-6: 90 [ENTER] 2 [×] 11 [ENTER] 7 [+] [÷]

A8: 88 ft² |Stats(60, 5, 9, 4)|
SC-6: 60 |Σ+| 5 |ENTER| 9 |+| 4 |g|[TRI] |Σ+| |f|[Σx]

A9: 104 cm² (80 + 24 = 104) |A = ½((10 + 16) × 8)|
SC-6: 10 [ENTER] 16 [+] 8 [g][TRI]

A10: 251 ft² |Stats(300, 24, 25)|
SC-6: 300 |Σ+| 24 |f|[Σ-] 25 |f|[Σ-] |f|[Σx]

PRACTICE B (p.51): REAL LIFE (#1-#10)

ExB: 1,008 square inches

B1: 440 in² |A = ½((16 + 28) × 20)|
SC-6: 16 [ENTER] 28 [+] 20 [g][TRI]

B2: 40 in² per side; 160 in² total |A = ½((4 + 6) × 8)|
SC-6: 4 [ENTER] 6 [+] 8 [g][TRI]

B3: 84 + 50 = 134 ft² |(14×6) + (10×5)|
SC-6: 14 [ENTER] 6 [f][RECT] 10 [ENTER] 5 [f][RECT] [+]

B4: 42 cm² (1 fin); 126 cm² (3 fins) |A = ½((5 + 9) × 6)|
SC-6: 5 [ENTER] 9 [+] 6 [g][TRI]

B5: 90 + 48 = 138 in² |(15×6) + ½(12×8)|
SC-6: 15 [ENTER] 6 [f][RECT] 12 [ENTER] 8 [g][TRI] [+]

B6: 1,152 - 36 = 1,116 in² |(48×24) - 6²|
SC-6: 48 [ENTER] 24 [f][RECT] 6 [x²] [-]

B7: h = 8 ft |104 ÷ 13|
SC-6: 104 [ENTER] 13 [÷]

B8: 280 + 60 = 340 ft² |(20×14) + ½(20×6)|
SC-6: 20 [ENTER] 14 [f][RECT] 20 [ENTER] 6 [g][TRI] [+]

B9: 416 - 200 = 216 m² |(26×16) - (20×10)|
SC-6: 26 [ENTER] 16 [f][RECT] 20 [ENTER] 10 [f][RECT] [-]

B10: 42 cm² |½((4 + 8)×3.5) × 2|
SC-6: 4 [ENTER] 8 [+] 3.5 [g][TRI] 2 [×]

SEC 5.4 (pp.52-53): POLYGONS IN THE COORDINATE PLANE & ELAPSED TIME

Methods: W1: Subtract Coordinates / Regroup 60 min = 1 hr | W2: Count Grid Squares / Time-Line Jumps

PRACTICE A (p.52): KEY IDEAS (#1-#10)

ExA: Width = 7, Height = 5, Perimeter = 24, Area = 35

A1: 8 units |11 - 3|
SC-6: 11 [ENTER] 3 [-]

A2: 54 square units |A = (10 - 1) × (8 - 2)|
SC-6: 10 [ENTER] 1 [-] 8 [ENTER] 2 [-] [f][RECT]

A3: 10:58 AM |7.10 + 3.48|
SC-6: [CLK] 7.10 [ENTER] 3.48 [+]

A4: 2.5 hours; 4.75 hours |2.30 (to HR); 4.45 (to HR)|
SC-6: 2.30 [f][→HR]; 4.45 [f][→HR]

A5: 6 hours 50 minutes |15.05 - 8.15|
SC-6: [CLK] 15.05 [ENTER] 8.15 [-]

A6: 1 hour 45 minutes |1.75 (to HMS)|
SC-6: 1.75 [g][→HMS]

A7: 2 hours 35 minutes |11.25 - 8.50|
SC-6: [CLK] 11.25 [ENTER] 8.50 [-]

A8: 24 square units |A = ½(8 × 6)|
SC-6: 8 [ENTER] 6 [g][TRI]

A9: 5.5 hours (5 hr 30 min); $101.75 |5.5 × 18.50|
SC-6: 5.5 [ENTER] 18.50 [×]

A10: Perimeter = 420 yd; Area = 10,800 yd² |A = 12 × 9|
SC-6: 12 [ENTER] 9 [f][RECT]

PRACTICE B (p.53): REAL LIFE (#1-#10)

ExB: 12:30 PM

B1: W=10, H=6, P=32, Area=60 |A = (11 - 1) × (8 - 2)|
SC-6: 11 [ENTER] 1 [-] 8 [ENTER] 2 [-] [f][RECT]

B2: Base=8, Height=6, Area=24 |A = ½((10 - 2) × (7 - 1))|
SC-6: 10 [ENTER] 2 [-] 7 [ENTER] 1 [-] [g][TRI]

B3: 12:15 PM |8.35 + 3.40|
SC-6: [CLK] 8.35 [ENTER] 3.40 [+]

B4: 3 hours 35 minutes |14.20 - 10.45|
SC-6: [CLK] 14.20 [ENTER] 10.45 [-]

B5: 3 hours 50 minutes |1.52 + 1.58|
SC-6: [CLK] 1.52 [ENTER] 1.58 [+]

B6: 3.5 hr × $14 = $49.00 |3.5 × 14|
SC-6: 3.5 [ENTER] 14 [×]

B7: 2 hours 45 minutes |2.75 (to HMS)|
SC-6: 2.75 [g][→HMS]

B8: 4 hours 30 minutes |(1.25 + 2.15) + 0.50|
SC-6: [CLK] 1.25 [ENTER] 2.15 [+] 0.50 [+]

B9: (3, 10); Area = 36 |A = (9 - 3) × (10 - 4)|
SC-6: 9 [ENTER] 3 [-] 10 [ENTER] 4 [-] [f][RECT]

B10: 180 miles |2.25 × 80|
SC-6: 2.25 [ENTER] 80 [×]

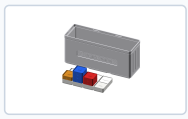
Unit 6: Integers, Rational Numbers & The Coordinate Plane — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 6 • Concept → Blocks (When Applicable: Place Value Blocks & Fraction Strips) → Calculator

1. Concept (Stack Approach): A signed number combines direction (+/-) and magnitude (|x|). In postfix entry, typing the magnitude first and pressing |+/-| attaches the sign to the number in X.

2. Blocks (Place Value & Fraction): Pair positive and negative Place Value blocks to form Zero Pairs, and compare rational magnitudes with 84 mm Fraction Strips.

3. Calculator (SC-6): Distinguish |+/-| (change sign of X) from |-| (subtract X from Y), animate signed hops on |f|[LINE], find distance with |g||x||, and test order with |f|[>]/|g||<|.



SEC 6.1 (pp.55-56): POSITIVE AND NEGATIVE INTEGERS & OPPOSITES

Methods: W1: Equal Distance on Opposite Sides of 0 (Number Line) | W2: Additive Inverse & Zero Pairs

PRACTICE A (p.55): KEY IDEAS (#1-#10) — ExA: Opposite of -15 is +15; -15 + 15 = 0		PRACTICE B (p.56): REAL LIFE (#1-#10) — ExB: +$5	
A1: -25; Opposite = +25 |-25; +/-| SC-6: 25 [+/-] → -25; [+/-] → 25	A2: +42; and +42 |42| SC-6: 42 [+/-] [+/-]	B1: 0 ft (opposites / additive inverses) |-120 + 120| SC-6: 120 [+/-] [ENTER] 120 [+]	B2: -282 ft; opposite is +282 ft |-282| SC-6: 282 [+/-]
A3: 0 m (Sea level) |-18 + 18| SC-6: 18 [+/-] [ENTER] 18 [+]	A4: 0 |6 + -6| SC-6: 6 [ENTER] 6 [+/-] [+]	B3: +19 yards |(12 + -8) + 15| SC-6: 12 [ENTER] 8 [+/-] [+] 15 [+]	B4: -28°C; opposite is +28°C |-28| SC-6: 28 [+/-]
A5: 0 ft |-120 + 120| SC-6: 120 [+/-] [ENTER] 120 [+]	A6: 0 (0 is 0 units from itself) |-0| SC-6: 0 [+/-]	B5: +$7 |-18 + 25| SC-6: 18 [+/-] [ENTER] 25 [+]	B6: -5 (5 under par) |(-4 + 2) + -3| SC-6: 4 [+/-] [ENTER] 2 [+] 3 [+/-] [+]
A7: +7°F |-12 + 19| SC-6: 12 [+/-] [ENTER] 19 [+]	A8: 0 (Even par) |-3 + 3| SC-6: 3 [+/-] [ENTER] 3 [+]	B7: -x = +35; -(-x) = -35 |35| SC-6: 35 [+/-] [+/-]	B8: +$10 |(25 - 25) + 10| SC-6: 25 [ENTER] 25 [-] 10 [+]
A9: Floor 3 |(14 + -20) + 9| SC-6: 14 [ENTER] 20 [+/-] [+] 9 [+]	A10: +$18,800 |((14500 - 3200) + 8900) - 1400| SC-6: 14500 [ENTER] 3200 [-] 8900 [+] 1400 [-]	B9: +35 meters |(-40 + 95) - 20| SC-6: 40 [+/-] [ENTER] 95 [+] 20 [-]	B10: +23 |-14 + -9| SC-6: 14 [+/-] [ENTER] 9 [+/-] [+] [+/-]

SEC 6.2 (pp.57-58): COMPARING AND ORDERING INTEGERS & RATIONAL NUMBERS

Methods: W1: Convert to Decimals & Plot on Number Line | W2: Rewrite with Least Common Denominator (LCD)

PRACTICE A (p.57): KEY IDEAS (#1-#10) — ExA: -5 > -18		PRACTICE B (p.58): REAL LIFE (#1-#10) — ExB: -0.75 > -4/5 (Diver B is closer to the surface)	
A1: -5°C (-12 < -5) |Compare -5 < -12| SC-6: 5 [+/-] [ENTER] 12 [+/-] [g][<]	A2: -9 < -2 |Compare -2 < -9| SC-6: 2 [+/-] [ENTER] 9 [+/-] [g][<]	B1: -11°C > -18°C (Tuesday was warmer) |Compare -18 > -11| SC-6: 18 [+/-] [ENTER] 11 [+/-] [f][>]	B2: Puffin (-45.8 m) is deeper; -45.25 > -45.8 |Compare -45.8 > -45.25| SC-6: 45.8 [+/-] [ENTER] 45.25 [+/-] [f][>]
A3: -15 < 0 |Compare 0 < -15| SC-6: 0 [ENTER] 15 [+/-] [g][<]	A4: -10, -7, -2, 0, 4 |fLINE| SC-6: Left-to-right on [f][LINE]	B3: -2/3 > -5/6 (-4/6 > -5/6) |Compare -5/6 > -2/3| SC-6: [.] 5 [.] 6 [+/-] [ENTER] [.] 2 [.] 3 [+/-] [f][>]	B4: -9, -6, -1, +2 |Compare -6 < -9| SC-6: 6 [+/-] [ENTER] 9 [+/-] [g][<]
A5: -3/4 < -0.6 (-0.6 is greater) |Compare -0.6 < -0.75| SC-6: 0.6 [+/-] [ENTER] 0.75 [+/-] [g][<]	A6: -2 1/2 < -2.3 |Compare -2.3 < -2.5| SC-6: 2.3 [+/-] [ENTER] 2.5 [+/-] [g][<]	B5: -89°C, -39°C, -6°C, 0°C |Compare -39 < -89| SC-6: 39 [+/-] [ENTER] 89 [+/-] [g][<]	B6: Sea turtle (-3.1 > -3.25) |Compare -3.1 < -3.25| SC-6: 3.1 [+/-] [ENTER] 3.25 [+/-] [g][<]
A7: -5/8 > -7/8 |Compare -7/8 > -5/8| SC-6: [.] 7 [.] 8 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [f][>]	A8: -4.1 is closer to 0 (further to the right) than -4.5 |Compare -4.5 < -4.1| SC-6: 4.5 [+/-] [ENTER] 4.1 [+/-] [g][<]	B7: Stock B (-1.08% > -1.8%) |Compare -1.8 > -1.08| SC-6: 1.8 [+/-] [ENTER] 1.08 [+/-] [f][>]	B8: Examples: -3 1/2 and -3.5 |-3.5| SC-6: 3.5 [+/-]
A9: -1 3/4, -1.5, -0.5, 1 1/4 |-1.75, -1.5, -0.5, 1.25| SC-6: Mental / Table Check	A10: Deepest: -15.2 m; Shallowest: -14 1/2 m |-15.2 < -14.8 < -14.5| SC-6: Mental / Table Check	B9: -2, -7/4 (-1.75), -1.6, 3/2 (1.5) |-7/4 → Dec/Frac| SC-6: [.] 7 [.] 4 [+/-] [FDISP]	B10: a is farther from 0 (e.g., -10 < -2) |-10| SC-6: 10 [+/-] [g][|x|]

SEC 6.3 (pp.59-60): ABSOLUTE VALUE AND MAGNITUDE

Methods: W1: Measure Distance from 0 on a Number Line | W2: Evaluate |x| Grouping Bars Before Outside Signs

PRACTICE A (p.59): KEY IDEAS (#1-#10) — ExA: |-14| = 14, |14| = 14, -|-14| = -14		PRACTICE B (p.60): REAL LIFE (#1-#10) — ExB: 46 m plunge; and 50 - |-18| = 32	
A1: 125 ||-125|| SC-6: 125 [+/-] [g][|x|]	A2: 30 ||-18| + 12| SC-6: 18 [+/-] [g][|x|] [ENTER] 12 [+]	B1: |32| + |-14| = 46 meters ||32| + |-14|| SC-6: 32 [g][|x|] 14 [+/-] [g][|x|] [+]	B2: -$40 is greater balance; Account A owes more ($65) ||-65|| SC-6: 65 [+/-] [g][|x|]
A3: 8.0 ||-4.8| + |-3.2|| SC-6: 4.8 [+/-] [g][|x|] [ENTER] 3.2 [+/-] [g][|x|] [+]	A4: 16 ||-35| - |-19|| SC-6: 35 [+/-] [g][|x|] [ENTER] 19 [+/-] [g][|x|] [-]	B3: |-18| + |14| = 32 meters ||-18| + |14|| SC-6: 18 [+/-] [g][|x|] 14 [g][|x|] [+]	B4: 35 and -35 (outside minus applies after | |) ||-35|| SC-6: 35 [+/-] [g][|x|] [+/-]
A5: |-12| > |9| (12 > 9) |12 > 9| SC-6: Mental / Table Check	A6: -24 ||-24|| SC-6: 24 [+/-] [g][|x|] [+/-]	B5: -1.5 cm (|-1.5| = 1.5 cm is smallest) ||-1.5|| SC-6: 1.5 [+/-] [g][|x|]	B6: 42 ÷ 6 + 11 = 18 |(|-42| ÷ |-6|) + |-11|| SC-6: 42 [+/-] [g][|x|] 6 [+/-] [g][|x|] [+] 11 [+/-] [g][|x|] [+]
A7: 32 |50 - |-18|| SC-6: 50 [ENTER] 18 [+/-] [g][|x|] [-]	A8: -|-9|, -6, 0, |-3|, |-8| |Values: 8, -6, 3, 0, -9| SC-6: Mental / Table Check	B7: |25| + |-140| = 165 feet ||25| + |-140|| SC-6: 25 [g][|x|] 140 [+/-] [g][|x|] [+]	B8: 19 and -19 ||-19|| SC-6: 19 [+/-] [g][|x|]
A9: Seagull is closer by 7 feet (25 < 32) ||-32| - 25| SC-6: 32 [+/-] [g][|x|] [ENTER] 25 [-]	A10: Greater debt: -$45 ($45); Greater balance: -$28 ||-45|=45 > |-28|=28| SC-6: Mental / Table Check	B9: -|-9| (-9), -6, |5| (5), |-8| (8) ||-9|| SC-6: 9 [+/-] [g][|x|] [+/-]	B10: 4 5/8 |(3 1/2 + 2 1/4) - 1 1/8| SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [+] 1 [.] 1 [.] 8 [-]

SEC 6.4 (pp.61-62): THE FOUR-QUADRANT COORDINATE PLANE & REFLECTIONS

Methods: W1: Plot & Count Distance to the Mirror Axis | W2: Coordinate Reflection Sign Rules

PRACTICE A (p.61): KEY IDEAS (#1-#10) — ExA: A: Quadrant II, B: Quadrant IV, C: Quadrant III		PRACTICE B (p.62): REAL LIFE (#1-#10) — ExB: (5, -8) in Quadrant IV	
A1: Quadrant IV |(+, -) → Right, Down| SC-6: Mental / Table Check	A2: Quadrant III |(-, -) → Left, Down| SC-6: Mental / Table Check	B1: Cave: II, Lighthouse: I, Shipwreck: III, Pier: IV |4| SC-6: 8 [+/-] [ENTER] 4	B2: (0, -7) is on y-axis; (9, 0) is on x-axis |-7| SC-6: 0 [ENTER] 7 [+/-]
A3: Quadrant II |(-, +) → Left, Up| SC-6: Mental / Table Check	A4: (-6, 9) |-6| SC-6: 6 [+/-] → -6	B3: (8, -12) in Quadrant IV |-12| SC-6: 8 [ENTER] 12 [+/-]	B4: (9, 5) in Quadrant I |5| SC-6: 9 [+/-] [+/-] [ENTER] 5
A5: (6, -9) |-9| SC-6: 9 [+/-] → -9	A6: (-3, 7) |7| SC-6: 7 [+/-] [+/-] → +7	B5: Step 1: (-4, 9); Final: (4, 9) in Quad I |9| SC-6: 4 [+/-] [+/-] [ENTER] 9 [+/-] [+/-]	B6: (4, -8) in Quadrant IV |-6 + 10| SC-6: 6 [+/-] [ENTER] 10 [+]
A7: (8, -4) |Flip x → (8,4); flip y → (8,-4)| SC-6: Mental / Table Check	A8: (0, -9) on y-axis; (-7, 0) on x-axis |0 sideways vs 0 up/down| SC-6: Mental / Table Check	B7: P: Quadrant II; Q: Quadrant III |4.5| SC-6: 2.5 [+/-] [ENTER] 4.5	B8: y-axis (x-coordinate changed sign, y stayed -3) |-7| SC-6: 7 [+/-]
A9: (-3, 3) in Quadrant II |4 - 7; -5 + 8| SC-6: 4 [ENTER] 7 [-]; 5 [+/-] [ENTER] 8 [+]	A10: D(-4, -5) in Quadrant III |Match x of A (-4) & y of C (-5)| SC-6: Mental / Table Check	B9: (-5, 5), (-5, -5), and (5, -5) |-5| SC-6: 5 [+/-]	B10: (x, y) is in Quad II; (-x, -y) is in Quad IV |-1| SC-6: 1 [+/-]

SEC 6.5 (pp.63-64): DISTANCES IN THE COORDINATE PLANE

Methods: W1: Add/Subtract Absolute Values to the Axis | W2: Absolute Difference of Coordinates |x1 - x2|

PRACTICE A (p.63): KEY IDEAS (#1-#10) — ExA: 15 units; and 8 units		PRACTICE B (p.64): REAL LIFE (#1-#10) — ExB: Perimeter = 34 units; Area = 72 square units	
A1: 27 units ||(-12 - 15)|| SC-6: 12 [+/-] [ENTER] 15 [-] [g][|x|]	A2: 11 units ||(-4 - -15)|| SC-6: 4 [+/-] [ENTER] 15 [+/-] [-] [g][|x|]	B1: |-8| + |11| = 19 units ||(-8 - 11)|| SC-6: 8 [+/-] [ENTER] 11 [-] [g][|x|]	B2: 9 units = 90 meters ||(-14 - -5)|| SC-6: 14 [+/-] [ENTER] 5 [+/-] [-] [g][|x|]
A3: 15 units ||(9 - -6)|| SC-6: 9 [ENTER] 6 [+/-] [-] [g][|x|]	A4: 9 units ||(-14 - -5)|| SC-6: 14 [+/-] [ENTER] 5 [+/-] [-] [g][|x|]	B3: |15| + |-35| = 50 units ||(15 - -35)|| SC-6: 15 [ENTER] 35 [+/-] [-] [g][|x|]	B4: W = 10, H = 6, Perimeter = 32 units |(|(-3 - 7)| + |(4 - -2)|) × 2| SC-6: 3 [+/-] [ENTER] 7 [-] [g][|x|] 4 [ENTER] 2 [+/-] [-] [g][|x|] [+] 2 [×]
A5: 10 units ||(-3.5 - 6.5)|| SC-6: 3.5 [+/-] [ENTER] 6.5 [-] [g][|x|]	A6: 8 units ||(-4.5 - 3.5)|| SC-6: 4.5 [+/-] [ENTER] 3.5 [-] [g][|x|]	B5: 60 square units |A = 10 × 6| SC-6: 10 [ENTER] 6 [f][RECT]	B6: 7.75 miles |3.5 + 4.25| SC-6: 3.5 [ENTER] 4.25 [+]
A7: (4, 3) |-6 + 10| SC-6: 6 [+/-] [ENTER] 10 [+]	A8: 13 units (They lie on opposite sides of the y-axis: |-5| + |8| = 13) ||(-5 - 8)|| SC-6: 5 [+/-] [ENTER] 8 [-] [g][|x|]	B7: 5 3/4 units |3 1/2 + 2 1/4| SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [+]	B8: 7 + 10 = 17 units |(5 + 2) + (4 + 6)| SC-6: 5 [ENTER] 2 [+] 4 [ENTER] 6 [+] [+]
A9: Perimeter = 34 units, Area = 72 sq units |Width 9, Height 8| SC-6: Mental / Table Check	A10: 20 grid units = 10 miles |10 + 10 = 20 units × 0.5| SC-6: Mental / Table Check	B9: (11, 4) and (-5, 4) |3 - 8| SC-6: 3 [ENTER] 8 [+] 3 [ENTER] 8 [-]	B10: Base = 10, Height = 8, Area = 40 units² |A = ½((4 + 6) × (7 + 1))| SC-6: 4 [ENTER] 6 [+] 7 [ENTER] 1 [+] [g][TRI]

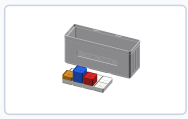
Unit 7: Equations & Inequalities — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 7 • Concept → Blocks (When Applicable: Place Value Blocks & Fraction Strips) → Calculator

1. **Concept (Stack Approach): An equation is a balanced two-register comparison (Y = X). Solving isolates the unknown by applying inverse operations in reverse order.**

2. Blocks (Place Value & Fraction): Model both sides of an equation on the Place Value tray or with Fraction Strips so removing/dividing equal amounts preserves balance.

3. **Calculator (SC-6): Solve with inverse operations on the stack, then push the left and right sides into Y and X and press |!=| (flashes NO when both sides are equal).**



SEC 7.1 (pp.66-67): WRITING EQUATIONS IN ONE VARIABLE & TESTING SOLUTIONS

Methods: W1: Substitute Candidates to Test Equality | W2: Balance Scale / Mental Unwrapping

PRACTICE A (p.66): KEY IDEAS (#1-#10) — ExA: x = 9 is the solution (41 = 41)		PRACTICE B (p.67): REAL LIFE (#1-#10) — ExB: n - 8 = 25; n = 33 drones	
A1: x = 8 (5(8)-7 = 33) |(5 × 8) - 7| SC-6: 33 [ENTER] 5 [ENTER] 8 [×] 7 [-] [!=]	A2: y = 15 (15÷3 + 4 = 9) |(15 ÷ 3) + 4| SC-6: 9 [ENTER] 15 [ENTER] 3 [÷] 4 [+] [!=]	B1: x = 8 (5·8 - 7 = 33) |(5 × 8) - 7| SC-6: 33 [ENTER] 5 [ENTER] 8 [×] 7 [-] [!=]	B2: 15t + 5 = 65; yes, t = 4 is a solution |(15 × 4) + 5| SC-6: 65 [ENTER] 15 [ENTER] 4 [×] 5 [+] [!=]
A3: x + 14 = 52; x = 38 |x + 14 = 52| SC-6: Mental / Table Check	A4: 6k = 84; k = 14 |6k = 84| SC-6: Mental / Table Check	B3: k - 12 = 29; k = 41 |29 + 12| SC-6: 29 [ENTER] 12 [+]	B4: n ÷ 6 = 14; n = 84 |14 × 6| SC-6: 14 [ENTER] 6 [×]
A5: w - 9 = 17; w = 26 |w - 9 = 17| SC-6: Mental / Table Check	A6: m / 5 = 13; m = 65 |m / 5 = 13| SC-6: Mental / Table Check	B5: 3s = 54; s = 18 cm |54 ÷ 3| SC-6: 54 [ENTER] 3 [÷]	B6: m = 15 (100 - 60 = 40) |100 - (4 × 15)| SC-6: 40 [ENTER] 100 [ENTER] 4 [ENTER] 15 [×] [-] [!=]
A7: 4p + 6 = 54; p = $12 |(4 × 12) + 6| SC-6: 54 [ENTER] 4 [ENTER] 12 [×] 6 [+] [!=]	A8: x - 12 = 20; x = 32 |x - 12 = 20| SC-6: Mental / Table Check	B7: p ÷ 8 = 3; p = 24 slices |24 ÷ 8| SC-6: 24 [ENTER] 8 [÷]	B8: 45 + d = 120; d = $75 |120 - 45| SC-6: 120 [ENTER] 45 [-]
A9: 10 + 3m = 28; m = 6 movies |(28 - 10) ÷ 3| SC-6: 28 [ENTER] 10 [-] 3 [÷]	A10: b / 3 = 14.50; b = $43.50 |14.50 × 3| SC-6: 14.50 [ENTER] 3 [×]	B9: Yes (both sides equal 28) |(5 × 6) - 2| SC-6: 3 [ENTER] 6 [×] 10 [+] 5 [ENTER] 6 [×] 2 [-] [!=]	B10: b - 4 = 13; b = 17 years old |13 + 4| SC-6: 13 [ENTER] 4 [+]

SEC 7.2 (pp.68-69): SOLVING EQUATIONS USING ADDITION OR SUBTRACTION

Methods: W1: Properties of Equality (Inverse Operations) | W2: Balance Scale / Part-Part-Whole Bar Model

PRACTICE A (p.68): KEY IDEAS (#1-#10) — ExA: x = 45		PRACTICE B (p.69): REAL LIFE (#1-#10) — ExB: w = 54.4 g; m = 11/24	
A1: x = 45 |73 - 28| SC-6: 73 [ENTER] 28 [-]	A2: y = 92 |58 + 34| SC-6: 58 [ENTER] 34 [+]	B1: s + 45 = 210; s = 165 points |210 - 45| SC-6: 210 [ENTER] 45 [-]	B2: p - 18 = 67; p = $85 |67 + 18| SC-6: 67 [ENTER] 18 [+]
A3: n = 45 |64 - 19| SC-6: 64 [ENTER] 19 [-]	A4: k = 76 |29 + 47| SC-6: 29 [ENTER] 47 [+]	B3: h + 1450 = 5200; h = 3,750 feet |5200 - 1450| SC-6: 5200 [ENTER] 1450 [-]	B4: f - 64.8 = 185.2; f = 250.0 grams |185.2 + 64.8| SC-6: 185.2 [ENTER] 64.8 [+]
A5: w = 54.4 |34.8 + 19.6| SC-6: 34.8 [ENTER] 19.6 [+]	A6: p = 16.65 |25.4 - 8.75| SC-6: 25.4 [ENTER] 8.75 [-]	B5: t - 0.48 = 12.65; t = 13.13 seconds |12.65 + 0.48| SC-6: 12.65 [ENTER] 0.48 [+]	B6: m + 3/8 = 7/8; m = 1/2 cup |7/8 - 3/8| SC-6: [.] 7 [.] 8 [ENTER] [.] 3 [.] 8 [-]
A7: m = 11/24 |5/6 - 3/8| SC-6: [.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]	A8: a = 4 3/4 |3 1/2 + 1 1/4| SC-6: 3 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 4 [+]	B7: 2 3/4 + d = 4 1/2; d = 1 3/4 miles |4 1/2 - 2 3/4| SC-6: 4 [.] 1 [.] 2 [ENTER] 2 [.] 3 [.] 4 [-]	B8: b + 38 = 125; b = 87° |125 - 38| SC-6: 125 [ENTER] 38 [-]
A9: m - 18.50 = 26.75; m = $45.25 |26.75 + 18.50| SC-6: 26.75 [ENTER] 18.50 [+]	A10: h + 42.5 = 110; h = 67.5 ft |110 - 42.5| SC-6: 110 [ENTER] 42.5 [-]	B9: -40 + d = -115; d = -75 meters |-115 - -40| SC-6: 115 [+/-] [ENTER] 40 [+/-] [-]	B10: L = 44 cm; 1 length = 22 cm |(64 - 20) ÷ 2| SC-6: 64 [ENTER] 20 [-] 2 [÷]

SEC 7.3 (pp.70-71): SOLVING EQUATIONS USING MULTIPLICATION OR DIVISION

Methods: W1: Divide/Multiply by Coefficient | W2: Multiply by the Reciprocal | W3: Equal-Groups Tape Diagram

PRACTICE A (p.70): KEY IDEAS (#1-#10) — ExA: x = 17; k = 101.5		PRACTICE B (p.71): REAL LIFE (#1-#10) — ExB: 40 songs	
A1: x = 17 |136 ÷ 8| SC-6: 136 [ENTER] 8 [÷]	A2: y = 15 |180 ÷ 12| SC-6: 180 [ENTER] 12 [÷]	B1: 12s = 168; s = $14 per shirt |168 ÷ 12| SC-6: 168 [ENTER] 12 [÷]	B2: p ÷ 9 = 3; p = 27 slices |3 × 9| SC-6: 3 [ENTER] 9 [×]
A3: m = 114 |19 × 6| SC-6: 19 [ENTER] 6 [×]	A4: k = 101.5 |14.5 × 7| SC-6: 14.5 [ENTER] 7 [×]	B3: 65t = 325; t = 5 hours |325 ÷ 65| SC-6: 325 [ENTER] 65 [÷]	B4: 2.5p = 16.25; p = $6.50 per pound |16.25 ÷ 2.5| SC-6: 16.25 [ENTER] 2.5 [÷]
A5: w = 14 |63 ÷ 4.5| SC-6: 63 [ENTER] 4.5 [÷]	A6: x = 24 |18 ÷ 3/4| SC-6: 18 [ENTER] [.] 3 [.] 4 [÷]	B5: s ÷ 4 = 9; s = $36 |9 × 4| SC-6: 9 [ENTER] 4 [×]	B6: (3/4)c = 21; c = 28 students |21 ÷ 3/4| SC-6: 21 [ENTER] [.] 3 [.] 4 [÷]
A7: n = 40 |16 ÷ 2/5| SC-6: 16 [ENTER] [.] 2 [.] 5 [÷]	A8: p = 4/5 (0.8) |2/3 ÷ 5/6| SC-6: [.] 2 [.] 3 [ENTER] [.] 5 [.] 6 [÷]	B7: (5/6)p = 150; p = $180 |150 ÷ 5/6| SC-6: 150 [ENTER] [.] 5 [.] 6 [÷]	B8: 13.5w = 108; w = 8 feet |108 ÷ 13.5| SC-6: 108 [ENTER] 13.5 [÷]
A9: 1.5t = 27; t = 18 minutes |27 ÷ 1.5| SC-6: 27 [ENTER] 1.5 [÷]	A10: 3/5 s = 27; s = 45 students |27 ÷ 3/5| SC-6: 27 [ENTER] [.] 3 [.] 5 [÷]	B9: h ÷ 15 = 3.2; h = 48 feet |3.2 × 15| SC-6: 3.2 [ENTER] 15 [×]	B10: 0.40n = 56; n = 140 |56 ÷ 0.40| SC-6: 56 [ENTER] 0.40 [÷]

SEC 7.4 (pp.72-73): WRITING AND GRAPHING INEQUALITIES

Methods: W1: Inverse Property of Inequality & Number Line Graph | W2: Boundary & Test-Point Substitution

PRACTICE A (p.72): KEY IDEAS (#1-#10) — ExA: x >= 19 (Solid dot at 19, arrow right)		PRACTICE B (p.73): REAL LIFE (#1-#10) — ExB: g <= 15 (at most 15 gold bars)	
A1: x > 13 (Open circle at 13, arrow right) |21 - 8| SC-6: 21 [ENTER] 8 [-]	A2: y <= 24 (Solid dot at 24, arrow left) |15 + 9| SC-6: 15 [ENTER] 9 [+]	B1: h >= 54 (solid dot at 54, shade right); at least 5 in |54 - 49| SC-6: 54 [ENTER] 49 [-]	B2: p + 5 <= 12 → p <= 7 passengers |12 - 5| SC-6: 12 [ENTER] 5 [-]
A3: n >= 13 (Solid dot at 13, arrow right) |78 ÷ 6| SC-6: 78 [ENTER] 6 [÷]	A4: m < 36 (Open circle at 36, arrow left) |9 × 4| SC-6: 9 [ENTER] 4 [×]	B3: 22.50 + x >= 35 → x >= $12.50 |35 - 22.50| SC-6: 35 [ENTER] 22.50 [-]	B4: 3.50g <= 28 → g <= 8 games |28 ÷ 3.50| SC-6: 28 [ENTER] 3.50 [÷]
A5: h >= 48 |'At least' = >=| SC-6: Mental / Table Check	A6: w <= 2,000 |'At most' = <=| SC-6: Mental / Table Check	B5: 15r > 240 → r > 16 (at least 17 rows) |240 ÷ 15| SC-6: 240 [ENTER] 15 [÷]	B6: b - 25 < 15 → b < 40% |15 + 25| SC-6: 15 [ENTER] 25 [+]
A7: Only 6 (At n = 5, 23 > 23 is false) |Compare 23 > ((4 × 6) + 3)| SC-6: 23 [ENTER] 4 [ENTER] 6 [×] 3 [+] [f][>]	A8: 5 and 6 |(4 × 5) - 3| SC-6: 17 [ENTER] 4 [ENTER] 5 [×] 3 [-] [f][>=]	B7: 8 and 10 |(5 × 8) - 10| SC-6: 30 [ENTER] 5 [ENTER] 8 [×] 10 [-] [f][>=]	B8: 2c - 30 >= 70 → c >= 50 cupcakes |(70 + 30) ÷ 2| SC-6: 70 [ENTER] 30 [+] 2 [÷]
A9: 8b <= 50; at most 6 books ($2 left over) |50 ÷ 8 (Q & R)| SC-6: 50 [ENTER] 8 [g][÷R]	A10: p + 340 >= 500; p >= 160 points |500 - 340| SC-6: 500 [ENTER] 340 [-]	B9: s <= 20 (solid dot at 20, shade left down to 0) |Compare 20 <= 20| SC-6: 20 [ENTER] 20 [g][<=]	B10: 45c + 170 <= 800 → c <= 14 crates |(800 - 170) ÷ 45| SC-6: 800 [ENTER] 170 [-] 45 [÷]

SEC 7.5 (pp.74-75): WRITING EQUATIONS IN TWO VARIABLES (Y = KX AND Y = X + B)

Methods: W1: Test Constant Ratio y/x vs. Constant Difference y - x | W2: Input-Output Table & Coordinate Graph

PRACTICE A (p.74): KEY IDEAS (#1-#10) — ExA: Table A: y = 7x; Table B: y = x - 4		PRACTICE B (p.75): REAL LIFE (#1-#10) — ExB: p = 12.50h; $81.25	
A1: y = 8x; y = 80 |16 ÷ 2; 8×10| SC-6: 16 [ENTER] 2 [÷] → 8; 8×10	A2: y = x + 8; y = 28 |9 - 1; 20+8| SC-6: 9 [ENTER] 1 [-] → +8; 20+8	B1: y = x + 5 is additive; y = 8m is multiplicative |8 × 1| SC-6: 8 [ENTER] 1 [×]	B2: g = 18h (h = indep, g = dep); 81 grams |18 × 4.5| SC-6: 18 [ENTER] 4.5 [×]
A3: y = 7x; y = 84 |14 ÷ 2; 7×12| SC-6: 14 [ENTER] 2 [÷] → 7; 7×12	A4: y = x - 4; y = 26 |30 - 4| SC-6: 30 [ENTER] 4 [-]	B3: m = z + 4; m = 12, 16, 19 |8 + 4| SC-6: 8 [ENTER] 4 [+]	B4: s = 15p; p = 12 sheets |180 ÷ 15| SC-6: 180 [ENTER] 15 [÷]
A5: (1, 6), (2, 12), (3, 18), (4, 24) |Repeat(+6)| SC-6: 6 [g][REP] [+] [+] [+]	A6: Independent: t; Dependent: d |d depends on t| SC-6: Mental / Table Check	B5: y = x - 6; $18, $29, and $44 |24 - 6| SC-6: 24 [ENTER] 6 [-]	B6: y = 3.5x; 140 wheel turns |3.5 × 40| SC-6: 3.5 [ENTER] 40 [×]
A7: 292.5 miles |65 × 4.5| SC-6: 65 [ENTER] 4.5 [×]	A8: y = 4x passes through (0, 0); y = x + 4 crosses at (0, 4) |Test x = 0 in both| SC-6: Mental / Table Check	B7: y = 9x; at x = 20, y = 180 |(27 ÷ 3) × 20| SC-6: 27 [ENTER] 3 [÷] 20 [×]	B8: c = 20g + 7; $127 |(20 × 6) + 7| SC-6: 20 [ENTER] 6 [×] 7 [+]
A9: 8 hours |100 ÷ 12.50| SC-6: 100 [ENTER] 12.50 [÷]	A10: c = 15m + 25; $205 |(15 × 12) + 25| SC-6: 15 [ENTER] 12 [×] 25 [+]	B9: Multiplicative y = kx (when x = 0, y = k·0 = 0) |0 × 7| SC-6: 0 [ENTER] 7 [×]	B10: d = 0.2s; 3 miles away |0.2 × 15| SC-6: 0.2 [ENTER] 15 [×]

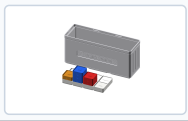
Unit 8: Surface Area & Volume of 3D Solids — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 8 • Concept → Blocks (When Applicable: Place Value Blocks & Fraction Strips) → Calculator

1. **Concept (Stack Approach): Surface area sums 2D face areas across an unfolded net, whereas 3D volume multiplies three orthogonal dimensions (l, w, h) held simultaneously in registers Z, Y, and X.**

2. Blocks (When Applicable — Place Value & Fraction): Pack Place Value unit blocks to see V = l·w·h in layers, and use Fraction Strips to model fractional edge lengths.

3. **Calculator (SC-6): Accumulate opposite face pairs with |Σ+| → |f||Σx| for surface area, multiply Z·Y·X with |f||PRISM| for volume, and verify |f||CIRC|, |g||CONE|, and |g||HYP|.**



SEC 8.1 (pp.77-78): THREE-DIMENSIONAL FIGURES, NETS & RECTANGULAR PRISMS

Methods: W1: Unfold into a 6-Face 2D Net | W2: Opposite Face Pairs SA = 2(lw + lh + wh) | W3: Lateral Strip Ph + 2B

PRACTICE A (p.77): KEY IDEAS (#1-#10) — ExA: 158 cm³		PRACTICE B (p.78): REAL LIFE (#1-#10) — ExB: Cube = 294 in²; Open-Top Bin = 260 cm³	
A1: 96 cm² |4² × 6| SC-6: 4 [x²] 6 [×]	**A2: 486 in.²** |9² × 6| SC-6: 9 [x²] 6 [×]	B1: 2(160 + 600 + 240) = 2,000 cm² |(((20 × 8) + (20 × 30)) + (8 × 30)) × 2| SC-6: 20 [ENTER] 8 [×] 20 [ENTER] 30 [×] [+] 8 [ENTER] 30 [×] [+] 2 [×]	**B2: 1 face = 81 in²; Total SA = 486 in²** |9² × 6| SC-6: 9 [x²] 6 [×]
A3: 108 cm² |2(24)+2(18)+2(12)| SC-6: Mental / Table Check	**A4: 160 in.²** |2(50)+2(20)+2(10)| SC-6: Mental / Table Check	B3: 288 + 720 + 360 = 1,368 in² |((24 × 12) + ((24 × 15) × 2)) + ((12 × 15) × 2)| SC-6: 24 [ENTER] 12 [×] 24 [ENTER] 15 [×] 2 [×] [+] 12 [ENTER] 15 [×] 2 [×] [+]	B4: SA = 500 in²; yes (100 in² to spare) |(((15 × 10) + (15 × 4)) + (10 × 4)) × 2| SC-6: 15 [ENTER] 10 [×] 15 [ENTER] 4 [×] [+] 10 [ENTER] 4 [×] [+] 2 [×]
A5: 392 ft² |2(96)+2(60)+2(40)| SC-6: Mental / Table Check	**A6: 85 m²** |2(20)+2(12.5)+2(10)| SC-6: Mental / Table Check	B5: 120 + 192 + 160 = 472 ft² |((12 × 10) + ((12 × 8) × 2)) + ((10 × 8) × 2)| SC-6: 12 [ENTER] 10 [×] 12 [ENTER] 8 [×] 2 [×] [+] 10 [ENTER] 8 [×] 2 [×] [+]	**B6: 1 face = 25 cm²; edge = 5 cm** |√((150 ÷ 6))| SC-6: 150 [ENTER] 6 [÷] |f|[√x]
A7: One face = 25 cm²; s = 5 cm |√((150 ÷ 6))| SC-6: 150 [ENTER] 6 [÷] |f|[√x]	A8: 94 cm² (He only summed 3 faces instead of doubling for all 6 faces) |47 × 2| SC-6: 47 [ENTER] 2 [×]	**B7: 32 + 8 = 40 ft²** |((4 × 2) × 4) + ((2 × 2) × 2)| SC-6: 4 [ENTER] 2 [×] 4 [×] 2 [ENTER] 2 [×] 2 [×] [+]	B8: 2(24 + 60 + 40) = 248 cm² |(((6 × 4) + (6 × 10)) + (4 × 10)) × 2| SC-6: 6 [ENTER] 4 [×] 6 [ENTER] 10 [×] [+] 4 [ENTER] 10 [×] [+] 2 [×]
A9: 2,100 cm² |600 + 2(450) + 2(300)| SC-6: Mental / Table Check	A10: 500 in.³ needed; Yes, with 50 in.³ to spare |2(150)+2(60)+2(40)| SC-6: Mental / Table Check	B9: 54 cm² vs. 216 cm² (4 times as great) |(6² × 6) ÷ 54| SC-6: 6 [x²] 6 [×] 54 [÷]	**B10: 160 cm²** |4² × 10| SC-6: 4 [x²] 10 [×]

SEC 8.2 (pp.79-80): SURFACE AREAS OF TRIANGULAR PRISMS AND PYRAMIDS

Methods: W1: Add All Individual Faces on the 2D Net | W2: Base Area + Combined Lateral Area

PRACTICE A (p.79): KEY IDEAS (#1-#10) — ExA: 340 m³		PRACTICE B (p.80): REAL LIFE (#1-#10) — ExB: 288 square feet	
A1: 96 cm² |6² + (½(6×5) × 4)| SC-6: 6 [x²] 6 [ENTER] 5 [g][TRI] 4 [×] [+]	**A2: 224 in.²** |8² + (½(8×10) × 4)| SC-6: 8 [x²] 8 [ENTER] 10 [g][TRI] 4 [×] [+]	B1: 24 + 80 + 48 = 152 ft² |((½(6×4) × 2) + ((5×8) × 2)) + (6×8)| SC-6: 6 [ENTER] 4 [g][TRI] 2 [×] 5 [ENTER] 8 [f][RECT] 2 [×] [+] 6 [ENTER] 8 [f][RECT]..	**B2: 144 + 240 = 384 cm²** |12² + (½(12×10) × 4)| SC-6: 12 [x²] 12 [ENTER] 10 [g][TRI] 4 [×] [+]
A3: 360 ft² |12² + (½(12×9) × 4)| SC-6: 12 [x²] 12 [ENTER] 9 [g][TRI] 4 [×] [+]	**A4: 62.4 cm²** |½(6×5.2) × 4| SC-6: 6 [ENTER] 5.2 [g][TRI] 4 [×]	**B3: 96 ft²** |½(8×6) × 4| SC-6: 8 [ENTER] 6 [g][TRI] 4 [×]	**B4: 13.8 + 240 = 253.8 cm²** |(6.9 × 2) + ((4×20) × 3)| SC-6: 6.9 [ENTER] 2 [×] 4 [ENTER] 20 [f][RECT] 3 [×] [+]
A5: 132 cm² |½(4×3) × 2| SC-6: 4 [ENTER] 3 [g][TRI] 2 [×] (12) + 12×10 (120)	**A6: 150 ft²** |½(8×3) × 2| SC-6: 8 [ENTER] 3 [g][TRI] 2 [×] (24) + 18×7 (126)	B5: 48 + 120 = 168 in² |(½(8×6) × 2) + (((8 + 6) + 10) × 5)| SC-6: 8 [ENTER] 6 [g][TRI] 2 [×] 8 [ENTER] 6 [+] 10 [×] 5 [×] [+]	**B6: 15.6 cm²** |½(3×2.6) × 4| SC-6: 3 [ENTER] 2.6 [g][TRI] 4 [×]
A7: 110.88 cm² |½(8×6.93) × 4| SC-6: 8 [ENTER] 6.93 [g][TRI] 4 [×]	A8: Triangular prism: 2 triangles + 3 rectangles; Square pyramid: 1 square + 4 triangles |Count net fa.. SC-6: Mental / Table Check	**B7: 96 + 600 = 696 ft²** |(½(16×6) × 2) + ((10×30) × 2)| SC-6: 16 [ENTER] 6 [g][TRI] 2 [×] 10 [ENTER] 30 [f][RECT] 2 [×] [+]	**B8: 15 in² per face; slant height = 5 in** |(96 - 36) ÷ 4| SC-6: 96 [ENTER] 36 [-] 4 [÷]
A9: 152 ft² |24 + 48 + 80| SC-6: Mental / Table Check	**A10: 108 ft²** |½(9×6) × 4| SC-6: 9 [ENTER] 6 [g][TRI] 4 [×]	B9: 96 + 480 = 576 cm² |(½(12×8) × 2) + (((10 + 10) + 12) × 15)| SC-6: 12 [ENTER] 8 [g][TRI] 2 [×] 10 [ENTER] 10 [+] 12 [×] 15 [×] [+]	B10: Pyramid A (260 cm² > 224 cm²) |8² + (½(8×10) × 4)| SC-6: 10 [x²] 10 [ENTER] 8 [g][TRI] 4 [×] [+] 8 [x²] 8 [ENTER] 10 [g][TRI] 4 [×] [+]

SEC 8.3 (pp.81-82): VOLUMES OF RECTANGULAR PRISMS

Methods: W1: Base Area × Height (V = Bh) | W2: Multiply Length × Width × Height (Fractions or Decimals)

PRACTICE A (p.81): KEY IDEAS (#1-#10) — ExA: Crate = 480 in.³; Aquarium = 10 1/2 ft³		PRACTICE B (p.82): REAL LIFE (#1-#10) — ExB: 8 inches deep	
A1: 3,024 cm³ |Prism(18×6×28)| SC-6: 18 [ENTER] 6 [ENTER] 28 [f][PRISM]	**A2: 125 in.³** |5³| SC-6: 5 [x³]	**B1: 500 m³** |Prism(25×10×2)| SC-6: 25 [ENTER] 10 [ENTER] 2 [f][PRISM]	**B2: 48 ft³** |Prism(8×4×1.5)| SC-6: 8 [ENTER] 4 [ENTER] 1.5 [f][PRISM]
A3: 500 m³ |Prism(25×10×2)| SC-6: 25 [ENTER] 10 [ENTER] 2 [f][PRISM]	**A4: 224 m³** |32 × 7| SC-6: 32 [ENTER] 7 [×]	**B3: 15,625 cm³** |25³| SC-6: 25 [x³]	**B4: 65 in³** |Prism(6.5×4×2.5)| SC-6: 6.5 [ENTER] 4 [ENTER] 2.5 [f][PRISM]
A5: 10 1/2 ft³ (10.5) |Prism(3.5×1.5×2)| SC-6: 3.5 [ENTER] 1.5 [ENTER] 2 [f][PRISM]	**A6: 27 in.³** |Prism(4.5×3×2)| SC-6: 4.5 [ENTER] 3 [ENTER] 2 [f][PRISM]	**B5: 8 per in³; 96 half-inch cubes** |12 × 8| SC-6: 12 [ENTER] 8 [×]	**B6: 3 cm rise** |120 ÷ 40| SC-6: 120 [ENTER] 40 [÷]
A7: 6 in. |270 ÷ (9 × 5)| SC-6: 270 [ENTER] 9 [ENTER] 5 [×] [÷]	**A8: 8 half-inch cubes** |2³| SC-6: 2 [x³]	**B7: 77 ft³** |Prism(5.5×4×3.5)| SC-6: 5.5 [ENTER] 4 [ENTER] 3.5 [f][PRISM]	**B8: 36 ft³** |(12 × 9) ÷ 3| SC-6: 12 [ENTER] 9 [×] 3 [÷]
A9: 8 inches |360 ÷ 45| SC-6: 360 [ENTER] 45 [÷]	**A10: 216 half-inch cubes** |27 × 2³| SC-6: 27 [ENTER] 2 [x³] [×]	**B9: 8 inches** |360 ÷ 45| SC-6: 360 [ENTER] 45 [÷]	B10: 4,608 in³ ≈ 19.95 (about 20) gallons |Prism(24×12×16) ÷ 231| SC-6: 24 [ENTER] 12 [ENTER] 16 [f][PRISM] 231 [÷]

SEC 8.4 (pp.83-84): CIRCLES, CONES, AND THE PYTHAGOREAN THEOREM

Methods: W1: Exact Form in Terms of π & $a^2 + b^2 = c^2$ | W2: Decimal Approximation (π ≈ 3.14) & Scaled Triples

PRACTICE A (p.83): KEY IDEAS (#1-#10) — ExA: Pizza = 113.10 in.²; Ramp diagonal = 13 ft		PRACTICE B (p.84): REAL LIFE (#1-#10) — ExB: 94.25 cm³ (exact: 30π cm³)	
A1: 78.54 cm² |π × 5²| SC-6: 5 [f][CIRC]	**A2: 706.86 cm²** |π × 15²| SC-6: 15 [f][CIRC]	**B1: 153.94 ft²** |π × 7²| SC-6: 7 [f][CIRC]	**B2: r = 20 m; Area ≈ 1,256.64 m²** |π × 20²| SC-6: 20 [f][CIRC]
A3: 5 cm |√(3² + 4²)| SC-6: 3 [ENTER] 4 [g][HYP]	**A4: 10 meters** |√(6² + 8²)| SC-6: 6 [ENTER] 8 [g][HYP]	**B3: 87.96 inches** |(2 × π) × 14| SC-6: 2 [ENTER] |π| [×] 14 [×]	**B4: 25 meters** |√(15² + 20²)| SC-6: 15 [ENTER] 20 [g][HYP]
A5: 13 ft |√(12² + 5²)| SC-6: 12 [ENTER] 5 [g][HYP]	**A6: r = 8 m; Area = 201.06 m²** |π × (16 ÷ 2)²| SC-6: 16 [ENTER] 2 [÷] [f][CIRC]	**B5: 20 inches** |√(16² + 12²)| SC-6: 16 [ENTER] 12 [g][HYP]	**B6: 150.80 cm³ (48π)** |1/3π(4²)(9)| SC-6: 4 [ENTER] 9 [g][CONE]
A7: 18.85 m³ |1/3π(3²)(2)| SC-6: 3 [ENTER] 2 [g][CONE]	**A8: 188.50 in.³** |1/3π(6²)(5)| SC-6: 6 [ENTER] 5 [g][CONE]	**B7: 565.49 in³ (180π)** |1/3π(6²)(15)| SC-6: 6 [ENTER] 15 [g][CONE]	B8: One 16-in pizza (201.06 in² > 157.08 in²) |π × 5² × 2| SC-6: 8 [f][CIRC] 5 [f][CIRC] 2 [×]
A9: Cylinder = 282.74 cm³; Cone = 94.25 cm³ |1/3π(3²)(10)| SC-6: 3 [ENTER] 10 [g][CONE]	**A10: 40 inches** |√(32² + 24²)| SC-6: 32 [ENTER] 24 [g][HYP]	**B9: 56.55 ft²** |π × 6² ÷ 2| SC-6: 6 [f][CIRC] 2 [÷]	**B10: 17 feet** |√(8² + 15²)| SC-6: 8 [ENTER] 15 [g][HYP]

Unit 9: Statistical Measures & Data Displays — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapters 9 & 10 (Statistical Measures and Data Displays) • Concept → Blocks (When Applicable: Place Value Blocks & Fraction Strips) → Calculator

1. Concept (Stack Approach): Statistical summaries condense a list of numbers into measures of center (Mean, Median) and spread (IQR, MAD) by accumulating data into dedicated statistical registers.

2. Blocks (Place Value & Fraction): Level unequal Place Value digit stacks to model the Mean as a fair share, and use 4 equal Fraction Strips to visualize 25% quartiles.

3. Calculator (SC-6): Clear with |g||CLΣ|, enter data once with |Σ+| (or |FRQ|), and read |g||x-bar|, |f||MED|, |g||MODE|, |RNG|, |f||1Q|, |g||3Q|, |IQR|, and |g||MAD|.



SEC 9.1 (pp.86-87): STATISTICAL QUESTIONS, DOT PLOTS, AND THE MEAN

Methods: W1: Mean Formula (Sum ÷ Count) | W2: Fair-Share Block Leveling | W3: Balance Point Deviations

PRACTICE A (p.86): KEY IDEAS (#1-#10) — ExA: Sum = 105, Count = 5, Mean = 21

A1: Sum = 20, Mean = 5 |Mean(2, 4, 6, 8)|
SC-6: |g||CLΣ| 2 |Σ+| 4 |Σ+| 6 |Σ+| 8 |Σ+| |g||x-bar|

A2: Sum = 70, Mean = 14 |Sort/Load 5 values → Stats|
SC-6: Load 5 nums into |Σ+| → |g||x-bar|

A3: 156 cm |Mean(150, 162, 156)|
SC-6: 150 |Σ+| 162 |Σ+| 156 |Σ+| |g||x-bar|

A4: Mean = 4.0 g |Load data → Stats|
SC-6: Load into |Σ+| → |g||x-bar|

A5: Sum = 55, Mean = 18.33 |Mean(35, 25)|
SC-6: ... 35 |f||Σ-| 25 |Σ+| |g||x-bar|

A6: 13.33 (80 ÷ 6) |Mean(10, 4, 20, 2)|
`SC-6: 10 [ENTER] 4 [FRQ] 20 [ENTER] 2 [FRQ] [g][x-bar]`

A7: 70°F |Sort/Load 6 values → Stats|
SC-6: Load 6 temps into |Σ+| → |g||x-bar|

A8: 60 total blocks |5 × 12|
`SC-6: 5 [ENTER] 12 [×]`

A9: 95 |360 - 265|
`SC-6: 360 [ENTER] 265 [-]`

A10: 17 points |(90 + 29) ÷ 7|
`SC-6: 90 [ENTER] 29 [+] 7 [÷]`

PRACTICE B (p.87): REAL LIFE (#1-#10) — ExB: 94

B1: Sum = 85; Mean = 17 points |Mean(14, 18, 12, 22, 19)|
SC-6: |g||CLΣ| 14 |Σ+| 18 |Σ+| 12 |Σ+| 22 |Σ+| 19 |Σ+| |g||x-bar|

B2: 42 grams |Mean(41, 43, 42, 40, 44...)|
SC-6: |g||CLΣ| 41 |Σ+| 43 |Σ+| 42 |Σ+| 40 |Σ+| 44 |Σ+| 42 |Σ+| |g||x-bar|

B3: 89 ÷ 10 = 8.9 |(((3 × 8) + (5 × 9)) + (2 × 10)) ÷ 10|
`SC-6: 3 [ENTER] 8 [×] 5 [ENTER] 9 [×] [+] 2 [ENTER] 10 [×] [+] 10 [÷]`

B4: 91 |(5 × 85) - (((80 + 88) + 82) + 84)|
`SC-6: 5 [ENTER] 85 [×] 80 [ENTER] 88 [+] 82 [+] 84 [+] [-]`

B5: Total = 36,000; Mean = 9,000 steps |(((8200 + 9500) + 7800) + 10500) ÷ 4|
`SC-6: 8200 [ENTER] 9500 [+] 7800 [+] 10500 [+] 4 [÷]`

B6: 82.5 → 90.0 (+7.5 points) |((88 + 92) + 90) ÷ 3|
`SC-6: 88 [ENTER] 92 [+] 90 [+] 3 [÷]`

B7: 12 books per stack |(((6 + 10) + 14) + 18) ÷ 4|
`SC-6: 6 [ENTER] 10 [+] 14 [+] 18 [+] 4 [÷]`

B8: 4.8 g |(((4.2 + 5.1) + 4.8) + 5.5) + 4.4) ÷ 5|
`SC-6: 4.2 [ENTER] 5.1 [+] 4.8 [+] 5.5 [+] 4.4 [+] 5 [÷]`

B9: Reading time varies from student to student |60 × 1|
`SC-6: 60 [ENTER] 1 [×]`

B10: 260 ÷ 10 = 26 points |(80 + 180) ÷ 10|
`SC-6: 80 [ENTER] 180 [+] 10 [÷]`

SEC 9.2 (pp.88-89): MEASURES OF CENTER: MEDIAN, MODE, AND RANGE

Methods: W1: Ordered List (Least to Greatest) | W2: Dot Plot / Frequency Table Analysis

PRACTICE A (p.88): KEY IDEAS (#1-#10) — ExA: Mean = 7.875, Median = 7.5, Mode = 7, Range = 4

A1: Mode = 7 |Sort/Load 8 values → Mode|
`SC-6: Load 8 nums → [g][MODE]`

A2: Median = 14, Mode = 12, Range = 8 |Sort/Load 6 values → Med/Mode/Range|
`SC-6: Load 6 nums → [f][MED], [g][MODE], [RNG]`

A3: Mean = 9, Median = 8, Mode = 8, Range = 9 |Sort/Load 5 values → Stats|
SC-6: Load 5 nums into |Σ+|

A4: Min = 62°F, Max = 85°F, Range = 23°F |Sort/Load 7 values → Range|
`SC-6: Load 7 nums → [f][MIN], [g][MAX], [RNG]`

A5: 75°F |Sort/Load 6 values → Med|
`SC-6: Load 6 nums → [f][MED]`

A6: Mean = $29.17, Median = $17.50 (Median is best) |Sort/Load 6 values → Mean/Med|
`SC-6: Load 6 nums → [g][x-bar], [f][MED]`

A7: Mean increases from 14 to 30.8; Median stays 14 |Compare both sets|
`SC-6: Mental / Table Check`

A8: 95 |64 + 31|
`SC-6: 64 [ENTER] 31 [+]`

A9: Median = 14, Range = 13 |Sorted: 8, 11, 14, 16, 21|
`SC-6: Mental / Table Check`

A10: Mean = $370k, Median = $220k (Use Median due to the $980k outlier) |Sort/Load 5 values →...
`SC-6: Load 5 nums → [g][x-bar], [f][MED]`

PRACTICE B (p.89): REAL LIFE (#1-#10) — ExB: Mean = $29.17, Median = $17.50, Mode = $15 (Median is best)

B1: Median = 62, Mode = 62, Range = 22 |Med/Mode/Range(55, 62, 48, 70, 62...)|
SC-6: |g||CLΣ| 55 |Σ+| 62 |Σ+| 48 |Σ+| 70 |Σ+| 62 |Σ+| 50 |Σ+| 66 |Σ+| |f||MED| |g||MODE| |RNG|

B2: Median = 12.5, Mode = 12, Range = 5 |Med(11, 13, 12, 14, 12...)|
SC-6: |g||CLΣ| 11 |Σ+| 13 |Σ+| 12 |Σ+| 14 |Σ+| 12 |Σ+| 16 |Σ+| |f||MED|

B3: Mean=$40, Med=$25, Mode=$20; Median (outlier $105 skews mean) |Mean/Med(20, 25, 20, 30, 1...
SC-6: |g||CLΣ| 20 |Σ+| 25 |Σ+| 20 |Σ+| 30 |Σ+| 105 |Σ+| |g||x-bar| |f||MED|

B4: Modes = 6 and 8 (bimodal); Median = 7 |8|
`SC-6: 6 [ENTER] 8`

B5: Median = 11.8 s, Range = 1.2 s, No mode |12.4 - 11.2|
`SC-6: 12.4 [ENTER] 11.2 [-]`

B6: Min = 64, Max = 81, Range = 17°F, Median = 72°F |Med/Ranges(72, 68, 75, 81, 64...)|
SC-6: |g||CLΣ| 72 |Σ+| 68 |Σ+| 75 |Σ+| 81 |Σ+| 64 |Σ+| 70 |Σ+| 74 |Σ+| |f||MED| |RNG|

B7: Mean = 11, Median = 11, Mode = 12, Range = 7 |Mean/Med/Mode/Range(8, 12, 9, 15, 12...)|
SC-6: |g||CLΣ| 8 |Σ+| 12 |Σ+| 9 |Σ+| 15 |Σ+| 12 |Σ+| 10 |Σ+| |g||x-bar| |f||MED| |g||MODE| |RNG|

B8: Example: 8, 8, 10, 12, 15 |15 - 8|
`SC-6: 15 [ENTER] 8 [-]`

B9: Median increases by 5 (to 85); Range stays 20 |95 - 75|
`SC-6: 95 [ENTER] 75 [-]`

B10: Mean = 11.75, Median = 11.5, Mode = 10, Range = 6 |Mean/Med(9, 12, 10, 14, 11...)|
SC-6: |g||CLΣ| 9 |Σ+| 12 |Σ+| 10 |Σ+| 14 |Σ+| 11 |Σ+| 15 |Σ+| 10 |Σ+| 13 |Σ+| |g||x-bar| |f||MED|

SEC 9.3 (pp.90-91): QUARTILES, INTERQUARTILE RANGE (IQR) & BOX PLOTS

Methods: W1: Split Ordered Data into Lower & Upper Halves | W2: Box-and-Whisker Plot & 25% Quartile Sections

PRACTICE A (p.90): KEY IDEAS (#1-#10) — ExA: Min = 68, Q1 = 73.5, Med = 82, Q3 = 90, Max = 96; IQR = 16.5

A1: Min = 10, Q1 = 15, Med = 30, Q3 = 45, Max = 50, IQR = 30 |Sort/Load 5 values → Q1/Q3/IQR|
`SC-6: Load 5 nums → [f][1Q], [g][3Q], [IQR]`

A2: Q1 = 6, Med = 10, Q3 = 14, IQR = 8 |Sort/Load 7 values → Q1/Q3/IQR|
`SC-6: Load 7 nums → [f][1Q], [g][3Q], [IQR]`

A3: 34 |58 - 24|
`SC-6: 58 [ENTER] 24 [-]`

A4: 75% |3 quarters = 3 × 25%|
`SC-6: Mental / Table Check`

A5: Q1 = 73.5, Med = 82, Q3 = 90, IQR = 16.5 |Sort/Load 8 values → Stats|
SC-6: Load 8 nums into |Σ+|

A6: Q1 = 46.5, Med = 51, Q3 = 56.5, IQR = 10 |Sort/Load 8 values → Stats|
SC-6: Load 8 nums into |Σ+|

A7: Player A (smaller IQR means the middle 50% of scores are closer together) |Compare box widths|
`SC-6: Mental / Table Check`

A8: 90 (Any score > 90 is an outlier) |60 + (1.5 × 20)|
`SC-6: 60 [ENTER] 1.5 [ENTER] 20 [×] [+]`

A9: Q1 = 15, Med = 22, Q3 = 28, IQR = 13 |Sort/Load 7 values → Q1/Q3/IQR|
`SC-6: Load 7 nums → [IQR]`

A10: About 100 students (50% of 200) |50% of 200|
`SC-6: 200 [ENTER] 50 [%]`

PRACTICE B (p.91): REAL LIFE (#1-#10) — ExB: IQR = 16; 20 students (50%)

B1: Min=40, Q1=45, Med=55, Q3=65, Max=70; IQR = 20 |Q1/Q3/IQR(40, 45, 50, 55, 60...)|
SC-6: |g||CLΣ| 40 |Σ+| 45 |Σ+| 50 |Σ+| 55 |Σ+| 60 |Σ+| 65 |Σ+| 70 |Σ+| |IQR|

B2: Q1 = 7.5, Med = 10, Q3 = 13, IQR = 5.5 |Q1/Q3/IQR(6, 7, 8, 9, 11...)|
SC-6: |g||CLΣ| 6 |Σ+| 7 |Σ+| 8 |Σ+| 9 |Σ+| 11 |Σ+| 12 |Σ+| 14 |Σ+| 15 |Σ+| |IQR|

B3: Q1 = 22.5, Med = 30, Q3 = 42.5, IQR = 20 |Q1/Q3/IQR(30, 15, 45, 25, 35...)|
SC-6: |g||CLΣ| 30 |Σ+| 15 |Σ+| 45 |Σ+| 25 |Σ+| 35 |Σ+| 20 |Σ+| 50 |Σ+| 40 |Σ+| 30 |Σ+| |IQR|

B4: Range = 28 ft; IQR = 13 ft |31 - 18|
`SC-6: 40 [ENTER] 12 [-] 31 [ENTER] 18 [-]`

B5: 75% = 45 trees |75% of 60|
`SC-6: 60 [ENTER] 75 [%]`

B6: 25% of 60 = 15 trees |25% of 60|
`SC-6: 60 [ENTER] 25 [%]`

B7: Class A (smaller IQR means less spread in the middle 50%) |19 - 8|
`SC-6: 19 [ENTER] 8 [-]`

B8: 30 + 15 = 45 |30 + (1.5 × 10)|
`SC-6: 30 [ENTER] 1.5 [ENTER] 10 [×] [+]`

B9: Min=18, Q1=22, Med=27, Q3=34, Max=40; IQR = 12 |Q1/Q3/IQR(18, 22, 25, 29, 34...)|
SC-6: |g||CLΣ| 18 |Σ+| 22 |Σ+| 25 |Σ+| 29 |Σ+| 34 |Σ+| 40 |Σ+| |IQR|

B10: Range = 90, IQR = 8 (IQR is not distorted by the outlier 100) |Q1/Q3/IQR(10, 12, 14, 16, 18...)|
SC-6: |g||CLΣ| 10 |Σ+| 12 |Σ+| 14 |Σ+| 16 |Σ+| 18 |Σ+| 20 |Σ+| 100 |Σ+| |IQR|

SEC 9.4 (pp.92-93): MEAN ABSOLUTE DEVIATION (MAD)

Methods: W1: 3-Step MAD Distance Table | W2: Number-Line Deviation Balance

PRACTICE A (p.92): KEY IDEAS (#1-#10) — ExA: Mean = 14, MAD = 2.4

A1: Mean = 156 cm, MAD = 4.0 cm |MAD(150, 162, 156)|
SC-6: 150 |Σ+| 162 |Σ+| 156 |Σ+| |g||MAD|

A2: Mean = 14, MAD = 2.4 |Sort/Load 5 values → Mean/MAD|
`SC-6: Load 5 nums → [g][x-bar], [g][MAD]`

A3: Mean = 20, MAD = 6.0 |Sort/Load 5 values → Mean/MAD|
`SC-6: Load 5 nums → [g][x-bar], [g][MAD]`

A4: Mean = 7, MAD = 0 (Zero spread) |All distances from 7 are 0|
`SC-6: Mental / Table Check`

A5: Mean = 28, MAD = 4.8 |Sort/Load 5 values → Mean/MAD|
`SC-6: Load 5 nums → [g][x-bar], [g][MAD]`

A6: Mean = 13.0 s, MAD = 0.40 s |Sort/Load 6 values → Mean/MAD|
`SC-6: Load 6 decimals → [g][x-bar], [g][MAD]`

A7: Mean = 88, MAD = 3.6 |Sort/Load 5 values → Mean/MAD|
`SC-6: Load 5 nums → [g][x-bar], [g][MAD]`

A8: Team B (a larger MAD indicates greater variation from the mean) |Compare MADs|
`SC-6: Mental / Table Check`

A9: Mean = 10; deviations: 4, 0, 1, 3; MAD = 2.0 |Mean=10; dists 4+0+1+3=8 ÷ 4|
`SC-6: Mental / Table Check`

A10: Mean = 9, Med = 8, Mode = 8, Range = 9, IQR = 5.5, MAD = 2.4 |1 entry → 6 stats|
`SC-6: Mental / Table Check`

PRACTICE B (p.93): REAL LIFE (#1-#10) — ExB: Mean = 13.0 s; MAD = 0.40 s

B1: Mean = 210°C; MAD = 3.0°C |MAD(206, 208, 212, 214)|
SC-6: |g||CLΣ| 206 |Σ+| 208 |Σ+| 212 |Σ+| 214 |Σ+| |g||MAD|

B2: Mean = 10; MAD = 1.6 shots |Mean/MAD(7, 9, 10, 11, 13)|
SC-6: |g||CLΣ| 7 |Σ+| 9 |Σ+| 10 |Σ+| 11 |Σ+| 13 |Σ+| |g||x-bar| |g||MAD|

B3: Mean = 50 g, MAD = 0 g (no deviation from the mean) |0|
`SC-6: 0 [ENTER]`

B4: Mean = 9 s, MAD = 2.0 s; Raven A is more consistent |MAD(6, 8, 10, 12)|
SC-6: |g||CLΣ| 6 |Σ+| 8 |Σ+| 10 |Σ+| 12 |Σ+| |g||MAD|

B5: Mean = 85; MAD = 4.8 |Mean/MAD(76, 82, 85, 88, 94)|
SC-6: |g||CLΣ| 76 |Σ+| 82 |Σ+| 85 |Σ+| 88 |Σ+| 94 |Σ+| |g||x-bar| |g||MAD|

B6: Mean = 15 m, MAD = 1.0 m |Mean/MAD(15, 16, 14, 15, 17...)|
SC-6: |g||CLΣ| 15 |Σ+| 16 |Σ+| 14 |Σ+| 15 |Σ+| 17 |Σ+| 13 |Σ+| |g||x-bar| |g||MAD|

B7: Mean = 85°F; deviations 5, 1, 1, 5; MAD = 3.0°F |MAD(80, 84, 86, 90)|
SC-6: |g||CLΣ| 80 |Σ+| 84 |Σ+| 86 |Σ+| 90 |Σ+| |g||MAD|

B8: The signed deviations (-5 + 0 + 5) always add to 0 |-5 + 5|
`SC-6: 5 [+/-] [ENTER] 5 [+]`

B9: Mean = 5 min; MAD = 2.4 min |Mean/MAD(2, 3, 4, 5, 11)|
SC-6: |g||CLΣ| 2 |Σ+| 3 |Σ+| 4 |Σ+| 5 |Σ+| 11 |Σ+| |g||x-bar| |g||MAD|

B10: Mean=16, Med=15, Mode=15, Rng=8, IQR=5.5, MAD=2.4 |Mean/Med/Mode/Range/Q1/Q3/I...
SC-6: |g||CLΣ| 12 |Σ+| 15 |Σ+| 15 |Σ+| 18 |Σ+| 20 |Σ+| |g||x-bar| |f||MED| |g||MODE| |RNG| |IQR| |g||...

Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability — Practice A & B Complete Key

Big Ideas Math Grade 7 Bridge — Chapters 1–3 & Chapter 7 • Concept → Blocks (When Applicable: Place Value, Fraction & Prime Blocks) → Calculator

1. **Concept (Stack Approach): Two-step equations unwrap in reverse order of operations (undo ±b first, then ÷a second), and squaring a negative in X (|+/-|[x²]) squares the entire signed value.**

2. Blocks (Place Value, Fraction & Prime): Model zero pairs and 2-step balance with Place Value blocks and probability part-to-whole ratios with Fraction Strips.

3. **Calculator (SC-6): Verify signed powers (5 |+/-|[x²] = +25), check 2-step equations with |!=|, and evaluate factorials |x!|, combinations |f|[nCr], and permutations |g|[nPr].**



SEC 10.1 (pp.95-96): ADDING AND SUBTRACTING RATIONAL NUMBERS

Methods: W1: Zero-Pair Cancellation | W2: Number-Line Jumps | W3: Absolute Value Sign Rules & Add-the-Opposite

PRACTICE A (p.95): KEY IDEAS (#1-#10)

ExA: -8 + 5 = -3; -6 + (-4) = -10

A1: (a) -6; (b) -24 |-15 + 9; -9|
SC-6: 15 [+/-] [ENTER] 9 [+] / 9 [+/-] [+]

A2: -9 |14 + -23|
SC-6: 14 [ENTER] 23 [+/-] [+]

A3: (a) 30; (b) -4 |12 - -18; -12|
SC-6: 12 [ENTER] 18 [+/-] [-] / 12 [+/-]...

A4: -25 |-9 - 16|
SC-6: 9 [+/-] [ENTER] 16 [-]

A5: -14.2°C |(0 - 7.8) - 6.4|
SC-6: 0 [ENTER] 7.8 [-] 6.4 [-]

A6: -4.8 |6.4 - 11.2|
SC-6: 6.4 [ENTER] 11.2 [-]

A7: +2.7 |-7.5 - -10.2|
SC-6: 7.5 [+/-] [ENTER] 10.2 [+/-] [-]

A8: -1/8 |-3/4 - -5/8|
SC-6: [.] 3 [.] 4 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [-]

A9: 36°F |14 - -22|
SC-6: 14 [ENTER] 22 [+/-] [-]

A10: 2 7/8 (2.875) |(-2.5 + 4.25) + 1.125|
SC-6: 2.5 [+/-] [ENTER] 4.25 [+] 1.125 [+]

PRACTICE B (p.96): REAL LIFE (#1-#10)

ExB: 14°C; and -1/8

B1: +8°F |-14 + 22|
SC-6: 14 [+/-] [ENTER] 22 [+]

B2: -83 meters |-45 + -38|
SC-6: 45 [+/-] [ENTER] 38 [+/-] [+]

B3: 4,800 - (-200) = 5,000 meters |4800 - -200|
SC-6: 4800 [ENTER] 200 [+/-] [-]

B4: -15 - (-25) = +10 points |-15 - -25|
SC-6: 15 [+/-] [ENTER] 25 [+/-] [-]

B5: +$8.75 |(-12.50 + 30) - 8.75|
SC-6: 12.50 [+/-] [ENTER] 30 [+] 8.75 [-]

B6: 1/8 |-5/8 + 3/4|
SC-6: [.] 5 [.] 8 [+/-] [ENTER] [.] 3 [.] 4 [+]

B7: -9/6 = -1 1/2 |-2/3 - 5/6|
SC-6: [.] 2 [.] 3 [+/-] [ENTER] [.] 5 [.] 6 [-]

B8: -16.0 meters |(-18.4 + 7.9) - 5.5|
SC-6: 18.4 [+/-] [ENTER] 7.9 [+] 5.5 [-]

B9: -15.0°C |-8.5 - 6.5|
SC-6: 8.5 [+/-] [ENTER] 6.5 [-]

B10: -8 |((-14 - -20) + -9) - 5|
SC-6: 14 [+/-] [ENTER] 20 [+/-] [-] 9 [+/-] [+] 5 [-]

SEC 10.2 (pp.97-98): MULTIPLYING AND DIVIDING RATIONAL NUMBERS

Methods: W1: Same/Different Sign Rules | W2: Repeated Addition / Equal Number-Line Jumps

PRACTICE A (p.97): KEY IDEAS (#1-#10)

ExA: 6×(-4) = -24, (-6)×(-4) = +24, (-5)³ = +25

A1: (a) -56; (b) +56 |7 × -8; -7|
SC-6: 7 [ENTER] 8 [+/-] [×] / 7 [+/-]...

A2: (a) -5; (b) +5 |-45 ÷ 9; -9|
SC-6: 45 [+/-] [ENTER] 9 [÷] / 9 [+/-] [÷]

A3: +25 (2 negative factors = positive) |-5²|
SC-6: 5 [+/-] [x²]

A4: -8 (3 negative factors = odd count = negative) |-2³|
SC-6: 2 [+/-] [x³]

A5: +8.4 |-3.5 × -2.4|
SC-6: 3.5 [+/-] [ENTER] 2.4 [+/-] [×]

A6: -12 |-14.4 ÷ 1.2|
SC-6: 14.4 [+/-] [ENTER] 1.2 [÷]

A7: -3/5 (-0.6) |-2/3 × 9/10|
SC-6: [.] 2 [.] 3 [+/-] [ENTER] [.] 9 [.] 10 [×]

A8: +2 |-5/6 ÷ -5/12|
SC-6: [.] 5 [.] 6 [+/-] [ENTER] [.] 5 [.] 12 [+/-] [÷]

A9: -120 |(-4 × -5) × -6|
SC-6: 4 [+/-] [ENTER] 5 [+/-] [×] 6 [+/-] [×]

A10: -76 |(-18 × 4) - (-36 ÷ -9)|
SC-6: 18 [+/-] [ENTER] 4 [×] 36 [+/-] [ENTER] 9 [+/-] [÷] [-]

PRACTICE B (p.98): REAL LIFE (#1-#10)

ExB: 2 minutes

B1: -120 feet |8 × -15|
SC-6: 8 [ENTER] 15 [+/-] [×]

B2: +40 points (same signs give a positive product) |-4 × -10|
SC-6: 4 [+/-] [ENTER] 10 [+/-] [×]

B3: -4.5°F per hour |-27 ÷ 6|
SC-6: 27 [+/-] [ENTER] 6 [÷]

B4: (-4)² = +16; (-4)³ = -64 |-4³|
SC-6: 4 [+/-] [x²] 4 [+/-] [x³]

B5: +2/3 |-3/4 × -8/9|
SC-6: [.] 3 [.] 4 [+/-] [ENTER] [.] 8 [.] 9 [+/-] [×]

B6: -2 |-7/8 ÷ 7/16|
SC-6: [.] 7 [.] 8 [+/-] [ENTER] [.] 7 [.] 16 [÷]

B7: -12 |(-2.5 × -1.6) × -3|
SC-6: 2.5 [+/-] [ENTER] 1.6 [+/-] [×] 3 [+/-] [×]

B8: -3.2 meters per year |-14.4 ÷ 4.5|
SC-6: 14.4 [+/-] [ENTER] 4.5 [÷]

B9: +120 |((-2 × -3) × -4) × -5|
SC-6: 2 [+/-] [ENTER] 3 [+/-] [×] 4 [+/-] [×] 5 [+/-] [×]

B10: 36 - (-20) + 6 = 62 |(-6² - (4 × -5)) + (-18 ÷ -3)|
SC-6: 6 [+/-] [x²] 4 [ENTER] 5 [+/-] [×] [-] 18 [+/-] [ENTER] 3 [+/-] [÷] [+]

SEC 10.3 (pp.99-100): SOLVING TWO-STEP EQUATIONS (AX + B = C)

Methods: W1: Two-Step Inverse Operations | W2: Balance Scale / Bar Model Unwrapping

PRACTICE A (p.99): KEY IDEAS (#1-#10)

ExA: x = 7 (Step 1: 3x = 21)

A1: 4x = 36; x = 9 |(45 - 9) ÷ 4|
SC-6: 45 [ENTER] 9 [-] 4 [÷]

A2: 5x = 50; x = 10 |(38 + 12) ÷ 5|
SC-6: 38 [ENTER] 12 [+] 5 [÷]

A3: 6y = 48; y = 8 |(59 - 11) ÷ 6|
SC-6: 59 [ENTER] 11 [-] 6 [÷]

A4: 8n = 64; n = 8 |(49 + 15) ÷ 8|
SC-6: 49 [ENTER] 15 [+] 8 [÷]

A5: m / 4 = 9; m = 36 |(15 - 6) × 4|
SC-6: 15 [ENTER] 6 [-] 4 [×]

A6: k / 3 = 12; k = 36 |(7 + 5) × 3|
SC-6: 7 [ENTER] 5 [+] 3 [×]

A7: -3x = -12; x = 4 |(2 - 14) ÷ -3|
SC-6: 2 [ENTER] 14 [-] 3 [+/-] [÷]

A8: 2.5w = 15; w = 6 |(23 - 8) ÷ 2.5|
SC-6: 23 [ENTER] 8 [-] 2.5 [÷]

A9: 25h + 35 = 160; h = 5 hours |(160 - 35) ÷ 25|
SC-6: 160 [ENTER] 35 [-] 25 [÷]

A10: 15w + 45 = 120; w = 5 weeks |(120 - 45) ÷ 15|
SC-6: 120 [ENTER] 45 [-] 15 [÷]

PRACTICE B (p.100): REAL LIFE (#1-#10)

ExB: 4 hours

B1: 6r + 8 = 38 → r = 5 laps |(38 - 8) ÷ 6|
SC-6: 38 [ENTER] 8 [-] 6 [÷]

B2: 4h + 12 = 48 → h = 9 hours |(48 - 12) ÷ 4|
SC-6: 48 [ENTER] 12 [-] 4 [÷]

B3: 15w + 35 = 140 → w = 7 weeks |(140 - 35) ÷ 15|
SC-6: 140 [ENTER] 35 [-] 15 [÷]

B4: 4b - 10 = 46 → b = $14 per book |(46 + 10) ÷ 4|
SC-6: 46 [ENTER] 10 [+] 4 [÷]

B5: x = 6 (-4·6 + 18 = -6) |(-6 - 18) ÷ -4|
SC-6: 6 [+/-] [ENTER] 18 [-] 4 [+/-] [÷]

B6: n = 35 |(16 - 9) × 5|
SC-6: 16 [ENTER] 9 [-] 5 [×]

B7: 3000 - 40m = 600 → m = 60 minutes |(600 - 3000) ÷ -40|
SC-6: 600 [ENTER] 3000 [-] 40 [+/-] [÷]

B8: x = 24 |(11 + 5) ÷ 2/3|
SC-6: 11 [ENTER] 5 [+] [.] 2 [.] 3 [÷]

B9: 2w + 36 = 56 → w = 10 cm |(56 - 36) ÷ 2|
SC-6: 56 [ENTER] 36 [-] 2 [÷]

B10: 2.25m + 3.50 = 21.50 → m = 8 miles |(21.50 - 3.50) ÷ 2.25|
SC-6: 21.50 [ENTER] 3.50 [-] 2.25 [÷]

SEC 10.4 (pp.101-102): THEORETICAL PROBABILITY, COMBINATIONS & PERMUTATIONS

Methods: W1: Probability Ratio & Complement Rule | W2: Tree Diagram / Outcome List | W3: Fundamental Counting Principle

PRACTICE A (p.101): KEY IDEAS (#1-#10)

ExA: P(Blue) = 2/5 = 0.4 = 40%; 5! = 120 ways

A1: P(5) = 1/6; P(not 6) = 5/6 |1/6; and 1 - 1/6 = 5/6|
SC-6: Mental / Table Check

A2: 0.65 (65% or 13/20) |1 - 0.35|
SC-6: 1 [ENTER] 0.35 [-]

A3: P(double 6s) = 1/36; P(sum 11 or 12) = 3/36 = 1/12 |1/36; and 3/36 = 1/12|
SC-6: Mental / Table Check

A4: 24 ways |4!|
SC-6: 4 [x!]

A5: 720 ways |6!|
SC-6: 6 [x!]

A6: 56 teams |C(8,3)|
SC-6: 8 [ENTER] 3 [f][nCr]

A7: 336 podium arrangements |P(8,3)|
SC-6: 8 [ENTER] 3 [g][nPr]

A8: 1/8 (12.5%) |1/2 × 1/2 × 1/2 = 1/8|
SC-6: Mental / Table Check

A9: 1/3 (6/10 × 5/9 = 30/90) |6/10 × 5/9 = 30/90|
SC-6: Mental / Table Check

A10: 120 combinations vs. 720 permutations (differ by 3! = 6) |C(10,3); gnPr|
SC-6: 10 [ENTER] 3 [f][nCr] / [g][nPr]

PRACTICE B (p.102): REAL LIFE (#1-#10)

ExB: 720 ordered podiums; 120 unordered 3-runner crews

B1: 4/20 = 1/5 = 0.20 = 20% |Simp(4/20)|
SC-6: [.] 4 [.] 20 [g][SIMP] [FDISP]

B2: 65% (13/20 = 0.65) |100 - 35|
SC-6: 100 [ENTER] 35 [-]

B3: P(black) = 2/5; P(not white) = 11/15 |Simp(6/15)|
SC-6: [.] 6 [.] 15 [g][SIMP]

B4: 120 custom skateboards |(5 × 4) × 6|
SC-6: 5 [ENTER] 4 [×] 6 [×]

B5: 5! = 120 orders |5!|
SC-6: 5 [x!]

B6: 210 batting orders (permutations) |P(7,3)|
SC-6: 7 [ENTER] 3 [g][nPr]

B7: 35 lab groups (combinations) |C(7,3)|
SC-6: 7 [ENTER] 3 [f][nCr]

B8: 28 fruit combinations |C(8,2)|
SC-6: 8 [ENTER] 2 [f][nCr]

B9: 60 times |160 × 3/8|
SC-6: 160 [ENTER] [.] 3 [.] 8 [×]

B10: 12 outcomes; P(H and 6) = 1/12 |2 × 6|
SC-6: 2 [ENTER] 6 [×]

Middle School 10-Unit Misconception & Hands-On Remediation Matrix

UNIT & HANDS-ON MODEL	TARGET CLASSROOM MISCONCEPTION	HANDS-ON BLOCK & SC-6 REMEDIATION
Unit 1: Numerical Expressi Place Value & Prime Fact	Students interpret remainders as meaningless decimal tails (e.g., .875 of a person), read 2^3 as 2×3=6, lose track of PEMDAS parentheses, and confuse GCF (overlap) with LCM (union).	Have students model division remainders on the Place Value Tray, stack Prime Factor Blocks ([2][2][2] = 2^3) for exponents, use the 4-line elevator (Y holds Room 1 while X solves Room 2) instead of rewriting PEMDAS lines, and stand logarithmic Prime Factor Towers side-by-side on the 12-socket lid.
Unit 2: Fractions & Decima Fraction Strips & Place	Students treat fractions, mixed numbers, and decimals as unrelated rules, expect multiplication to always make numbers bigger, forget which fraction to flip in division, and misalign decimal columns.	Use the 84.0 mm Fraction Studio towers so equivalence and common denominators are physical millimeter height matches. Use [f] [1/x] on Floor 1 (X) so students see only the divisor flip, and use [f] [IP] / [g] [FP] to split whole dollars from decimal cents.
Unit 3: Ratios, Rates & Pe Fraction Strips & Place	Students confuse Part-to-Part (3:2) with Part-to-Whole (3/5), add instead of multiply in ratio tables, divide unit rates upside-down, and memorize 6 disconnected rules for moving decimal points in percents.	Use equal Fraction Studio blocks as physical Tape Diagram units, step through ratio tables rhythmically with [g] [REP], use [x swap y] when unit-rate division is entered upside-down, and use [f] [%] + [FDISP] to show Fraction/Decimal/Percent are three costumes of one number.
Unit 4: Algebraic Expressi Place Value & Prime Fact	Students panic when letters appear, erroneously combine $3x + 2$ into $5x$ or $x^2 + x$ into x^3, and don't see why $3(x + 4) = 3x + 12$.	Show that a variable x is simply an empty Place Value tray socket waiting for a Digit Block to be substituted in, and that different place-value columns (x^2, x, and 1s) cannot be combined into one column! Use Prime Factor Studio to find the GCF when factoring expressions.
Unit 5: Areas of Polygons Fraction Strips & Place	Students multiply the slanted side of a parallelogram instead of the vertical height, forget to divide by 2 for triangles, lose track of multi-piece composite areas, and subtract clock times in base-10 instead of base-60.	Slide the corner triangle of a parallelogram to form a straight [f] [RECT] rectangle, use the dedicated [g] [TRI] key ([g][7]) which divides by 2 automatically, accumulate composite chunks with [Σ+], and use the Place Value Studio + [CLK] key for base-60 time.
Unit 6: Integers, Rational Place Value Blocks & Fra	Students confuse the subtraction operator [-] with the negative sign [+/-], think -18 > -5 because 18 > 5, confuse Opposite ([+/-]) with Absolute Value ([g] [|x|]), and reverse (x, y) coordinates.	Use Place Value Studio tokens and the SC-6 Number Line ([f] [LINE]) to model opposites and zero pairs, test inequalities with [f] [>] and [g] [<], contrast [+/-] (flip sign) with [g] [|x|] (always positive distance from 0), and use the Elevator Rule (walk across x first, ride up/down y second).
Unit 7: Equations & Inequa Place Value Blocks & Fra	Students treat '=' as 'write the answer', translate '8 less than n' backwards as 8 - n, resent writing -14 under both sides of $x + 14 = 39$, and confuse open vs. solid inequality circles.	Model '=' as a Balance Scale using Place Value Studio and Fraction Studio. Solve 1-step equations with inverse operations on both sides, verify your solution with the SC-6 Equality Test [!=] (which flashes NO when Left = Right!), and test inequalities on [f] [LINE] with [f] [>=] and [g] [<=] (which flash YES or NO).
Unit 8: Surface Area & Vol Place Value Blocks & Fra	Students drop one of the 6 faces when finding rectangular prism surface area, tangle prism and pyramid formulas, and confuse 2D Surface Area (cardboard outside) with 3D Volume (cubes inside).	Have students count the 3 pairs of opposite faces on a rectangular prism while logging each pair into [Σ+] (n=1, 2, 3), pack unit blocks or use [f] [PRISM] ([f][9]) to multiply 3 stack floors (l, w, h) at once for volume, and verify [f] [CIRC], [g] [CONE], and [g] [HYP].
Unit 9: Statistical Measur Place Value Blocks & Fra	Students forget to sort data before finding Median/Quartiles, lose their place adding long lists for Mean, and dread Mean Absolute Deviation (MAD) because doing it by hand takes 20 minutes of subtraction.	Show that Mean is physically leveling unequal Place Value digit stacks to the same height. Then show that entering a list ONCE with [Σ+] unlocks Mean ([g] [x-bar]), Median ([f] [MED]), Mode ([g] [MODE]), Range ([RNG]), Quartiles ([f] [1Q], [g] [3Q], [IQR]), and MAD ([g] [MAD]) in 1 click each!
Unit 10: Grade 7 Bridge — S Place Value, Fraction &	Students guess signs when adding/subtracting negatives, get -25 instead of +25 for $(-5)^2$ on algebraic calculators, undo ×3 before +7 in $3x + 7 = 28$, and confuse Permutations (order matters) with Combinations (order doesn't).	Use Place Value blocks and the SC-6 Number Line ([f] [LINE]) to show that removing a negative debt (10 - (-4)) equals +14. Notice how RPN 5 [+/-] [x^2] squares the entire register $(-5)^2 = +25$ without parentheses! Unwrap 2-step equations in reverse (undo ±b first, then ÷a second), and finish with [DICE], [f] [nCr], [g] [nPr], and [f] [?].

Photocopiable 10-Unit Middle School Student Mastery Tracker

STUDENT NAME / PERIOD	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10
01. _____										
02. _____										
03. _____										
04. _____										
05. _____										
06. _____										
07. _____										
08. _____										
09. _____										
10. _____										
11. _____										
12. _____										
13. _____										
14. _____										
15. _____										
16. _____										
17. _____										
18. _____										
19. _____										
20. _____										

TEACHER OBSERVATION NOTES & INTERVENTION LOG

- **Spiral-Bound Student Workbooks & Teacher Guides (Lulu): https://www.lulu.com/spotlight/stackcalc**
- Companion Book — From Blocks to the Stack (Amazon): https://link.amazon/B0f5cP0Ur | Manipulatives: https://stackcalc.io

General Student Lab Workbook Key — Part 1

Complete worked solutions and canonical SC-6 / SC-32 keystrokes for Warm-Up A/B and Investigations 1–4.

Domain / Unit 01: Counting, Cardinality & The One-More Step Counter (Pre-K – Kindergarten • SC-6)

Core Investigation (Warm-Up A): 5 ([5] Digit Block). Keystrokes: [4] [ENTER] [1] [+]

Core Keys: `4` `ENTER` `1` `+`

Extension Challenge (Warm-Up B): 4 more. Keystrokes: [6] [ENTER] [2] [-]

Ext. Keys: `6` `ENTER` `2` `-`

Lab Worksheet (pp. 2): Investigation 1: 5 | Investigation 2: 3 | Investigation 3: 0 (Zero) | Investigation 4: 4

Domain / Unit 02: Part-Part-Whole & Number Bonds on the Place Value Tray (Grades K–1 • SC-6)

Core Investigation (Warm-Up A): 9 ([9] Digit Block). Keystrokes: [5] [ENTER] [4] [+]

Core Keys: `5` `ENTER` `4` `+`

Extension Challenge (Warm-Up B): Missing Part is 3. Keystrokes: [8] [ENTER] [5] [-]

Ext. Keys: `8` `ENTER` `5` `-`

Lab Worksheet (pp. 4): Investigation 1: 4 | Investigation 2: 9 | Investigation 3: 9 | Investigation 4: 3

Domain / Unit 03: Complements to 10 & Make-10 Regrouping with Carry Tokens (Grade 1 • SC-6)

Core Investigation (Warm-Up A): 4 ones. Keystrokes: [1][0] [ENTER] [6] [-]

Core Keys: `1` `0` `ENTER` `6` `-`

Extension Challenge (Warm-Up B): 15. Keystrokes: [9] [ENTER] [1] [+] [5] [+]

Ext. Keys: `9` `ENTER` `1` `+` `5` `+`

Lab Worksheet (pp. 6): Investigation 1: 6 | Investigation 2: 12 | Investigation 3: 1 | Investigation 4: 14

Domain / Unit 04: Doubles, Near-Doubles & Equal Groups on the Place Value Tray (Grades 1–2 • SC-6)

Core Investigation (Warm-Up A): 14 ([1][4] on the tray). Keystrokes: [7] [ENTER] [+]

Core Keys: `7` `ENTER` `+`

Extension Challenge (Warm-Up B): 13. Keystrokes: [6] [ENTER] [+] [1] [+]

Ext. Keys: `6` `ENTER` `+` `1` `+`

Lab Worksheet (pp. 8): Investigation 1: 16 | Investigation 2: 17 | Investigation 3: 18 | Investigation 4: 11

Domain / Unit 05: The 4-Register Stack: Four Operations & Automatic Lift (Grades 3–4 • SC-6)

Core Investigation (Warm-Up A): 3 cookies per bag. Keystrokes: [1][5] [ENTER] [6] [-] [3] [÷]

Core Keys: `1` `5` `ENTER` `6` `-` `3` `÷`

Extension Challenge (Warm-Up B): 19. Keystrokes: [2][4] [ENTER] [6] [÷] [1][5] [+]

Ext. Keys: `2` `4` `ENTER` `6` `÷` `1` `5` `+`

Lab Worksheet (pp. 10): Investigation 1: 27 | Investigation 2: 6 | Investigation 3: 4 | Investigation 4: 3 eggs

Domain / Unit 06: Fraction Studio: Height Equivalence & Two-Dot Notation (Grades 4–5 • SC-6)

Core Investigation (Warm-Up A): 3/4 (height 63 mm). Keystrokes: [.][1][.][2] [ENTER] [.][1][.][4] [+]

Core Keys: `•` `1` `•` `2` `ENTER` `•` `1` `•` `4` `+`

Extension Challenge (Warm-Up B): 1/4 (height 21 mm). Keystrokes: [.][2][.][3] [ENTER] [.][3][.][8] [×]

Ext. Keys: `•` `2` `•` `3` `ENTER` `•` `3` `•` `8` `×`

Lab Worksheet (pp. 12): Investigation 1: 1/2 (42 mm) | Investigation 2: 3/4 (63 mm) | Investigation 3: 0.375 | Investigation 4: 1/2 (42 mm)

Diagnostic Grading Rubric

Award credit for: (1) Concrete manipulative model sketch, (2) Written mathematical strategy & prediction before pressing keys, and (3) Stack verification on the SC-6 or SC-32.

General Student Lab Workbook Key — Part 2

Complete worked solutions and canonical SC-6 / SC-32 keystrokes for Warm-Up A/B and Investigations 1–4.

Domain / Unit 07: Postfix Evaluation, Factor Trees & Chained Operations (Grades 5–6 • SC-6)

Core Investigation (Warm-Up A): 50. Keystrokes: [8] [ENTER] [2] [+] [1][4] [ENTER] [9] [-] [×]

Core Keys: 8 ENTER 2 + 1 4 ENTER 9 - ×

Extension Challenge (Warm-Up B): 2. Keystrokes: [1][5] [ENTER] [5] [-] [3] [ENTER] [2] [+] [÷]

Ext. Keys: 1 5 ENTER 5 - 3 ENTER 2 + ÷

Lab Worksheet (pp. 14): Investigation 1: 28 | Investigation 2: 2 | Investigation 3: 4 | Investigation 4: 17

Domain / Unit 08: Algebraic Expressions, Base-x Polynomials & Quadratic Factoring (Grades 7–8 • SC-32)

Core Investigation (Warm-Up A): (x + 3)(x + 4), roots -3 and -4. Keystrokes: [1] [ENTER] [7] [ENTER] [1][2] [g:][POLY]

Core Keys: 1 ENTER 7 ENTER 1 2 (→) POLY

Extension Challenge (Warm-Up B): (x - 2)(x - 3), roots 2 and 3. Keystrokes: [1] [ENTER] [5] [+/-] [ENTER] [6] [g:][POLY]

Ext. Keys: 1 ENTER 5 +/- ENTER 6 (→) POLY

Lab Worksheet (pp. 16): Investigation 1: (x + 2)(x + 4); roots -2, -4 | Investigation 2: (x - 2)(x + 2); roots 2, -2 | Investigation 3: (x + 1)(x + 4); roots -1, -4 | Investigation 4: (x - 2)(x - 3); roots 2, 3

Domain / Unit 09: Trigonometry, Vectors & Polar/Rectangular Coordinates (Grades 9–11 • SC-32)

Core Investigation (Warm-Up A): Displacement: 100 m, Heading: 53.13°. Keystrokes: [6][0] [ENTER] [8][0] [f:][→ϑ,r] [x swap y]

Core Keys: 6 0 ENTER 8 0 ↶ ▸θ,r x⇄y

Extension Challenge (Warm-Up B): Magnitude: 6.71. Resultant: <3, 6>.

Ext. Keys: 5 ENTER 2 +/- + 2 ENTER 4 + ↶ ▸θ,r x⇄y

Lab Worksheet (pp. 18): Investigation 1: ϑ = 22.62°, r = 13 | Investigation 2: 25 | Investigation 3: 0.5 | Investigation 4: x = 7.07, y = 7.07

Domain / Unit 10: Exponential Growth, Compound Interest & Logarithms (Grades 10–12 • SC-32)

Core Investigation (Warm-Up A): 40,500 bacteria. Keystrokes: [3] [ENTER] [4] [y^x] [5][0][0] [×]

Core Keys: 3 ENTER 4 y^x 5 0 0 ×

Extension Challenge (Warm-Up B): 19.80 years. Keystrokes: [2] [LN] [0][.][0][3][5] [÷]

Ext. Keys: 2 LN 0 • 0 3 5 ÷

Lab Worksheet (pp. 20): Investigation 1: 7.3891 | Investigation 2: 8.66 years | Investigation 3: 3162.28 | Investigation 4: 2.6990

Domain / Unit 11: Combinatorics, Probability Ratios & Series Accumulation (Grades 11–12 / AP • SC-32)

Core Investigation (Warm-Up A): 210 combinations. Keystrokes: [1][0] [f:][x!] [4] [f:][x!] [÷] [6] [f:][x!] [÷]

Core Keys: 1 0 ↶ x! 4 ↶ x! ÷ 6 ↶ x! ÷

Extension Challenge (Warm-Up B): Count n = 4, Mean x-bar = 16.5.

Ext. Keys: 1 2 Σ+ 1 5 Σ+ 1 8 Σ+ 2 1 Σ+

Lab Worksheet (pp. 22): Investigation 1: 720 | Investigation 2: 60 | Investigation 3: 21 | Investigation 4: Mean = 10

Domain / Unit 12: Digital Logic, Base Conversions & Computer Architecture (Grades 9–12 / AP CS • SC-32)

Core Investigation (Warm-Up A): HEX: FF, BIN: 11111111 (8 ones). Keystrokes: [2][5][5] [f:][BASE]

Core Keys: 2 5 5 ↶ BASE

Extension Challenge (Warm-Up B): Binary: 1111, Decimal: 15. Keystrokes: [1][2] [ENTER] [3] [OR]

Ext. Keys: 1 2 ENTER 3 OR

Lab Worksheet (pp. 24): Investigation 1: 101010 (32 + 8 + 2) | Investigation 2: 42 | Investigation 3: 7 (0111) | Investigation 4: 6 (0110)

Diagnostic Grading Rubric

Award credit for: (1) Concrete manipulative model sketch, (2) Written mathematical strategy & prediction before pressing keys, and (3) Stack verification on the SC-6 or SC-32.

Proficiency Trials Master Key — Part 1

Canonical answers and RPN keystrokes for Part 1 (Trials 1–6 & Cumulative C1).

Unit 1 Time Trial: Step Counter & Accumulator Sprint (Max Stack 1)

1. 3 ([1] [ENTER] [1] [+] [1] [+])
2. 7 ([5] [ENTER] [1] [+] [1] [+])
3. 4 ([1] [ENTER] [+] [+] [+])
4. 8 ([1] [ENTER] [+] [+] [+] [+] [+])
5. 4 ([6] [ENTER] [1] [-] [1] [-])
6. 5 ([C] [1] [ENTER] [+] [+] [+] [+])
7. 6 ([2] [ENTER] [2] [+] [2] [+])
8. 10 ([7] [ENTER] [1] [+] [1] [+] [1])
9. 9 ([0] [ENTER] [9] [+])
10. 7 ([10] [ENTER] [1] [-] [1] [-] [1])

Unit 2 Time Trial: Part-Part-Whole & Stack Addition (Max Stack 2)

1. 7 ([4] [ENTER] [3] [+])
2. 10 ([5] [ENTER] [5] [+])
3. 3 ([9] [ENTER] [6] [-])
4. 6 ([10] [ENTER] [4] [-])
5. 15 ([8] [ENTER] [7] [+])
6. 6 ([14] [ENTER] [8] [-])
7. 13 ([6] [ENTER] [7] [+])
8. 9 ([18] [ENTER] [9] [-])
9. 20 ([12] [ENTER] [8] [+])
10. 14 ([25] [ENTER] [11] [-])

Unit 3 Time Trial: Complements to 10 & Carry Regrouping (Max Stack 2)

1. 3 ([10] [ENTER] [7] [-])
2. 6 ([10] [ENTER] [4] [-])
3. 13 ([8] [ENTER] [2] [+] [3] [+])
4. 15 ([9] [ENTER] [1] [+] [5] [+])
5. 13 ([7] [ENTER] [3] [+] [3] [+])
6. 9 ([10] [ENTER] [1] [-])
7. 13 ([20] [ENTER] [7] [-])
8. 15 ([8] [ENTER] [2] [+] [5] [+])
9. 16 ([10] [ENTER] [6] [+])
10. 8 ([10] [ENTER] [2] [-])

Unit 4 Time Trial: Twin-Row Doubles, Near-Doubles & Groups (Max Stack 2)

1. 12 ([6] [ENTER] [+])
2. 16 ([8] [ENTER] [+])
3. 15 ([7] [ENTER] [+] [1] [+])
4. 13 ([6] [ENTER] [+] [1] [+])
5. 18 ([9] [ENTER] [+])
6. 17 ([8] [ENTER] [+] [1] [+])
7. 12 ([4] [ENTER] [3] [×])
8. 20 ([5] [ENTER] [4] [×])
9. 24 ([12] [ENTER] [+])
10. 19 ([9] [ENTER] [+] [1] [+])

Cumulative Trial C1: Primary Stack Fundamentals Championship (Max Stack 2)

1. 7 ([1] [ENTER] [+] [+] [+] [+] [+])
2. 16 ([9] [ENTER] [7] [+])
3. 9 ([17] [ENTER] [8] [-])
4. 7 ([10] [ENTER] [3] [-])
5. 28 ([14] [ENTER] [+])
6. 23 ([11] [ENTER] [+] [1] [+])
7. 17 ([9] [ENTER] [1] [+] [7] [+])
8. 30 ([5] [ENTER] [6] [×])
9. 18 ([30] [ENTER] [12] [-])
10. 21 ([8] [ENTER] [+] [5] [+])

Unit 5 Time Trial: 4-Register Stack & Four Operations (Max Stack 4)

1. 36 ([18] [ENTER] [6] [-] [3] [×])
2. 5 ([24] [ENTER] [16] [+] [8] [÷])
3. 5 ([45] [ENTER] [12] [ENTER] [3] [-] [÷])
4. 69 ([15] [ENTER] [4] [×] [18] [ENTER] [2])
5. 4 ([50] [ENTER] [14] [-] [3] [ENTER] [6])
6. 28 ([12] [ENTER] [5] [×] [8] [ENTER] [4])
7. 8 ([20] [ENTER] [8] [-] [4] [×] [6])
8. 24 ([72] [ENTER] [4] [ENTER] [5] [+] [÷])
9. 32 ([100] [ENTER] [16] [ENTER] [5] [×] [12])
10. 6 ([8] [ENTER] [4] [+] [9] [ENTER] [3])

Unit 6 Time Trial: Fraction Studio & Mixed Numbers (Max Stack 4)

1. 0.75 ([.1.2] [ENTER] [.1.4] [+])
2. 0.25 ([.3.4] [ENTER] [.1.2] [-])
3. 3.75 ([2.1.2] [ENTER] [1.1.4] [+])
4. 1.25 ([.4.5] [1/x])
5. 0.625 ([.3.8] [ENTER] [.1.4] [+])
6. 0.5 ([.2.3] [ENTER] [.3.4] [×])
7. 1.5 ([.3.4] [ENTER] [.1.2] [÷])
8. 6 ([1.1.2] [ENTER] [4] [×])
9. 2.25 ([5] [ENTER] [2.3.4] [-])
10. 6 ([.1.3] [ENTER] [.1.6] [+] [12] [×])

Speed Trial Scoring & Stack-Depth Discipline

Score (+5 × Correct) − (2 × Incorrect), plus +1 bonus point per full minute remaining when all 10 problems are solved within the allowed Max Stack Depth.

Proficiency Trials Master Key — Part 2

Canonical answers and RPN keystrokes for Part 2 (Trials 7–12 & Cumulative C2–C3).

Unit 7 Time Trial: Postfix Evaluation & Chained Operations (Max Stack 4)

1. 50 ([8] [ENTER] [2] [+] [9] [ENTER] [4])
2. 30 ([14] [ENTER] [6] [-] [3] [×] [24])
3. 20 ([30] [ENTER] [4] [ENTER] [5] [×] [-])
4. 12 ([5] [ENTER] [3] [+] [8] [ENTER] [2])
5. 40 ([18] [ENTER] [3] [÷] [4] [+] [15])
6. 5 ([10] [ENTER] [5] [ENTER] [2] [×] [+])
7. 70 ([25] [ENTER] [5] [ENTER] [3] [×] [-])
8. 8 ([7] [ENTER] [3] [+] [4] [×] [8])
9. 36 ([48] [ENTER] [2] [ENTER] [6] [+] [÷])
10. 30 ([16] [ENTER] [4] [ENTER] [2] [+] [-])

Unit 8 Time Trial: Base-x Polynomials & Quadratic Factoring (Max Stack 4)

1. -2 ([1] [ENTER] [5] [ENTER] [6] [POLY])
2. -3 ([1] [ENTER] [7] [ENTER] [12] [POLY])
3. -3 ([1] [ENTER] [8] [ENTER] [15] [POLY])
4. 3 ([1] [ENTER] [1] [+/-] [ENTER] [6] [+/-])
5. 2 ([1] [ENTER] [5] [+/-] [ENTER] [6] [POLY])
6. 3 ([1] [ENTER] [0] [ENTER] [9] [+/-] [POLY])
7. 4 ([1] [ENTER] [0] [ENTER] [16] [+/-] [POLY])
8. 25 ([3] [ENTER] [x^2] [2] [×] [3] [ENTER])
9. -0.5 ([2] [ENTER] [5] [ENTER] [2] [POLY])
10. -5 ([1] [ENTER] [10] [ENTER] [25] [POLY])

Cumulative Trial C2: Intermediate Stack Mastery Championship (Max Stack 4)

1. 15 ([32] [ENTER] [14] [-] [5] [×] [6])
2. 0.875 ([.1.2] [ENTER] [.1.4] [×] [.3.4] [+])
3. 10 ([9] [ENTER] [3] [-] [x^2] [12] [ENTER])
4. 1.75 ([3.1.2] [ENTER] [1.3.4] [-])
5. 10 ([15] [ENTER] [5] [+] [8] [ENTER] [3])
6. 27 ([18] [ENTER] [+] [9] [-])
7. 3 ([10] [ENTER] [2] [+/-] [-] [3] [ENTER])
8. 32 ([80] [ENTER] [15] [ENTER] [4] [×] [-])
9. 2.4 ([.1.4] [ENTER] [.1.6] [+] [1/x])
10. 20 ([12] [x^2] [16] [x^2] [+] [√x])

Unit 9 Time Trial: Trigonometry, Vectors & Polar Pipeline (Max Stack 4)

1. 0.5 ([30] [SIN])
2. 0.5 ([60] [COS])
3. 1 ([45] [TAN])
4. 5 ([10] [ENTER] [30] [SIN] [×])
5. 6 ([12] [ENTER] [60] [COS] [×])
6. 10 ([10] [ENTER] [0] [yellow] [→y,x])
7. 5 ([3] [ENTER] [4] [yellow] [→ϑ,r])
8. 13 ([5] [ENTER] [12] [yellow] [→ϑ,r])
9. 10 ([8] [ENTER] [6] [yellow] [→ϑ,r])
10. 30 ([.5] [yellow] [ASIN])

Unit 10 Time Trial: Exponential Growth & Logarithms (Max Stack 4)

1. 32 ([2] [ENTER] [5] [y^x])
2. 81 ([3] [ENTER] [4] [y^x])
3. 1 ([1] [e^x] [LN])
4. 1 ([0] [e^x])
5. 3 ([1000] [yellow] [LOG])
6. 1102.5 ([1000] [ENTER] [1.05] [ENTER] [2] [y^x] [×])
7. 9.9021 ([2] [LN] [.07] [÷])
8. 12.5 ([100] [ENTER] [.5] [ENTER] [3] [y^x] [×])
9. 610.7014 ([.04] [ENTER] [5] [×] [e^x] [500] [×])
10. 1024 ([2] [ENTER] [10] [y^x])

Unit 11 Time Trial: Combinatorics, Stats & Probability (Max Stack 4)

1. 120 ([5] [yellow] [x!])
2. 720 ([6] [yellow] [x!])
3. 20 ([5] [yellow] [x!] [3] [yellow] [x!] [÷])
4. 120 ([6] [yellow] [x!] [3] [yellow] [x!] [÷])
5. 10 ([5] [yellow] [x!] [2] [yellow] [x!] [3])
6. 15 ([6] [yellow] [x!] [2] [yellow] [x!] [4])
7. 35 ([7] [yellow] [x!] [3] [yellow] [x!] [4])
8. 15 ([1] [Σ+] [2] [Σ+] [3] [Σ+] [4])
9. 24 ([4] [yellow] [x!])
10. 0.0667 ([6] [yellow] [x!] [2] [yellow] [x!] [4])

Unit 12 Time Trial: Digital Logic, Base Conversions & CS (Max Stack 4)

1. FF ([255] [yellow] [BASE] [HEX])
2. 16 ([HEX] [10] [yellow] [BASE] [DEC])
3. 1111 ([15] [yellow] [BASE] [BIN])
4. 10 ([BIN] [1010] [yellow] [BASE] [DEC])
5. 8 ([12] [ENTER] [10] [AND])
6. 14 ([12] [ENTER] [10] [OR])
7. 6 ([12] [ENTER] [10] [XOR])
8. 32 ([HEX] [20] [yellow] [BASE] [DEC])
9. 100 ([64] [yellow] [BASE] [OCT])
10. 15 ([255] [ENTER] [15] [AND])

Cumulative Trial C3: Grand RPN Invitational Championship (Max Stack 4)

1. 281.79 ([.06] [ENTER] [12] [÷] [1] [+] [24])
2. 17 ([5] [ENTER] [3] [+/-] [-] [x^2] [10])
3. 12.0208 ([12] [ENTER] [45] [COS] [×] [5] [ENTER])
4. 3.0667 ([3.1.3] [ENTER] [1.1.4] [÷] [.2.5] [+])
5. 56 ([8] [yellow] [x!] [3] [yellow] [x!] [5])
6. 2473.08 ([.05] [ENTER] [10] [×] [e^x] [1500] [×])
7. 7 ([35] [ENTER] [7] [-] [4] [×] [14])
8. 192 ([192] [ENTER] [240] [AND])
9. 29 ([6] [ENTER] [+] [8] [ENTER] [+] [1])
10. FFF ([4095] [yellow] [BASE] [HEX])

Speed Trial Scoring & Stack-Depth Discipline

Score (+5 × Correct) − (2 × Incorrect), plus +1 bonus point per full minute remaining when all 10 problems are solved within the allowed Max Stack Depth.

Manipulative Kits & Keycap Quick-Reference

How each physical manipulative models the 4-register RPN stack, plus essential SC-6 and SC-32 keycap sequences.

Classroom Manipulative Kit	Target Domains	Core Mathematical & RPN Stack Connection
Place Value Studio: Digit Blocks (0–9)	**Domains 01, 02, 04**	Subitizing, part-part-whole bonds (Y + X) & doubles ([ENTER][+])
Place Value Studio: Gold [+1] Carry Tokens	**Domain 03**	10:1 column overflow trade (10 ones → 1 Gold [+1] carry token)
Place Value Studio: Whole Tray (6×2)	**Domain 05**	Multi-digit place-value alignment, 10:1 trading & [PV] check
Place Value Studio: Decimal Point & Tray	**Domains 05, 06**	Tenths/Hundredths alignment, [RND]/[FIX] & fraction-to-decimal
Place Value Studio: Binary Bit Sockets	**Domain 12**	0 & 1 bit blocks in powers-of-2 columns (128..1) & bitwise logic
Fraction Studio: 84 mm Fraction Towers	**Domain 06**	1 Whole = 84 mm down to 1/12; two-dot fraction entry ([.][n][.][d])
Fraction Studio: 12-Socket Comparator Lid	**Domains 06, 09, 11**	Flush-top equivalence check, sine/cosine ratios & P(E) heights
Prime Factor Studio: Logarithmic Blocks	**Domains 07, 10**	Heights h ~ ln(p) for primes 2, 3, 5, 7; stacking adds logs to multiply
Prime Factor Studio: GCF/LCM Comparator	**Domains 07, 08**	Shared prime sub-tower = [GCF]; union of prime towers = [LCM]
Prime Factor Studio: Exponential Towers	**Domains 08, 10**	Identical prime stacks (p^n) & Fundamental Theorem uniqueness

CANONICAL SC-6 & SC-32 KEYCAP SEQUENCES (EXACT HARDWARE KEYCAPS)

Chained Expression (3×4)+(5×6) = 42	**SC-6**	3 ENTER 4 × 5 ENTER 6 × +
Two-Dot Fraction 2/3 + 1/4 = 11/12	**SC-6**	• 2 • 3 ENTER • 1 • 4 +
Mixed Number 2 3/8 → Improper 19/8	**SC-6**	2 • 3 • 8 ENTER f d/c
Quotient & Remainder 23 ÷ 4 (Y=5, X=3)	**SC-6**	2 3 ENTER 4 g ÷R
Prime Factorization & GCF(24, 36) = 12	**SC-6**	2 4 g 2 4 ENTER 3 6 GCF
Quadratic Roots of $x^2 + 5x + 6 = 0$	**SC-32**	1 ENTER 5 ENTER 6 R/S POLY
2D Polar Vector (x=3, y=4) → r=5, ϑ=53.13°	**SC-32**	4 ENTER 3 ← →θ,r
Continuous Growth 500 × e^(0.04×5)	**SC-32**	• 0 4 ENTER 5 × e^x 5 0 0 ×

Concept Pacing & Classroom Observation Log

Photocopiable yearly planning tracker for Concept Domains 01–12 and Middle School Units 01–10.

Concept Domain & Topic	Dates Taught	Components Completed	Student Mastery Notes
Domain 01: Cardinality, Counting & Step Accumulation	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 02: Part-Part-Whole Number Bonds & Missing Addends	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 03: Complements to 10 & Gold [+1] Carry Regrouping	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 04: Twin-Row Place Value Tray: Doubles & Equal Groups	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 05: Place Value & The 4-Level Stack Elevator	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 06: Fraction Studio: 84 mm Towers & Two-Dot Entry	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 07: Postfix Trees, Order of Operations & Prime Factors	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 08: Base-x Polynomials, Prime Factor Pairs & Roots	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 09: Coordinate Geometry, Trigonometry & Vectors	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 10: Exponents, Logarithms & Continuous Growth	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 11: Combinatorics, Probability Ratios & Series Statistics	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 12: Digital Logic, Base-N & Computer Architecture	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	

CLASSROOM MISCONCEPTION NOTES & SMALL-GROUP INTERVENTION LOG

PART III • REPRODUCIBLE FAMILY HANDOUT (PHOTOCOPY & SEND HOME ON DAY 1)

Welcome to Hands-On & Stack Mathematics!

A 2-Minute Parent & Family Guide: Why Our Classroom Uses Physical Blocks and a Calculator with No [=] Sign.

DEAR PARENTS & FAMILIES: WHY YOUR CHILD'S CALCULATOR HAS AN [ENTER] KEY INSTEAD OF [=]

This year, our math classroom pairs hands-on blocks (Place Value Digit & Carry Blocks, 84 mm Fraction Towers, and Logarithmic Prime Factor Blocks) with a 4-line StackCalc RPN (Reverse Polish Notation) calculator. If you grew up pressing 8 + 5 = on a standard calculator, you might wonder why our calculator has an [ENTER] button and no [=] key—and no parentheses keys!

• Standard 1-line calculators hide the math: Students often type long expressions blindly from left to right (or get lost in parentheses) without knowing which step happens first.

• The 4-Line Stack works just like paper-and-pencil math: On paper, you write the first number on the top line (Line Y), write the second number below it (Line X), and then add (+). That is exactly how our calculator works! Every step stays visible on the 4-line screen so your child always sees the 'why' behind the answer.

THE 3-EXAMPLE KITCHEN-TABLE CHEAT SHEET (TRY THIS WITH YOUR CHILD TONIGHT!)

Example 1: Everyday Addition — How to Calculate 8 + 5 = 13

1. Type [8] and press [ENTER] to place 8 on Waiting Shelf Y. 2. Type [5] on Workbench X. 3. Press [+] to combine Y + X = 13!

Keys to Press: [8] [ENTER] [5] [+]

Example 2: Order of Operations Without Parentheses — (8 + 4) × (9 − 3) = 72

1. Solve the first group: [8] [ENTER] [4] [+] → 12 waits visibly on Line Y! 2. Solve the second group: [9] [ENTER] [3] [-] → 6 sits on Line X. 3. Press [×] to multiply 12 × 6 = 72!

Keys to Press: [8] [ENTER] [4] [+] [9] [ENTER] [3] [-] [×]

Example 3: Real Fractions Using Two Dots — 1/3 + 1/4 = 7/12

Pressing [.] twice types a real fraction ([.] [1] [.] [3] = 1/3). Press [ENTER], type 1/4 ([.] [1] [.] [4]), and press [+] to see 7/12!

Keys to Press: [•] [1] [•] [3] [ENTER] [•] [1] [•] [4] [+]

3 GREAT QUESTIONS TO ASK YOUR CHILD DURING HOMEWORK

1. "Before you touch the calculator keys, can you show me or sketch what this problem looks like with blocks?"

2. "Which inner part of this problem do we need to solve first so it can wait on Shelf Y?"

3. "What number is sitting on Waiting Shelf Y right now, and what number is on Workbench X?"

Explore free printable family math activities & interactive guides at: https://stackcalc.io/curriculum/

OPTIONAL FAMILY RETURN SLIP (TRY THE 3 CHEAT-SHEET EXAMPLES TOGETHER!)

Student Name: _____ Parent/Guardian Signature: _____

[ ] We tried the 3 Stack Calculator examples together at home! Teacher / Class: _____